2

Evolution of the Major Programming Languages

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This chapter describes the development of a collection of programming languages. It explores the environment in which each was designed and focuses on the contributions of the language and the motivation for its development. Overall language descriptions are not included; rather, we discuss only some of the new features introduced by each language. Of particular interest are the features that most influenced subsequent languages or the field of computer science.

This chapter does not include an in-depth discussion of any language feature or concept; that is left for later chapters. Brief, informal explanations of features will suffice for our trek through the development of these languages.

This chapter discusses a wide variety of languages and language concepts that will not be familiar to many readers. These topics are discussed in detail only in later chapters. Those who find this unsettling may prefer to delay reading this chapter until the rest of the book has been studied.

The choice as to which languages to discuss here was subjective, and some readers will unhappily note the absence of one or more of their favorites. However, to keep this historical coverage to a reasonable size, it was necessary to leave out some languages that some regard highly. The choices were based on our estimate of each language's importance to language development and the computing world as a whole. We also include brief discussions of some other languages that are referenced later in the book.

The organization of this chapter is as follows: The initial versions of languages generally are discussed in chronological order. However, subsequent versions of languages appear with their initial version, rather than in later sections. For example, Fortran 2003 is discussed in the section with Fortran I (1956). Also, in some cases, languages of secondary importance that are related to a language that has its own section appear in that section.

This chapter includes listings of 14 complete example programs, each in a different language. These programs are not described in this chapter; they are meant simply to illustrate the appearance of programs in these languages. Readers familiar with any of the common imperative languages should be able to read and understand most of the code in these programs, except those in LISP, COBOL, and Smalltalk. (A Scheme function similar to the LISP example is discussed in Chapter 15.) The same problem is solved by the Fortran, ALGOL 60, PL/I, BASIC, Pascal, C, Perl, Ada, Java, JavaScript, and C# programs. Note that most of the contemporary languages in this list support dynamic arrays, but because of the simplicity of the example problem, we did not use them in the example programs. Also, in the Fortran 95 program, we avoided using the features that could have avoided the use of loops altogether, in part to keep the program simple and readable and in part just to illustrate the basic loop structure of the language.

Figure 2.1 is a chart of the genealogy of the high-level languages discussed in this chapter.

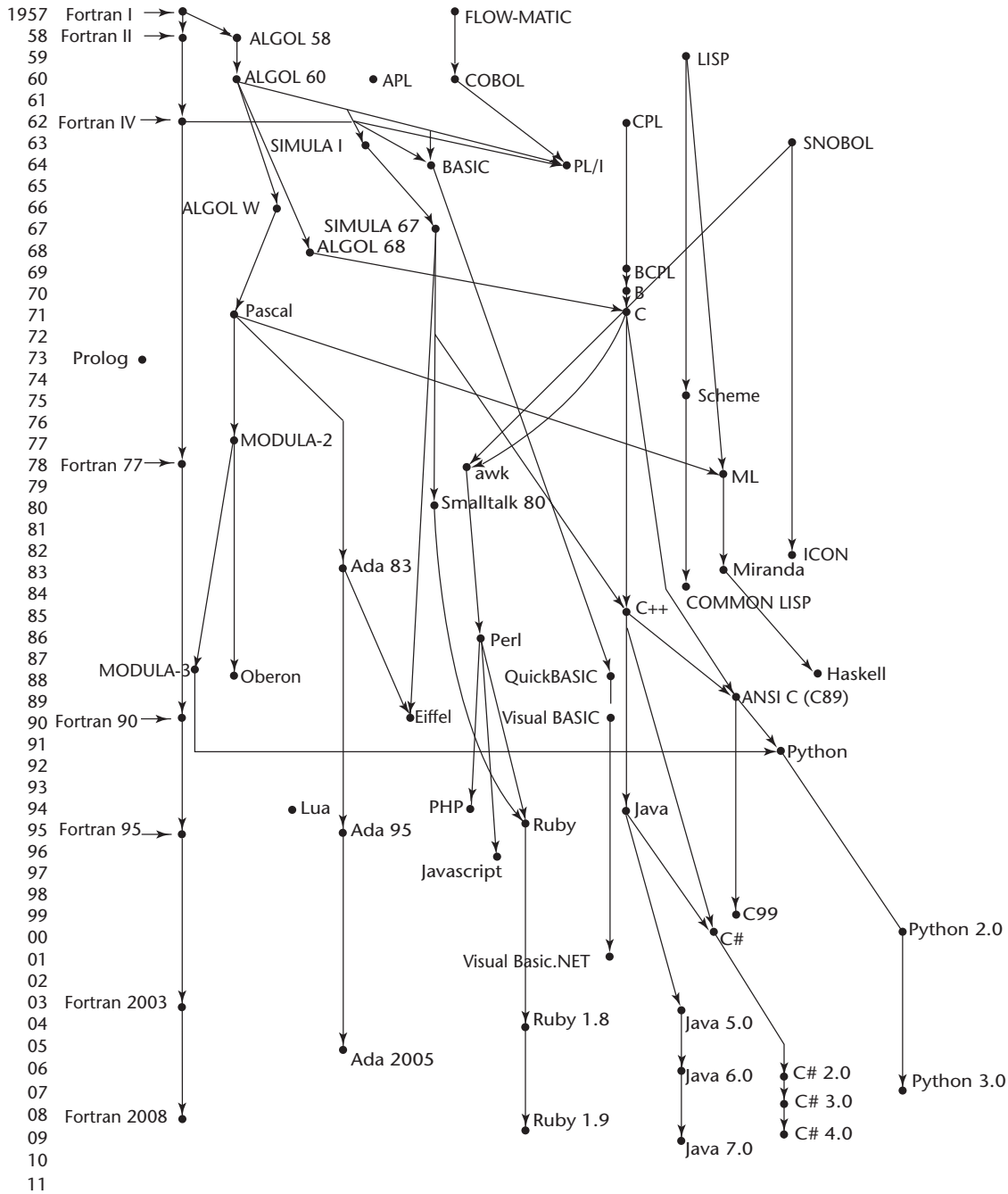

Figure 2.1

Genealogy of common high-level programming languages

2.1 Zuse's Plankalkül

The first programming language discussed in this chapter is highly unusual in several respects. For one thing, it was never implemented. Furthermore, although developed in 1945, its description was not published until 1972. Because so few people were familiar with the language, some of its capabilities did not appear in other languages until 15 years after its development.

2.1.1 Historical Background

Between 1936 and 1945, German scientist Konrad Zuse (pronounced "Tsoo-zuh") built a series of complex and sophisticated computers from electromechanical relays. By early 1945, Allied bombing had destroyed all but one of his latest models, the Z4, so he moved to a remote Bavarian village, Hinterstein, and his research group members went their separate ways.

Working alone, Zuse embarked on an effort to develop a language for expressing computations for the Z4, a project he had begun in 1943 as a proposal for his Ph.D. dissertation. He named this language Plankalkül, which means *program calculus.* In a lengthy manuscript dated 1945 but not published until 1972 (Zuse, 1972), Zuse defined Plankalkül and wrote algorithms in the language to solve a wide variety of problems.

2.1.2 Language Overview

Plankalkül was remarkably complete, with some of its most advanced features in the area of data structures. The simplest data type in Plankalkül was the single bit. Integer and floating-point numeric types were built from the bit type. The floating-point type used twos-complement notation and the "hidden bit" scheme currently used to avoid storing the most significant bit of the normalized fraction part of a floating-point value.

In addition to the usual scalar types, Plankalkül included arrays and records (called *structs* in the C-based languages). The records could include nested records.

Although the language had no explicit goto, it did include an iterative statement similar to the Ada **`for`**. It also had the command **`Fin`** with a superscript that specified an exit out of a given number of iteration loop nestings or to the beginning of a new iteration cycle. Plankalkül included a selection statement, but it did not allow an `else` clause.

One of the most interesting features of Zuse's programs was the inclusion of mathematical expressions showing the current relationships between program variables. These expressions stated what would be true during execution at the points in the code where they appeared. These are very similar to the assertions of Java and in those in axiomatic semantics, which is discussed in Chapter 3.

Zuse's manuscript contained programs of far greater complexity than any written prior to 1945. Included were programs to sort arrays of numbers; test the connectivity of a given graph; carry out integer and floating-point operations, including square root; and perform syntax analysis on logic formulas that had parentheses and operators in six different levels of precedence. Perhaps most remarkable were his 49 pages of algorithms for playing chess, a game in which he was not an expert.

If a computer scientist had found Zuse's description of Plankalkül in the early 1950s, the single aspect of the language that would have hindered its implementation as defined would have been the notation. Each statement consisted of either two or three lines of code. The first line was most like the statements of current languages. The second line, which was optional, contained the subscripts of the array references in the first line. The same method of indicating subscripts was used by Charles Babbage in programs for his Analytical Engine in the middle of the nineteenth century. The last line of each Plankalkül statement contained the type names for the variables mentioned in the first line. This notation is quite intimidating when first seen.

The following example assignment statement, which assigns the value of the expression `A[4] +1` to `A[5]`, illustrates this notation. The row labeled `V` is for subscripts, and the row labeled `S` is for the data types. In this example, `1.n` means an integer of *n* bits:

```
  | A + 1 => A
V | 4        5
S | 1.n      1.n
```

We can only speculate on the direction that programming language design might have taken if Zuse's work had been widely known in 1945 or even 1950. It is also interesting to consider how his work might have been different had he done it in a peaceful environment surrounded by other scientists, rather than in Germany in 1945 in virtual isolation.

2.2 Pseudocodes

First, note that the word *pseudocode* is used here in a different sense than its contemporary meaning. We call the languages discussed in this section pseudocodes because that's what they were named at the time they were developed and used (the late 1940s and early 1950s). However, they are clearly not pseudocodes in the contemporary sense.

The computers that became available in the late 1940s and early 1950s were far less usable than those of today. In addition to being slow, unreliable, expensive, and having extremely small memories, the machines of that time were difficult to program because of the lack of supporting software.

There were no high-level programming languages or even assembly languages, so programming was done in machine code, which is both tedious and

error prone. Among its problems is the use of numeric codes for specifying instructions. For example, an ADD instruction might be specified by the code 14 rather than a connotative textual name, even if only a single letter. This makes programs very difficult to read. A more serious problem is absolute addressing, which makes program modification tedious and error prone. For example, suppose we have a machine language program stored in memory. Many of the instructions in such a program refer to other locations within the program, usually to reference data or to indicate the targets of branch instructions. Inserting an instruction at any position in the program other than at the end invalidates the correctness of all instructions that refer to addresses beyond the insertion point, because those addresses must be increased to make room for the new instruction. To make the addition correctly, all instructions that refer to addresses that follow the addition must be found and modified. A similar problem occurs with deletion of an instruction. In this case, however, machine languages often include a "no operation" instruction that can replace deleted instructions, thereby avoiding the problem.

These are standard problems with all machine languages and were the primary motivations for inventing assemblers and assembly languages. In addition, most programming problems of that time were numerical and required floating-point arithmetic operations and indexing of some sort to allow the convenient use of arrays. Neither of these capabilities, however, was included in the architecture of the computers of the late 1940s and early 1950s. These deficiencies naturally led to the development of somewhat higher-level languages.

2.2.1 Short Code

The first of these new languages, named Short Code, was developed by John Mauchly in 1949 for the BINAC computer, which was one of the first successful stored-program electronic computers. Short Code was later transferred to a UNIVAC I computer (the first commercial electronic computer sold in the United States) and, for several years, was one of the primary means of programming those machines. Although little is known of the original Short Code because its complete description was never published, a programming manual for the UNIVAC I version did survive (Remington-Rand, 1952). It is safe to assume that the two versions were very similar.

The words of the UNIVAC I's memory had 72 bits, grouped as 12 six-bit bytes. Short Code consisted of coded versions of mathematical expressions that were to be evaluated. The codes were byte-pair values, and many equations could be coded in a word. The following operation codes were included:

```
01  -     06  abs value    1n  (n+2)nd power
02  )     07  +            2n  (n+2)nd root
03  =     08  pause        4n  if <= n
04  /     09  (            58  print and tab
```

Variables were named with byte-pair codes, as were locations to be used as constants. For example, `X0` and `Y0` could be variables. The statement

```
X0 = SQRT(ABS(Y0))
```

would be coded in a word as `00 X0 03 20 06 Y0`. The initial `00` was used as padding to fill the word. Interestingly, there was no multiplication code; multiplication was indicated by simply placing the two operands next to each other, as in algebra.

Short Code was not translated to machine code; rather, it was implemented with a pure interpreter. At the time, this process was called *automatic programming*. It clearly simplified the programming process, but at the expense of execution time. Short Code interpretation was approximately 50 times slower than machine code.

2.2.2 Speedcoding

In other places, interpretive systems were being developed that extended machine languages to include floating-point operations. The Speedcoding system developed by John Backus for the IBM 701 is an example of such a system (Backus, 1954). The Speedcoding interpreter effectively converted the 701 to a virtual three-address floating-point calculator. The system included pseudoinstructions for the four arithmetic operations on floating-point data, as well as operations such as square root, sine, arc tangent, exponent, and logarithm. Conditional and unconditional branches and input/output conversions were also part of the virtual architecture. To get an idea of the limitations of such systems, consider that the remaining usable memory after loading the interpreter was only 700 words and that the add instruction took 4.2 milliseconds to execute. On the other hand, Speedcoding included the novel facility of automatically incrementing address registers. This facility did not appear in hardware until the UNIVAC 1107 computers of 1962. Because of such features, matrix multiplication could be done in 12 Speedcoding instructions. Backus claimed that problems that could take two weeks to program in machine code could be programmed in a few hours using Speedcoding.

2.2.3 The UNIVAC "Compiling" System

Between 1951 and 1953, a team led by Grace Hopper at UNIVAC developed a series of "compiling" systems named A-0, A-1, and A-2 that expanded a pseudocode into machine code subprograms in the same way as macros are expanded into assembly language. The pseudocode source for these "compilers" was still quite primitive, although even this was a great improvement over machine code because it made source programs much shorter. Wilkes (1952) independently suggested a similar process.

2.2.4 Related Work

Other means of easing the task of programming were being developed at about the same time. At Cambridge University, David J. Wheeler (1950) developed a method of using blocks of relocatable addresses to solve, at least partially, the problem of absolute addressing, and later, Maurice V. Wilkes (also at Cambridge) extended the idea to design an assembly program that could combine chosen subroutines and allocate storage (Wilkes et al., 1951, 1957). This was indeed an important and fundamental advance.

We should also mention that assembly languages, which are quite different from the pseudocodes discussed, evolved during the early 1950s. However, they had little impact on the design of high-level languages.

2.3 The IBM 704 and Fortran

Certainly one of the greatest single advances in computing came with the introduction of the IBM 704 in 1954, in large measure because its capabilities prompted the development of Fortran. One could argue that if it had not been IBM with the 704 and Fortran, it would soon thereafter have been some other organization with a similar computer and related high-level language. However, IBM was the first with both the foresight and the resources to undertake these developments.

2.3.1 Historical Background

One of the primary reasons why the slowness of interpretive systems was tolerated from the late 1940s to the mid-1950s was the lack of floating-point hardware in the available computers. All floating-point operations had to be simulated in software, a very time-consuming process. Because so much processor time was spent in software floating-point processing, the overhead of interpretation and the simulation of indexing were relatively insignificant. As long as floating-point had to be done by software, interpretation was an acceptable expense. However, many programmers of that time never used interpretive systems, preferring the efficiency of hand-coded machine (or assembly) language. The announcement of the IBM 704 system, with both indexing and floating-point instructions in hardware, heralded the end of the interpretive era, at least for scientific computation. The inclusion of floating-point hardware removed the hiding place for the cost of interpretation.

Although Fortran is often credited with being the first compiled high-level language, the question of who deserves credit for implementing the first such language is somewhat open. Knuth and Pardo (1977) give the credit to Alick E. Glennie for his Autocode compiler for the Manchester Mark I computer. Glennie developed the compiler at Fort Halstead, Royal Armaments Research Establishment, in England. The compiler was operational by September 1952. However, according to John Backus (Wexelblat, 1981, p. 26),

Glennie's Autocode was so low level and machine oriented that it should not be considered a compiled system. Backus gives the credit to Laning and Zierler at the Massachusetts Institute of Technology.

The Laning and Zierler system (Laning and Zierler, 1954) was the first algebraic translation system to be implemented. By algebraic, we mean that it translated arithmetic expressions, used separately coded subprograms to compute transcendental functions (e.g., sine and logarithm), and included arrays. The system was implemented on the MIT Whirlwind computer, in experimental prototype form, in the summer of 1952 and in a more usable form by May 1953. The translator generated a subroutine call to code each formula, or expression, in the program. The source language was easy to read, and the only actual machine instructions included were for branching. Although this work preceded the work on Fortran, it never escaped MIT.

In spite of these earlier works, the first widely accepted compiled high-level language was Fortran. The following subsections chronicle this important development.

2.3.2 Design Process

Even before the 704 system was announced in May 1954, plans were begun for Fortran. By November 1954, John Backus and his group at IBM had produced the report titled "The IBM Mathematical FORmula TRANslating System: FORTRAN" (IBM, 1954). This document described the first version of Fortran, which we refer to as Fortran 0, prior to its implementation. It also boldly stated that Fortran would provide the efficiency of hand-coded programs and the ease of programming of the interpretive pseudocode systems. In another burst of optimism, the document stated that Fortran would eliminate coding errors and the debugging process. Based on this premise, the first Fortran compiler included little syntax error checking.

The environment in which Fortran was developed was as follows: (1) Computers had small memories and were slow and relatively unreliable; (2) the primary use of computers was for scientific computations; (3) there were no existing efficient and effective ways to program computers; and (4) because of the high cost of computers compared to the cost of programmers, speed of the generated object code was the primary goal of the first Fortran compilers. The characteristics of the early versions of Fortran follow directly from this environment.

2.3.3 Fortran I Overview

Fortran 0 was modified during the implementation period, which began in January 1955 and continued until the release of the compiler in April 1957. The implemented language, which we call Fortran I, is described in the first Fortran *Programmer's Reference Manual*, published in October 1956 (IBM, 1956). Fortran I included input/output formatting, variable names of up to six characters (it had been just two in Fortran 0), user-defined subroutines, although they

could not be separately compiled, the `If` selection statement, and the `Do` loop statement.

All of Fortran I's control statements were based on 704 instructions. It is not clear whether the 704 designers dictated the control statement design of Fortran I or whether the designers of Fortran I suggested these instructions to the 704 designers.

There were no data-typing statements in the Fortran I language. Variables whose names began with `I`, `J`, `K`, `L`, `M`, and `N` were implicitly integer type, and all others were implicitly floating-point. The choice of the letters for this convention was based on the fact that at that time scientists and engineers used letters as variable subscripts, usually *i*, *j*, and *k*. In a gesture of generosity, Fortran's designers threw in the three additional letters.

The most audacious claim made by the Fortran development group during the design of the language was that the machine code produced by the compiler would be about half as efficient as what could be produced by hand.[1] This, more than anything else, made skeptics of potential users and prevented a great deal of interest in Fortran before its actual release. To almost everyone's surprise, however, the Fortran development group nearly achieved its goal in efficiency. The largest part of the 18 worker-years of effort used to construct the first compiler had been spent on optimization, and the results were remarkably effective.

The early success of Fortran is shown by the results of a survey made in April 1958. At that time, roughly half of the code being written for 704s was being written in Fortran, in spite of the skepticism of most of the programming world only a year earlier.

2.3.4 Fortran II

The Fortran II compiler was distributed in the spring of 1958. It fixed many of the bugs in the Fortran I compilation system and added some significant features to the language, the most important being the independent compilation of subroutines. Without independent compilation, any change in a program required that the entire program be recompiled. Fortran I's lack of independent-compilation capability, coupled with the poor reliability of the 704, placed a practical restriction on the length of programs to about 300 to 400 lines (Wexelblat, 1981, p. 68). Longer programs had a poor chance of being compiled completely before a machine failure occurred. The capability of including precompiled machine language versions of subprograms shortened the compilation process considerably and made it practical to develop much larger programs.

1. In fact, the Fortran team believed that the code generated by their compiler could be no less than half as fast as handwritten machine code, or the language would not be adopted by users.

2.3.5 Fortrans IV, 77, 90, 95, 2003, and 2008

A Fortran III was developed, but it was never widely distributed. Fortran IV, however, became one of the most widely used programming languages of its time. It evolved over the period 1960 to 1962 and was standardized as Fortran 66 (ANSI, 1966), although that name was rarely used. Fortran IV was an improvement over Fortran II in many ways. Among its most important additions were explicit type declarations for variables, a logical `If` construct, and the capability of passing subprograms as parameters to other subprograms.

Fortran IV was replaced by Fortran 77, which became the new standard in 1978 (ANSI, 1978a). Fortran 77 retained most of the features of Fortran IV and added character string handling, logical loop control statements, and an `If` with an optional `Else` clause.

Fortran 90 (ANSI, 1992) was dramatically different from Fortran 77. The most significant additions were dynamic arrays, records, pointers, a multiple selection statement, and modules. In addition, Fortran 90 subprograms could be recursively called.

A new concept that was included in the Fortran 90 definition was that of removing some language features from earlier versions. While Fortran 90 included all of the features of Fortran 77, the language definition included a list of constructs that were recommended for removal in the next version of the language.

Fortran 90 included two simple syntactic changes that altered the appearance of both programs and the literature describing the language. First, the required fixed format of code, which required the use of specific character positions for specific parts of statements, was dropped. For example, statement labels could appear only in the first five positions and statements could not begin before the seventh position. This rigid formatting of code was designed around the use of punch cards. The second change was that the official spelling of FORTRAN became Fortran. This change was accompanied by the change in convention of using all uppercase letters for keywords and identifiers in Fortran programs. The new convention was that only the first letter of keywords and identifiers would be uppercase.

Fortran 95 (INCITS/ISO/IEC, 1997) continued the evolution of the language, but only a few changes were made. Among other things, a new iteration construct, `Forall`, was added to ease the task of parallelizing Fortran programs.

Fortran 2003 (Metcalf et al., 2004), added support for object-oriented programming, parameterized derived types, procedure pointers, and interoperability with the C programming language.

The latest version of Fortran, Fortran 2008 (ISO/IEC 1539-1, 2010) added support for blocks to define local scopes, co-arrays, which provide a parallel execution model, and the `DO CONCURRENT` construct, to specify loops without interdependencies.

2.3.6 Evaluation

The original Fortran design team thought of language design only as a necessary prelude to the critical task of designing the translator. Furthermore, it never occurred to them that Fortran would be used on computers not

manufactured by IBM. Indeed, they were forced to consider building Fortran compilers for other IBM machines only because the successor to the 704, the 709, was announced before the 704 Fortran compiler was released. The effect that Fortran has had on the use of computers, along with the fact that all subsequent programming languages owe a debt to Fortran, is indeed impressive in light of the modest goals of its designers.

One of the features of Fortran I, and all of its successors before 90, that allows highly optimizing compilers was that the types and storage for all variables are fixed before run time. No new variables or space could be allocated during execution time. This was a sacrifice of flexibility to simplicity and efficiency. It eliminated the possibility of recursive subprograms and made it difficult to implement data structures that grow or change shape dynamically. Of course, the kinds of programs that were being built at the time of the development of the early versions of Fortran were primarily numerical in nature and were simple in comparison with more recent software projects. Therefore, the sacrifice was not a great one.

The overall success of Fortran is difficult to overstate: It dramatically changed the way computers are used. This is, of course, in large part due to its being the first widely used high-level language. In comparison with concepts and languages developed later, early versions of Fortran suffer in a variety of ways, as should be expected. After all, it would not be fair to compare the performance and comfort of a 1910 Model T Ford with the performance and comfort of a 2013 Ford Mustang. Nevertheless, in spite of the inadequacies of Fortran, the momentum of the huge investment in Fortran software, among other factors, has kept it in use for more than a half century.

Alan Perlis, one of the designers of ALGOL 60, said of Fortran in 1978, "Fortran is the *lingua franca* of the computing world. It is the language of the streets in the best sense of the word, not in the prostitutional sense of the word. And it has survived and will survive because it has turned out to be a remarkably useful part of a very vital commerce" (Wexelblat, 1981, p. 161).

The following is an example of a Fortran 95 program:

```
! Fortran 95 Example program
!  Input:  An integer, List_Len, where List_Len is less
!          than 100, followed by List_Len-Integer values
!  Output: The number of input values that are greater
!          than the average of all input values
Implicit none
Integer Dimension(99) :: Int_List
Integer :: List_Len, Counter, Sum, Average, Result
Result= 0
Sum = 0
Read *, List_Len
If ((List_Len > 0) .AND. (List_Len < 100)) Then
! Read input data into an array and compute its sum
   Do Counter = 1, List_Len
      Read *, Int_List(Counter)
      Sum = Sum + Int_List(Counter)
```

```
   End Do
! Compute the average
   Average = Sum / List_Len
! Count the values that are greater than the average
   Do Counter = 1, List_Len
      If (Int_List(Counter) > Average) Then
         Result = Result + 1
      End If
   End Do
! Print the result
   Print *, 'Number of values > Average is:', Result
Else
   Print *, 'Error - list length value is not legal'
End If
End Program Example
```

2.4 Functional Programming: LISP

The first functional programming language was invented to provide language features for list processing, the need for which grew out of the first applications in the area of artificial intelligence (AI).

2.4.1 The Beginnings of Artificial Intelligence and List Processing

Interest in AI appeared in the mid-1950s in a number of places. Some of this interest grew out of linguistics, some from psychology, and some from mathematics. Linguists were concerned with natural language processing. Psychologists were interested in modeling human information storage and retrieval, as well as other fundamental processes of the brain. Mathematicians were interested in mechanizing certain intelligent processes, such as theorem proving. All of these investigations arrived at the same conclusion: Some method must be developed to allow computers to process symbolic data in linked lists. At the time, most computation was on numeric data in arrays.

The concept of list processing was developed by Allen Newell, J. C. Shaw, and Herbert Simon at the RAND Corporation. It was first published in a classic paper that describes one of the first AI programs, the Logic Theorist,[2] and a language in which it could be implemented (Newell and Simon, 1956). The language, named IPL-I (Information Processing Language I), was never implemented. The next version, IPL-II, was implemented on a RAND Johnniac computer. Development of IPL continued until 1960, when the description of IPL-V was published (Newell and Tonge, 1960). The low level of the IPL languages prevented their widespread use. They were actually assembly languages for a hypothetical computer, implemented with an interpreter, in which

2. Logic Theorist discovered proofs for theorems in propositional calculus.

list-processing instructions were included. Another factor that kept the IPL languages from becoming popular was their implementation on the obscure Johnniac machine.

The contributions of the IPL languages were in their list design and their demonstration that list processing was feasible and useful.

IBM became interested in AI in the mid-1950s and chose theorem proving as a demonstration area. At the time, the Fortran project was still underway. The high cost of the Fortran I compiler convinced IBM that their list processing should be attached to Fortran, rather than in the form of a new language. Thus, the Fortran List Processing Language (FLPL) was designed and implemented as an extension to Fortran. FLPL was used to construct a theorem prover for plane geometry, which was then considered the easiest area for mechanical theorem proving.

2.4.2 LISP Design Process

John McCarthy of MIT took a summer position at the IBM Information Research Department in 1958. His goal for the summer was to investigate symbolic computations and to develop a set of requirements for doing such computations. As a pilot example problem area, he chose differentiation of algebraic expressions. From this study came a list of language requirements. Among them were the control flow methods of mathematical functions: recursion and conditional expressions. The only available high-level language of the time, Fortran I, had neither of these.

Another requirement that grew from the symbolic-differentiation investigation was the need for dynamically allocated linked lists and some kind of implicit deallocation of abandoned lists. McCarthy simply would not allow his elegant algorithm for differentiation to be cluttered with explicit deallocation statements.

Because FLPL did not support recursion, conditional expressions, dynamic storage allocation, or implicit deallocation, it was clear to McCarthy that a new language was needed.

When McCarthy returned to MIT in the fall of 1958, he and Marvin Minsky formed the MIT AI Project, with funding from the Research Laboratory for Electronics. The first important effort of the project was to produce a software system for list processing. It was to be used initially to implement a program proposed by McCarthy called the Advice Taker.[3] This application became the impetus for the development of the list-processing language LISP. The first version of LISP is sometimes called "pure LISP" because it is a purely functional language. In the following section, we describe the development of pure LISP.

3. Advice Taker represented information with sentences written in a formal language and used a logical inferencing process to decide what to do.

2.4.3 Language Overview

2.4.3.1 Data Structures

Pure LISP has only two kinds of data structures: atoms and lists. Atoms are either symbols, which have the form of identifiers, or numeric literals. The concept of storing symbolic information in linked lists is natural and was used in IPL-II. Such structures allow insertions and deletions at any point, operations that were then thought to be a necessary part of list processing. It was eventually determined, however, that LISP programs rarely require these operations.

Lists are specified by delimiting their elements with parentheses. Simple lists, in which elements are restricted to atoms, have the form

```
(A B C D)
```

Nested list structures are also specified by parentheses. For example, the list

```
(A (B C) D (E (F G)))
```

is composed of four elements. The first is the atom `A`; the second is the sublist `(B C)`; the third is the atom `D`; the fourth is the sublist `(E (F G))`, which has as its second element the sublist `(F G)`.

Internally, lists are stored as single-linked list structures, in which each node has two pointers and represents a list element. A node containing an atom has its first pointer pointing to some representation of the atom, such as its symbol or numeric value, or a pointer to a sublist. A node for a sublist element has its first pointer pointing to the first node of the sublist. In both cases, the second pointer of a node points to the next element of the list. A list is referenced by a pointer to its first element.

The internal representations of the two lists shown earlier are depicted in Figure 2.2. Note that the elements of a list are shown horizontally. The last element of a list has no successor, so its link is `NIL`, which is represented in Figure 2.2 as a diagonal line in the element. Sublists are shown with the same structure.

2.4.3.2 Processes in Functional Programming

LISP was designed as a functional programming language. All computation in a purely functional program is accomplished by applying functions to arguments. Neither the assignment statements nor the variables that abound in imperative language programs are necessary in functional language programs. Furthermore, repetitive processes can be specified with recursive function calls, making iteration (loops) unnecessary. These basic concepts of functional programming make it significantly different from programming in an imperative language.

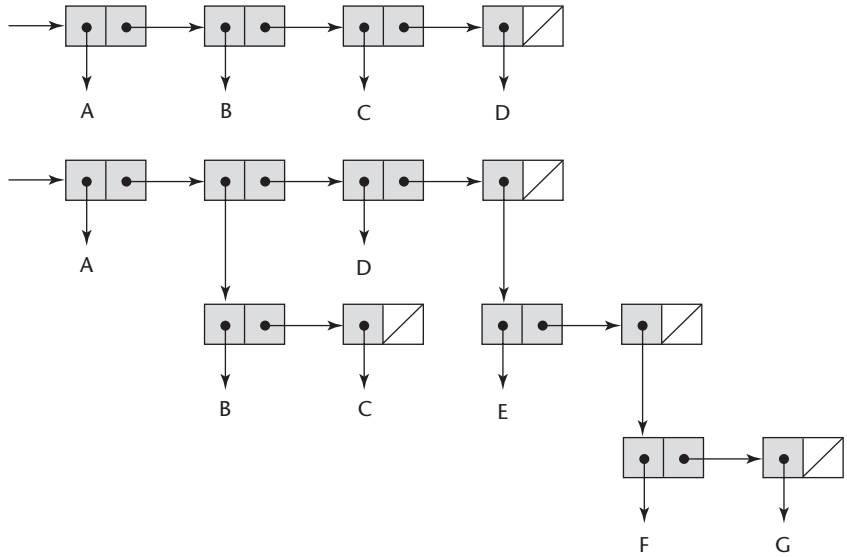

Figure 2.2

Internal representation of two LISP lists

2.4.3.3 The Syntax of LISP

LISP is very different from the imperative languages, both because it is a functional programming language and because the appearance of LISP programs is so different from those in languages like Java or C++. For example, the syntax of Java is a complicated mixture of English and algebra, while LISP's syntax is a model of simplicity. Program code and data have exactly the same form: parenthesized lists. Consider again the list

```
(A B C D)
```

When interpreted as data, it is a list of four elements. When viewed as code, it is the application of the function named `A` to the three parameters `B`, `C`, and `D`.

2.4.4 Evaluation

LISP completely dominated AI applications for a quarter century. Much of the cause of LISP's reputation for being highly inefficient has been eliminated. Many contemporary implementations are compiled, and the resulting code is much faster than running the source code on an interpreter. In addition to its success in AI, LISP pioneered functional programming, which has proven to be a lively area of research in programming languages. As stated in Chapter 1, many programming language researchers believe functional programming is a much better approach to software development than procedural programming using imperative languages.

The following is an example of a LISP program:

```
;  LISP Example function
;  The following code defines a LISP predicate function
;  that  takes two lists as arguments and returns True
;  if the two lists are equal, and NIL (false) otherwise
   (DEFUN equal_lists (lis1 lis2)
    (COND
      ((ATOM lis1) (EQ lis1 lis2))
      ((ATOM lis2) NIL)
      ((equal_lists (CAR lis1) (CAR lis2))
                 (equal_lists (CDR lis1) (CDR lis2)))
      (T NIL)
    )
)
```

2.4.5 Two Descendants of LISP

Two dialects of LISP are now widely used, Scheme and Common LISP. These are briefly discussed in the following subsections.

2.4.5.1 Scheme

The Scheme language emerged from MIT in the mid-1970s (Dybvig, 2003). It is characterized by its small size, its exclusive use of static scoping (discussed in Chapter 5), and its treatment of functions as first-class entities. As first-class entities, Scheme functions can be assigned to variables, passed as parameters, and returned as the values of function applications. They can also be the elements of lists. Early versions of LISP did not provide all of these capabilities, nor did they use static scoping.

As a small language with simple syntax and semantics, Scheme is well suited to educational applications, such as courses in functional programming and general introductions to programming. Scheme is described in some detail in Chapter 15.

2.4.5.2 Common LISP

During the 1970s and early 1980s, a large number of different dialects of LISP were developed and used. This led to the familiar problem of lack of portability among programs written in the various dialects. Common LISP (Graham, 1996) was created in an effort to rectify this situation. Common LISP was designed by combining the features of several dialects of LISP developed in the early 1980s, including Scheme, into a single language. Being such an amalgam, Common LISP is a relatively large and complex language. Its basis, however, is pure LISP, so its syntax, primitive functions, and fundamental nature come from that language.

Recognizing the flexibility provided by dynamic scoping as well as the simplicity of static scoping, Common LISP allows both. The default scoping for variables is static, but by declaring a variable to be **`special`**, that variable becomes dynamically scoped.

Common LISP has a large number of data types and structures, including records, arrays, complex numbers, and character strings. It also has a form of packages for modularizing collections of functions and data providing access control.

Common LISP is further described in Chapter 15.

2.4.6 Related Languages

ML (*M*eta*L*anguage; Ullman, 1998) was originally designed in the 1980s by Robin Milner at the University of Edinburgh as a metalanguage for a program verification system named Logic for Computable Functions (LCF; Milner et al., 1990). ML is primarily a functional language, but it also supports imperative programming. Unlike LISP and Scheme, the type of every variable and expression in ML can be determined at compile time. Types are associated with objects rather than names. Types of names and expressions are inferred from their context.

Unlike LISP and Scheme, ML does not use the parenthesized functional syntax that originated with lambda expressions. Rather, the syntax of ML resembles that of the imperative languages, such as Java and C++.

Miranda was developed by David Turner (1986) at the University of Kent in Canterbury, England, in the early 1980s. Miranda is based partly on the languages ML, SASL, and KRC. Haskell (Hudak and Fasel, 1992) is based in large part on Miranda. Like Miranda, it is a purely functional language, having no variables and no assignment statement. Another distinguishing characteristic of Haskell is its use of lazy evaluation. This means that no expression is evaluated until its value is required. This leads to some surprising capabilities in the language.

Caml (Cousineau et al., 1998) and its dialect that supports object-oriented programming, OCaml (Smith, 2006), descended from ML and Haskell. Finally, F# is a relatively new typed language based directly on OCaml. F# (Syme et al., 2010) is a .NET language with direct access to the whole .NET library. Being a .NET language also means it can smoothly interoperate with any other .NET language. F# supports both functional programming and procedural programming. It also fully supports object-oriented programming.

ML, Haskell, and F# are further discussed in Chapter 15.

2.5 The First Step Toward Sophistication: ALGOL 60

ALGOL 60 has had much influence on subsequent programming languages and is therefore of central importance in any historical study of languages.

2.5.1 Historical Background

ALGOL 60 was the result of efforts to design a universal programming language for scientific applications. By late 1954, the Laning and Zierler algebraic system had been in operation for over a year, and the first report on Fortran had been published. Fortran became a reality in 1957, and several other high-level languages were being developed. Most notable among them were IT, which was designed by Alan Perlis at Carnegie Tech, and two languages for the UNIVAC computers, MATH-MATIC and UNICODE. The proliferation of languages made program sharing among users difficult. Furthermore, the new languages were all growing up around single architectures, some for UNIVAC computers and some for IBM 700-series machines. In response to this blossoming of machine-dependent languages, several major computer user groups in the United States, including SHARE (the IBM scientific user group) and USE (UNIVAC Scientific Exchange, the large-scale UNIVAC scientific user group), submitted a petition to the Association for Computing Machinery (ACM) on May 10, 1957, to form a committee to study and recommend action to create a machine-independent scientific programming language. Although Fortran might have been a candidate, it could not become a universal language, because at the time it was solely owned by IBM.

Previously, in 1955, GAMM (a German acronym for Society for Applied Mathematics and Mechanics) had formed a committee to design one universal, machine-independent algorithmic language. The desire for this new language was in part due to the Europeans' fear of being dominated by IBM. By late 1957, however, the appearance of several high-level languages in the United States convinced the GAMM subcommittee that their effort had to be widened to include the Americans, and a letter of invitation was sent to ACM. In April 1958, after Fritz Bauer of GAMM presented the formal proposal to ACM, the two groups officially agreed to a joint language design project.

2.5.2 Early Design Process

GAMM and ACM each sent four members to the first design meeting. The meeting, which was held in Zurich from May 27 to June 1, 1958, began with the following goals for the new language:

- The syntax of the language should be as close as possible to standard mathematical notation, and programs written in it should be readable with little further explanation.
- It should be possible to use the language for the description of algorithms in printed publications.
- Programs in the new language must be mechanically translatable into machine language.

The first goal indicated that the new language was to be used for scientific programming, which was the primary computer application area at that time. The second was something entirely new to the computing business. The last goal is an obvious necessity for any programming language.

The Zurich meeting succeeded in producing a language that met the stated goals, but the design process required innumerable compromises, both among individuals and between the two sides of the Atlantic. In some cases, the compromises were not so much over great issues as they were over spheres of influence. The question of whether to use a comma (the European method) or a period (the American method) for a decimal point is one example.

2.5.3 ALGOL 58 Overview

The language designed at the Zurich meeting was named the International Algorithmic Language (IAL). It was suggested during the design that the language be named ALGOL, for ALGOrithmic Language, but the name was rejected because it did not reflect the international scope of the committee. During the following year, however, the name was changed to ALGOL, and the language subsequently became known as ALGOL 58.

In many ways, ALGOL 58 was a descendant of Fortran, which is quite natural. It generalized many of Fortran's features and added several new constructs and concepts. Some of the generalizations had to do with the goal of not tying the language to any particular machine, and others were attempts to make the language more flexible and powerful. A rare combination of simplicity and elegance emerged from the effort.

ALGOL 58 formalized the concept of data type, although only variables that were not floating-point required explicit declaration. It added the idea of compound statements, which most subsequent languages incorporated. Some features of Fortran that were generalized were the following: Identifiers were allowed to have any length, as opposed to Fortran I's restriction to six or fewer characters; any number of array dimensions was allowed, unlike Fortran I's limitation to no more than three; the lower bound of arrays could be specified by the programmer, whereas in Fortran it was implicitly 1; nested selection statements were allowed, which was not the case in Fortran I.

ALGOL 58 acquired the assignment operator in a rather unusual way. Zuse used the form

```
expression => variable
```

for the assignment statement in Plankalkül. Although Plankalkül had not yet been published, some of the European members of the ALGOL 58 committee were familiar with the language. The committee dabbled with the Plankalkül assignment form but, because of arguments about character set limitations,[4] the greater-than symbol was changed to a colon. Then, largely at the insistence of the Americans, the whole statement was turned around to the Fortran form

```
variable := expression
```

The Europeans preferred the opposite form, but that would be the reverse of Fortran.

4. The card punches of that time did not include the greater-than symbol.

2.5.4 Reception of the ALGOL 58 Report

In December 1958, publication of the ALGOL 58 report (Perlis and Samelson, 1958) was greeted with a good deal of enthusiasm. In the United States, the new language was viewed more as a collection of ideas for programming language design than as a universal standard language. Actually, the ALGOL 58 report was not meant to be a finished product but rather a preliminary document for international discussion. Nevertheless, three major design and implementation efforts used the report as their basis. At the University of Michigan, the MAD language was born (Arden et al., 1961). The U.S. Naval Electronics Group produced the NELIAC language (Huskey et al., 1963). At System Development Corporation, JOVIAL was designed and implemented (Shaw, 1963). JOVIAL, an acronym for Jules' Own Version of the International Algebraic Language, represents the only language based on ALGOL 58 to achieve widespread use (Jules was Jules I. Schwartz, one of JOVIAL's designers). JOVIAL became widely used because it was the official scientific language for the U.S. Air Force for a quarter century.

The rest of the U.S. computing community was not so kind to the new language. At first, both IBM and its major scientific user group, SHARE, seemed to embrace ALGOL 58. IBM began an implementation shortly after the report was published, and SHARE formed a subcommittee, SHARE IAL, to study the language. The subcommittee subsequently recommended that ACM standardize ALGOL 58 and that IBM implement it for all of the 700-series computers. The enthusiasm was short-lived, however. By the spring of 1959, both IBM and SHARE, through their Fortran experience, had had enough of the pain and expense of getting a new language started, both in terms of developing and using the first-generation compilers and in terms of training users in the new language and persuading them to use it. By the middle of 1959, both IBM and SHARE had developed such a vested interest in Fortran that they decided to retain it as *the* scientific language for the IBM 700-series machines, thereby abandoning ALGOL 58.

2.5.5 ALGOL 60 Design Process

During 1959, ALGOL 58 was furiously debated in both Europe and the United States. Large numbers of suggested modifications and additions were published in the European *ALGOL Bulletin* and in *Communications of the ACM*. One of the most important events of 1959 was the presentation of the work of the Zurich committee to the International Conference on Information Processing, for there Backus introduced his new notation for describing the syntax of programming languages, which later became known as BNF (Backus-Naur form). BNF is described in detail in Chapter 3.

In January 1960, the second ALGOL meeting was held, this time in Paris. The purpose of the meeting was to debate the 80 suggestions that had been formally submitted for consideration. Peter Naur of Denmark had become heavily involved in the development of ALGOL, even though he had not been

a member of the Zurich group. It was Naur who created and published the *ALGOL Bulletin.* He spent a good deal of time studying Backus's paper that introduced BNF and decided that BNF should be used to describe formally the results of the 1960 meeting. After making a few relatively minor changes to BNF, he wrote a description of the new proposed language in BNF and handed it out to the members of the 1960 group at the beginning of the meeting.

2.5.6 ALGOL 60 Overview

Although the 1960 meeting lasted only six days, the modifications made to ALGOL 58 were dramatic. Among the most important new developments were the following:

- The concept of block structure was introduced. This allowed the programmer to localize parts of programs by introducing new data environments, or scopes.
- Two different means of passing parameters to subprograms were allowed: pass by value and pass by name.
- Procedures were allowed to be recursive. The ALGOL 58 description was unclear on this issue. Note that although this recursion was new for the imperative languages, LISP had already provided recursive functions in 1959.
- Stack-dynamic arrays were allowed. A stack-dynamic array is one for which the subscript range or ranges are specified by variables, so that the size of the array is set at the time storage is allocated to the array, which happens when the declaration is reached during execution. Stack-dynamic arrays are described in detail in Chapter 6.

Several features that might have had a dramatic impact on the success or failure of the language were proposed and rejected. Most important among these were input and output statements with formatting, which were omitted because they were thought to be machine-dependent.

The ALGOL 60 report was published in May 1960 (Naur, 1960). A number of ambiguities still remained in the language description, and a third meeting was scheduled for April 1962 in Rome to address the problems. At this meeting the group dealt only with problems; no additions to the language were allowed. The results of this meeting were published under the title "Revised Report on the Algorithmic Language ALGOL 60" (Backus et al., 1963).

2.5.7 Evaluation

In some ways, ALGOL 60 was a great success; in other ways, it was a dismal failure. It succeeded in becoming, almost immediately, the only acceptable formal means of communicating algorithms in computing literature, and it remained that for more than 20 years. Every imperative programming language designed since 1960 owes something to ALGOL 60. In fact, most are direct

or indirect descendants; examples include PL/I, SIMULA 67, ALGOL 68, C, Pascal, Ada, C++, Java, and C#.

The ALGOL 58/ALGOL 60 design effort included a long list of firsts. It was the first time that an international group attempted to design a programming language. It was the first language that was designed to be machine independent. It was also the first language whose syntax was formally described. This successful use of the BNF formalism initiated several important fields of computer science: formal languages, parsing theory, and BNF-based compiler design. Finally, the structure of ALGOL 60 affected machine architecture. In the most striking example of this, an extension of the language was used as the systems language of a series of large-scale computers, the Burroughs B5000, B6000, and B7000 machines, which were designed with a hardware stack to implement efficiently the block structure and recursive subprograms of the language.

On the other side of the coin, ALGOL 60 never achieved widespread use in the United States. Even in Europe, where it was more popular than in the United States, it never became the dominant language. There are a number of reasons for its lack of acceptance. For one thing, some of the features of ALGOL 60 turned out to be too flexible; they made understanding difficult and implementation inefficient. The best example of this is the pass-by-name method of passing parameters to subprograms, which is explained in Chapter 9. The difficulties of implementing ALGOL 60 are evidenced by Rutishauser's statement in 1967 that few, if any, implementations included the full ALGOL 60 language (Rutishauser, 1967, p. 8).

The lack of input and output statements in the language was another major reason for its lack of acceptance. Implementation-dependent input/output made programs difficult to port to other computers.

Ironically, one of the most important contributions to computer science associated with ALGOL 60, BNF, was also a factor in its lack of acceptance. Although BNF is now considered a simple and elegant means of syntax description, in 1960 it seemed strange and complicated.

Finally, although there were many other problems, the entrenchment of Fortran among users and the lack of support by IBM were probably the most important factors in ALGOL 60's failure to gain widespread use.

The ALGOL 60 effort was never really complete, in the sense that ambiguities and obscurities were always a part of the language description (Knuth, 1967).

The following is an example of an ALGOL 60 program:

```
comment ALGOL 60 Example Program
 Input:  An integer, listlen, where listlen is less than
         100, followed by listlen-integer values
 Output: The number of input values that are greater than
         the average of all the input values  ;
begin
  integer array intlist [1:99];
```

```
  integer listlen, counter, sum, average, result;
  sum := 0;
  result := 0;
  readint (listlen);
  if (listlen > 0) ∧ (listlen < 100) then
    begin
comment Read input into an array and compute the average;
    for counter := 1 step 1 until listlen do
      begin
      readint (intlist[counter]);
      sum := sum + intlist[counter]
      end;
comment Compute the average;
    average := sum / listlen;
comment Count the input values that are > average;
    for counter := 1 step 1 until listlen do
      if intlist[counter] > average
        then result := result + 1;
comment Print result;
    printstring("The number of values > average is:");
    printint (result)
    end
  else
    printstring ("Error—input list length is not legal";
end
```

2.6 Computerizing Business Records: COBOL

The story of COBOL is, in a sense, the opposite of that of ALGOL 60. Although it has been used more than any other programming language, COBOL has had little effect on the design of subsequent languages, except for PL/I. It may still be the most widely used language,[5] although it is difficult to be sure one way or the other. Perhaps the most important reason why COBOL has had little influence is that few have attempted to design a new language for business applications since it appeared. That is due in part to how well COBOL's capabilities meet the needs of its application area. Another reason is that a great deal of growth in business computing over the past 30 years has occurred in small businesses. In these businesses, very little software development has taken place. Instead, most of the software used is purchased as off-the-shelf packages for various general business applications.

5. In the late 1990s, in a study associated with the Y2K problem, it was estimated that there were approximately 800 million lines of COBOL in use in the 22 square miles of Manhattan.

2.6.1 Historical Background

The beginning of COBOL is somewhat similar to that of ALGOL 60, in the sense that the language was designed by a committee of people meeting for relatively short periods of time. At the time, in 1959, the state of business computing was similar to the state of scientific computing several years earlier, when Fortran was being designed. One compiled language for business applications, FLOW-MATIC, had been implemented in 1957, but it belonged to one manufacturer, UNIVAC, and was designed for that company's computers. Another language, AIMACO, was being used by the U.S. Air Force, but it was only a minor variation of FLOW-MATIC. IBM had designed a programming language for business applications, COMTRAN (COMmercial TRANslator), but it had not yet been implemented. Several other language design projects were being planned.

2.6.2 FLOW-MATIC

The origins of FLOW-MATIC are worth at least a brief discussion, because it was the primary progenitor of COBOL. In December 1953, Grace Hopper at Remington-Rand UNIVAC wrote a proposal that was indeed prophetic. It suggested that "mathematical programs should be written in mathematical notation, data processing programs should be written in English statements" (Wexelblat, 1981, p. 16). Unfortunately, in 1953, it was impossible to convince nonprogrammers that a computer could be made to understand English words. It was not until 1955 that a similar proposal had some hope of being funded by UNIVAC management, and even then it took a prototype system to do the final convincing. Part of this selling process involved compiling and running a small program, first using English keywords, then using French keywords, and then using German keywords. This demonstration was considered remarkable by UNIVAC management and was instrumental in their acceptance of Hopper's proposal.

2.6.3 COBOL Design Process

The first formal meeting on the subject of a common language for business applications, which was sponsored by the Department of Defense, was held at the Pentagon on May 28 and 29, 1959 (exactly one year after the Zurich ALGOL meeting). The consensus of the group was that the language, then named CBL (Common Business Language), should have the following general characteristics: Most agreed that it should use English as much as possible, although a few argued for a more mathematical notation. The language must be easy to use, even at the expense of being less powerful, in order to broaden the base of those who could program computers. In addition to making the language easy to use, it was believed that the use of English would allow managers to read programs. Finally, the design should not be overly restricted by the problems of its implementation.

One of the overriding concerns at the meeting was that steps to create this universal language should be taken quickly, as a lot of work was already being done to create other business languages. In addition to the existing languages, RCA and Sylvania were working on their own business applications languages. It was clear that the longer it took to produce a universal language, the more difficult it would be for the language to become widely used. On this basis, it was decided that there should be a quick study of existing languages. For this task, the Short Range Committee was formed.

There were early decisions to separate the statements of the language into two categories—data description and executable operations—and to have statements in these two categories be in different parts of programs. One of the debates of the Short Range Committee was over the inclusion of subscripts. Many committee members argued that subscripts were too complex for the people in data processing, who were thought to be uncomfortable with mathematical notation. Similar arguments revolved around whether arithmetic expressions should be included. The final report of the Short Range Committee, which was completed in December 1959, described the language that was later named COBOL 60.

The language specifications for COBOL 60, published by the Government Printing Office in April 1960 (Department of Defense, 1960), were described as "initial." Revised versions were published in 1961 and 1962 (Department of Defense, 1961, 1962). The language was standardized by the American National Standards Institute (ANSI) group in 1968. The next three revisions were standardized by ANSI in 1974, 1985, and 2002. The language continues to evolve today.

2.6.4 Evaluation

The COBOL language originated a number of novel concepts, some of which eventually appeared in other languages. For example, the `DEFINE` verb of COBOL 60 was the first high-level language construct for macros. More important, hierarchical data structures (records), which first appeared in Plankalkül, were first implemented in COBOL. They have been included in most of the imperative languages designed since then. COBOL was also the first language that allowed names to be truly connotative, because it allowed both long names (up to 30 characters) and word-connector characters (hyphens).

Overall, the data division is the strong part of COBOL's design, whereas the procedure division is relatively weak. Every variable is defined in detail in the data division, including the number of decimal digits and the location of the implied decimal point. File records are also described with this level of detail, as are lines to be output to a printer, which makes COBOL ideal for printing accounting reports. Perhaps the most important weakness of the original procedure division was in its lack of functions. Versions of COBOL prior to the 1974 standard also did not allow subprograms with parameters.

Our final comment on COBOL: It was the first programming language whose use was mandated by the Department of Defense (DoD). This mandate came after its initial development, because COBOL was not designed specifically for the DoD. In spite of its merits, COBOL probably would not have

survived without that mandate. The poor performance of the early compilers simply made the language too expensive to use. Eventually, of course, compilers became more efficient and computers became much faster and cheaper and had much larger memories. Together, these factors allowed COBOL to succeed, inside and outside DoD. Its appearance led to the electronic mechanization of accounting, an important revolution by any measure.

The following is an example of a COBOL program. This program reads a file named `BAL-FWD-FILE` that contains inventory information about a certain collection of items. Among other things, each item record includes the number currently on hand (`BAL-ON-HAND`) and the item's reorder point (`BAL-REORDER-POINT`). The reorder point is the threshold number of items on hand at which more must be ordered. The program produces a list of items that must be reordered as a file named `REORDER-LISTING`.

```
IDENTIFICATION DIVISION.
PROGRAM-ID. PRODUCE-REORDER-LISTING.

ENVIRONMENT DIVISION.
CONFIGURATION SECTION.
SOURCE-COMPUTER. DEC-VAX.
OBJECT-COMPUTER. DEC-VAX.
INPUT-OUTPUT SECTION.
FILE-CONTROL.
    SELECT BAL-FWD-FILE   ASSIGN TO READER.
    SELECT REORDER-LISTING  ASSIGN TO LOCAL-PRINTER.

DATA DIVISION.
FILE SECTION.
FD  BAL-FWD-FILE
    LABEL RECORDS ARE STANDARD
    RECORD CONTAINS 80 CHARACTERS.

01  BAL-FWD-CARD.
    02 BAL-ITEM-NO        PICTURE IS 9(5).
    02 BAL-ITEM-DESC      PICTURE IS X(20).
    02 FILLER             PICTURE IS X(5).
    02 BAL-UNIT-PRICE     PICTURE IS 999V99.
    02 BAL-REORDER-POINT  PICTURE IS 9(5).
    02 BAL-ON-HAND        PICTURE IS 9(5).
    02 BAL-ON-ORDER       PICTURE IS 9(5).
    02 FILLER             PICTURE IS X(30).
FD  REORDER-LISTING
    LABEL RECORDS ARE STANDARD
    RECORD CONTAINS 132 CHARACTERS.

01  REORDER-LINE.
```

```
    02 RL-ITEM-NO         PICTURE IS Z(5).
    02 FILLER             PICTURE IS X(5).
    02 RL-ITEM-DESC       PICTURE IS X(20).
    02 FILLER             PICTURE IS X(5).
    02 RL-UNIT-PRICE      PICTURE IS ZZZ.99.
    02 FILLER             PICTURE IS X(5).
    02 RL-AVAILABLE-STOCK PICTURE IS Z(5).
    02 FILLER             PICTURE IS X(5).
    02 RL-REORDER-POINT   PICTURE IS Z(5).
    02 FILLER             PICTURE IS X(71).

WORKING-STORAGE SECTION.
01  SWITCHES.
    02 CARD-EOF-SWITCH    PICTURE IS X.
01  WORK-FIELDS.
    02 AVAILABLE-STOCK    PICTURE IS 9(5).

PROCEDURE DIVISION.
000-PRODUCE-REORDER-LISTING.
    OPEN INPUT BAL-FWD-FILE.
    OPEN OUTPUT REORDER-LISTING.
    MOVE "N" TO CARD-EOF-SWITCH.
    PERFORM 100-PRODUCE-REORDER-LINE
        UNTIL CARD-EOF-SWITCH IS EQUAL TO "Y".
    CLOSE BAL-FWD-FILE.
    CLOSE REORDER-LISTING.
    STOP RUN.

100-PRODUCE-REORDER-LINE.
    PERFORM 110-READ-INVENTORY-RECORD.
    IF CARD-EOF-SWITCH IS NOT EQUAL TO "Y"]
        PERFORM 120-CALCULATE-AVAILABLE-STOCK
        IF AVAILABLE-STOCK IS LESS THAN BAL-REORDER-POINT
            PERFORM 130-PRINT-REORDER-LINE.

110-READ-INVENTORY-RECORD.
    READ BAL-FWD-FILE RECORD
        AT END
            MOVE "Y" TO CARD-EOF-SWITCH.

120-CALCULATE-AVAILABLE-STOCK.
ADD BAL-ON-HAND BAL-ON-ORDER
    GIVING AVAILABLE-STOCK.

130-PRINT-REORDER-LINE.
    MOVE SPACE             TO REORDER-LINE.
```

```
MOVE BAL-ITEM-NO       TO RL-ITEM-NO.
MOVE BAL-ITEM-DESC     TO RL-ITEM-DESC.
MOVE BAL-UNIT-PRICE    TO RL-UNIT-PRICE.
MOVE AVAILABLE-STOCK   TO RL-AVAILABLE-STOCK.
MOVE BAL-REORDER-POINT TO RL-REORDER-POINT.
WRITE REORDER-LINE.
```

2.7 The Beginnings of Timesharing: BASIC

BASIC (Mather and Waite, 1971) is another programming language that has enjoyed widespread use but has gotten little respect. Like COBOL, it has largely been ignored by computer scientists. Also, like COBOL, in its earliest versions it was inelegant and included only a meager set of control statements.

BASIC was very popular on microcomputers in the late 1970s and early 1980s. This followed directly from two of the main characteristics of early versions of BASIC. It was easy for beginners to learn, especially those who were not science oriented, and its smaller dialects can be implemented on computers with very small memories.[6] When the capabilities of microcomputers grew and other languages were implemented, the use of BASIC waned. A strong resurgence in the use of BASIC began with the appearance of Visual Basic (Microsoft, 1991) in the early 1990s.

2.7.1 Design Process

BASIC (Beginner's All-purpose Symbolic Instruction Code) was originally designed at Dartmouth College (now Dartmouth University) in New Hampshire by two mathematicians, John Kemeny and Thomas Kurtz, who, in the early 1960s, developed compilers for a variety of dialects of Fortran and ALGOL 60. Their science students generally had little trouble learning or using those languages in their studies. However, Dartmouth was primarily a liberal arts institution, where science and engineering students made up only about 25 percent of the student body. It was decided in the spring of 1963 to design a new language especially for liberal arts students. This new language would use terminals as the method of computer access. The goals of the system were as follows:

1. It must be easy for nonscience students to learn and use.
2. It must be "pleasant and friendly."
3. It must provide fast turnaround for homework.

6. Some early microcomputers included BASIC interpreters that resided in 4096 bytes of ROM.

4. It must allow free and private access.
5. It must consider user time more important than computer time.

The last goal was indeed a revolutionary concept. It was based at least partly on the belief that computers would become significantly cheaper as time went on, which of course they did.

The combination of the second, third, and fourth goals led to the time-shared aspect of BASIC. Only with individual access through terminals by numerous simultaneous users could these goals be met in the early 1960s.

In the summer of 1963, Kemeny began work on the compiler for the first version of BASIC, using remote access to a GE 225 computer. Design and coding of the operating system for BASIC began in the fall of 1963. At 4:00 A.M. on May 1, 1964, the first program using the timeshared BASIC was typed in and run. In June, the number of terminals on the system grew to 11, and by the fall it had ballooned to 20.

2.7.2 Language Overview

The original version of BASIC was very small and, oddly, was not interactive: There was no way for an executing program to get input data from the user. Programs were typed in, compiled, and run, in a sort of batch-oriented way. The original BASIC had only 14 different statement types and a single data type—floating-point. Because it was believed that few of the targeted users would appreciate the difference between integer and floating-point types, the type was referred to as "numbers." Overall, it was a very limited language, though quite easy to learn.

2.7.3 Evaluation

The most important aspect of the original BASIC was that it was the first widely used language that was used through terminals connected to a remote computer.[7] Terminals had just begun to be available at that time. Before then, most programs were entered into computers through either punched cards or paper tape.

Much of the design of BASIC came from Fortran, with some minor influence from the syntax of ALGOL 60. Later, it grew in a variety of ways, with little or no effort made to standardize it. The American National Standards Institute issued a Minimal BASIC standard (ANSI, 1978b), but this represented only the bare minimum of language features. In fact, the original BASIC was very similar to Minimal BASIC.

Although it may seem surprising, Digital Equipment Corporation used a rather elaborate version of BASIC named BASIC-PLUS to write significant

7. LISP initially was used through terminals, but it was not widely used in the early 1960s.

portions of their largest operating system for the PDP-11 minicomputers, RSTS, in the 1970s.

BASIC has been criticized for the poor structure of programs written in it, among other things. By the evaluation criteria discussed in Chapter 1, specifically readability and reliability, the language does indeed fare very poorly. Clearly, the early versions of the language were not meant for and should not have been used for serious programs of any significant size. Later versions are much better suited to such tasks.

The resurgence of BASIC in the 1990s was driven by the appearance of Visual BASIC (VB). VB became widely used in large part because it provided a simple way of building graphical user interfaces (GUIs), hence the name Visual BASIC. Visual Basic .NET, or just VB.NET, is one of Microsoft's .NET languages. Although it is a significant departure from VB, it quickly displaced the older language. Perhaps the most important difference between VB and VB.NET is that VB.NET fully supports object-oriented programming.

The following is an example of a BASIC program:

```
REM  BASIC Example Program
REM  Input:  An integer, listlen, where listlen is less
REM          than 100, followed by listlen-integer values
REM  Output: The number of input values that are greater
REM          than the average of all input values
  DIM intlist(99)
  result = 0
  sum = 0
  INPUT listlen
  IF listlen > 0 AND listlen < 100 THEN
REM  Read input into an array and compute the sum
    FOR counter = 1 TO listlen
      INPUT intlist(counter)
      sum = sum + intlist(counter)
    NEXT counter
REM  Compute the average
    average = sum / listlen
REM  Count the number of input values that are > average
    FOR counter = 1 TO listlen
      IF intlist(counter) > average
        THEN result = result + 1
    NEXT counter
REM  Print the result
    PRINT "The number of values that are > average is:";
          result
  ELSE
    PRINT "Error—input list length is not legal"
  END IF
END
```

User Design and Language Design

ALAN COOPER

Best-selling author of *About Face: The Essentials of User Interface Design,* Alan Cooper also had a large hand in designing what can be touted as the language with the most concern for user interface design, Visual Basic. For him, it all comes down to a vision for humanizing technology.

SOME INFORMATION ON THE BASICS

How did you get started in all of this? I'm a high school dropout with an associate degree in programming from a California community college. My first job was as a programmer for American President Lines (one of the United States' oldest ocean transportation companies) in San Francisco. Except for a few months here and there, I've remained self-employed.

What is your current job? Founder and chairman of Cooper, the company that humanizes technology (www.cooper.com).

What is or was your favorite job? Interaction design consultant.

You are very well known in the fields of language design and user interface design. Any thoughts on designing languages versus designing software, versus designing anything else? It's pretty much the same in the world of software: Know your user.

ABOUT THAT EARLY WINDOWS RELEASE

In the 1980s, you started using Windows and have talked about being lured by its plusses: the graphical user interface support and the dynamically linked library that let you create tools that configured themselves. What about the parts of Windows that you eventually helped shape? I was very impressed by Microsoft's inclusion of support for practical multitasking in Windows. This included dynamic relocation and interprocess communications. MSDOS.exe was the shell program for the first few releases of Windows. It was a terrible program, and I believed that it could be improved dramatically, and I was the guy to do it. In my spare time, I immediately began to write a better shell program than the one Windows came with. I called it Tripod. Microsoft's original shell, MSDOS.exe, was one of the main stumbling blocks to the initial success of Windows. Tripod attempted to solve the problem by being easier to use and to configure.

When was that "Aha!" moment? It wasn't until late in 1987, when I was interviewing a corporate client, that the key design strategy for Tripod popped into my head. As the IS manager explained to me his need to create and publish a wide range of shell solutions to his disparate user base, I realized the conundrum that there is no such thing as an ideal shell. Every user would need their own personal shell, configured to their own needs and skill levels. In an instant, I perceived the solution to the shell design problem: It would be a shell construction set; a tool where each user would be able to construct exactly the shell that he or she needed for a unique mix of applications and training.

What is so compelling about the idea of a shell that can be individualized? Instead of me telling the users what the ideal shell was, they could design their own, personalized ideal shell. With a customizable shell, a programmer would create a shell that was powerful and wide ranging but also somewhat dangerous, whereas an IT manager would create a shell that could be given to a desk clerk that exposed only those few application-specific tools that the clerk used.

> *"MSDOS.exe was the shell program for the first few releases of Windows. It was a terrible program, and I believed that it could be improved dramatically, and I was the guy to do it. In my spare time, I immediately began to write a better shell program than the one Windows came with."*

How did you get from writing a shell program to collaborating with Microsoft? Tripod and Ruby are the same thing. After I signed a deal with Bill Gates, I changed the name of the prototype from Tripod to Ruby. I then used the Ruby prototype as prototypes should be used: as a disposable model for constructing release-quality code. Which is what I did. MS took the release version of Ruby and added QuickBASIC to it, creating VB. All of those original innovations were in Tripod/Ruby.

RUBY AS THE INCUBATOR FOR VISUAL BASIC

Let's revisit your interest in early Windows and that DLL feature. The DLL wasn't a thing, it was a facility in the OS. It allowed a programmer to build code objects that could be linked to at run time as opposed to only at compile time. This is what allowed me to invent the dynamically extensible parts of VB, where controls can be added by third-party vendors.

The Ruby product embodied many significant advances in software design, but two of them stand out as exceptionally successful. As I mentioned, the dynamic linking capability of Windows had always intrigued me, but having the tools and knowing what to do with them were two different things. With Ruby, I finally found two practical uses for dynamic linking, and the original program contained both. First, the language was both installable and could be extended dynamically. Second, the palette of gizmos could be added to dynamically.

Was your language in Ruby the first to have a dynamic linked library and to be linked to a visual front end? As far as I know, yes.

Using a simple example, what would this enable a programmer to do with his or her program? Purchase a control, such as a grid control, from a third-party vendor, install it on his or her computer, and have the grid control appear as an integral part of the language, including the visual programming front end.

Why do they call you "the father of Visual Basic"? Ruby came with a small language, one suited only for executing the dozen or so simple commands that a shell program needs. However, this language was implemented as a chain of DLLs, any number of which could be installed at run time. The internal parser would identify a verb and then pass it along the chain of DLLs until one of them acknowledged that it knew how to process the verb. If all of the DLLs passed, there was a syntax error. From our earliest discussions, both Microsoft and I had entertained the idea of growing the language, possibly even replacing it altogether with a "real" language. C was the candidate most frequently mentioned, but eventually, Microsoft took advantage of this dynamic interface to unplug our little shell language and replace it entirely with QuickBASIC. This new marriage of language to visual front end was static and permanent, and although the original dynamic interface made the coupling possible, it was lost in the process.

SOME FINAL COMMENTS ON NEW IDEAS

In the world of programming and programming tools, including languages and environments, what projects most interest you? I'm interested in creating programming tools that are designed to help users instead of programmers.

What's the most critical rule, famous quote, or design idea to keep in mind? Bridges are not built by engineers. They are built by ironworkers.

Similarly, software programs are not built by engineers. They are built by programmers.

2.8 Everything for Everybody: PL/I

PL/I represents the first large-scale attempt to design a language that could be used for a broad spectrum of application areas. All previous and most subsequent languages have focused on one particular application area, such as science, artificial intelligence, or business.

2.8.1 Historical Background

Like Fortran, PL/I was developed as an IBM product. By the early 1960s, the users of computers in industry had settled into two separate and quite different camps: scientific and business. From the IBM point of view, scientific programmers could use either the large-scale 7090 or the small-scale 1620 IBM computers. This group used floating-point data and arrays extensively. Fortran was the primary language, although some assembly language was also used. They had their own user group, SHARE, and had little contact with anyone who worked on business applications.

For business applications, people used the large 7080 or the small 1401 IBM computers. They needed the decimal and character string data types, as well as elaborate and efficient input and output facilities. They used COBOL, although in early 1963 when the PL/I story begins, the conversion from assembly language to COBOL was far from complete. This category of users also had its own user group, GUIDE, and seldom had contact with scientific users.

In early 1963, IBM planners perceived the beginnings of a change in this situation. The two widely separated computer user groups were moving toward each other in ways that were thought certain to create problems. Scientists began to gather large files of data to be processed. This data required more sophisticated and more efficient input and output facilities. Business applications people began to use regression analysis to build management information systems, which required floating-point data and arrays. It began to appear that computing installations would soon require two separate computers and technical staffs, supporting two very different programming languages.[8]

These perceptions naturally led to the concept of designing a single universal computer that would be capable of doing both floating-point and decimal arithmetic, and therefore both scientific and business applications. Thus was born the concept of the IBM System/360 line of computers. Along with this came the idea of a programming language that could be used for both business and scientific applications. For good measure, features to support systems programming and list processing were thrown in. Therefore, the new language was to replace Fortran, COBOL, LISP, and the systems applications of assembly language.

8. At the time, large computer installations required both full-time hardware and full-time system software maintenance staff.

2.8.2 Design Process

The design effort began when IBM and SHARE formed the Advanced Language Development Committee of the SHARE Fortran Project in October 1963. This new committee quickly met and formed a subcommittee called the 3 × 3 Committee, so named because it had three members from IBM and three from SHARE. The 3 × 3 Committee met for three or four days every other week to design the language.

As with the Short Range Committee for COBOL, the initial design was scheduled for completion in a remarkably short time. Apparently, regardless of the scope of a language design effort, in the early 1960s the prevailing belief was that it could be done in three months. The first version of PL/I, which was then named Fortran VI, was supposed to be completed by December, less than three months after the committee was formed. The committee pleaded successfully on two different occasions for extensions, moving the due date back to January and then to late February 1964.

The initial design concept was that the new language would be an extension of Fortran IV, maintaining compatibility, but that goal was dropped quickly along with the name Fortran VI. Until 1965, the language was known as NPL (New Programming Language). The first published report on NPL was given at the SHARE meeting in March 1964. A more complete description followed in April, and the version that would actually be implemented was published in December 1964 (IBM, 1964) by the compiler group at the IBM Hursley Laboratory in England, which was chosen to do the implementation. In 1965, the name was changed to PL/I to avoid the confusion of the name NPL with the National Physical Laboratory in England. If the compiler had been developed outside the United Kingdom, the name might have remained NPL.

2.8.3 Language Overview

Perhaps the best single-sentence description of PL/I is that it included what were then considered the best parts of ALGOL 60 (recursion and block structure), Fortran IV (separate compilation with communication through global data), and COBOL 60 (data structures, input/output, and report-generating facilities), along with an extensive collection of new constructs, all somehow cobbled together. Because PL/I is no longer a popular language, we will not attempt, even briefly, to discuss all the features of the language, or even its most controversial constructs. Instead, we will mention some of the language's contributions to the pool of knowledge of programming languages.

PL/I was the first programming language to have the following facilities:

- Programs were allowed to create concurrently executing subprograms. Although this was a good idea, it was poorly developed in PL/I.
- It was possible to detect and handle 23 different types of exceptions, or run-time errors.

- Subprograms were allowed to be used recursively, but the capability could be disabled, allowing more efficient linkage for nonrecursive subprograms.
- Pointers were included as a data type.
- Cross-sections of arrays could be referenced. For example, the third row of a matrix could be referenced as if it were a single-dimensioned array.

2.8.4 Evaluation

Any evaluation of PL/I must begin by recognizing the ambitiousness of the design effort. In retrospect, it appears naive to think that so many constructs could have been combined successfully. However, that judgment must be tempered by acknowledging that there was little language design experience at the time. Overall, the design of PL/I was based on the premise that any construct that was useful and could be implemented should be included, with insufficient concern about how a programmer could understand and make effective use of such a collection of constructs and features. Edsger Dijkstra, in his Turing Award Lecture (Dijkstra, 1972), made one of the strongest criticisms of the complexity of PL/I: "I absolutely fail to see how we can keep our growing programs firmly within our intellectual grip when by its sheer baroqueness the programming language—our basic tool, mind you!—already escapes our intellectual control."

In addition to the problem with the complexity due to its large size, PL/I suffered from a number of what are now considered to be poorly designed constructs. Among these were pointers, exception handling, and concurrency, although we must point out that in all cases, these constructs had not appeared in any previous language.

In terms of usage, PL/I must be considered at least a partial success. In the 1970s, it enjoyed significant use in both business and scientific applications. It was also widely used during that time as an instructional vehicle in colleges, primarily in several subset forms, such as PL/C (Cornell, 1977) and PL/CS (Conway and Constable, 1976).

The following is an example of a PL/I program:

```
/* PL/I PROGRAM EXAMPLE
 INPUT:  AN INTEGER, LISTLEN, WHERE LISTLEN IS LESS THAN
         100, FOLLOWED BY LISTLEN-INTEGER VALUES
 OUTPUT: THE NUMBER OF INPUT VALUES THAT ARE GREATER THAN
         THE AVERAGE OF ALL INPUT VALUES   */
PLIEX: PROCEDURE OPTIONS (MAIN);
  DECLARE INTLIST (1:99) FIXED.
  DECLARE (LISTLEN, COUNTER, SUM, AVERAGE, RESULT) FIXED;
  SUM = 0;
  RESULT = 0;
  GET LIST (LISTLEN);
  IF (LISTLEN > 0) & (LISTLEN < 100) THEN
```

```
    DO;
/* READ INPUT DATA INTO AN ARRAY AND COMPUTE THE SUM */
    DO COUNTER = 1 TO LISTLEN;
      GET LIST (INTLIST (COUNTER));
      SUM = SUM + INTLIST (COUNTER);
    END;
/* COMPUTE THE AVERAGE */
    AVERAGE = SUM / LISTLEN;
/* COUNT THE NUMBER OF VALUES THAT ARE > AVERAGE */
    DO COUNTER = 1 TO LISTLEN;
      IF INTLIST (COUNTER) > AVERAGE THEN
        RESULT = RESULT + 1;
    END;
/* PRINT RESULT */
    PUT SKIP LIST ('THE NUMBER OF VALUES > AVERAGE IS:');
    PUT LIST (RESULT);
    END;
  ELSE
    PUT SKIP LIST ('ERROR—INPUT LIST LENGTH IS ILLEGAL');
  END PLIEX;
```

2.9 Two Early Dynamic Languages: APL and SNOBOL

The structure of this section is different from that of the other sections because the languages discussed here are very different. Neither APL nor SNOBOL had much influence on later mainstream languages.[9] Some of the interesting features of APL are discussed later in the book.

In appearance and in purpose, APL and SNOBOL are quite different. They share two fundamental characteristics, however: dynamic typing and dynamic storage allocation. Variables in both languages are essentially untyped. A variable acquires a type when it is assigned a value, at which time it assumes the type of the value assigned. Storage is allocated to a variable only when it is assigned a value, because before that there is no way to know the amount of storage that will be needed.

2.9.1 Origins and Characteristics of APL

APL (Brown et al., 1988) was designed around 1960 by Kenneth E. Iverson at IBM. It was not originally designed to be an implemented programming language but rather was intended to be a vehicle for describing computer architecture.

9. However, they have some influence on some nonmainstream languages (J is based on APL, ICON is based on SNOBOL, and AWK is partially based on SNOBOL).

APL was first described in the book from which it gets its name, *A Programming Language* (Iverson, 1962). In the mid-1960s, the first implementation of APL was developed at IBM.

APL has a large number of powerful operators that are specified with a large number of symbols, which created a problem for implementors. Initially, APL was used through IBM printing terminals. These terminals had special print balls that provided the odd character set required by the language. One reason APL has so many operators is that it provides a large number of unit operations on arrays. For example, the transpose of any matrix is done with a single operator. The large collection of operators provides very high expressivity but also makes APL programs difficult to read. Therefore, people think of APL as a language that is best used for "throw-away" programming. Although programs can be written quickly, they should be discarded after use because they are difficult to maintain.

APL has been around for nearly 50 years and is still used today, although not widely. Furthermore, it has not changed a great deal over its lifetime.

2.9.2 Origins and Characteristics of SNOBOL

SNOBOL (pronounced "snowball"; Griswold et al., 1971) was designed in the early 1960s by three people at Bell Laboratories: D. J. Farber, R. E. Griswold, and I. P. Polonsky (Farber et al., 1964). It was designed specifically for text processing. The heart of SNOBOL is a collection of powerful operations for string pattern matching. One of the early applications of SNOBOL was for writing text editors. Because the dynamic nature of SNOBOL makes it slower than alternative languages, it is no longer used for such programs. However, SNOBOL is still a live and supported language that is used for a variety of text-processing tasks in several different application areas.

2.10 The Beginnings of Data Abstraction: SIMULA 67

Although SIMULA 67 never achieved widespread use and had little impact on the programmers and computing of its time, some of the constructs it introduced make it historically important.

2.10.1 Design Process

Two Norwegians, Kristen Nygaard and Ole-Johan Dahl, developed the language SIMULA I between 1962 and 1964 at the Norwegian Computing Center (NCC) in Oslo. They were primarily interested in using computers for simulation but also worked in operations research. SIMULA I was designed exclusively for system simulation and was first implemented in late 1964 on a UNIVAC 1107 computer.

As soon as the SIMULA I implementation was completed, Nygaard and Dahl began efforts to extend the language by adding new features and modifying some existing constructs in order to make the language useful for general-purpose applications. The result of this work was SIMULA 67, whose design was first presented publicly in March 1967 (Dahl and Nygaard, 1967). We will discuss only SIMULA 67, although some of the features of interest in SIMULA 67 are also in SIMULA I.

2.10.2 Language Overview

SIMULA 67 is an extension of ALGOL 60, taking both block structure and the control statements from that language. The primary deficiency of ALGOL 60 (and other languages at that time) for simulation applications was the design of its subprograms. Simulation requires subprograms that are allowed to restart at the position where they previously stopped. Subprograms with this kind of control are known as **coroutines** because the caller and called subprograms have a somewhat equal relationship with each other, rather than the rigid master/slave relationship they have in most imperative languages.

To provide support for coroutines in SIMULA 67, the class construct was developed. This was an important development because the concept of data abstraction began with it. Furthermore, data abstraction provides the foundation for object-oriented programming.

It is interesting to note that the important concept of data abstraction was not developed and attributed to the class construct until 1972, when Hoare (1972) recognized the connection.

2.11 Orthogonal Design: ALGOL 68

ALGOL 68 was the source of several new ideas in language design, some of which were subsequently adopted by other languages. We include it here for that reason, even though it never achieved widespread use in either Europe or the United States.

2.11.1 Design Process

The development of the ALGOL family did not end when the revised report on ALGOL 60 appeared in 1962, although it was six years until the next design iteration was published. The resulting language, ALGOL 68 (van Wijngaarden et al., 1969), was dramatically different from its predecessor.

One of the most interesting innovations of ALGOL 68 was one of its primary design criteria: orthogonality. Recall our discussion of orthogonality in Chapter 1. The use of orthogonality resulted in several innovative features of ALGOL 68, one of which is described in the following section.

2.11.2 Language Overview

One important result of orthogonality in ALGOL 68 was its inclusion of user-defined data types. Earlier languages, such as Fortran, included only a few basic data structures. PL/I included a larger number of data structures, which made it harder to learn and difficult to implement, but it obviously could not provide an appropriate data structure for every need.

The approach of ALGOL 68 to data structures was to provide a few primitive types and structures and allow the user to combine those primitives into a large number of different structures. This provision for user-defined data types was carried over to some extent into all of the major imperative languages designed since then. User-defined data types are valuable because they allow the user to design data abstractions that fit particular problems very closely. All aspects of data types are discussed in Chapter 6.

As another first in the area of data types, ALGOL 68 introduced the kind of dynamic arrays that will be termed *implicit heap-dynamic* in Chapter 5. A dynamic array is one in which the declaration does not specify subscript bounds. Assignments to a dynamic array cause allocation of required storage. In ALGOL 68, dynamic arrays are called **`flex`** arrays.

2.11.3 Evaluation

ALGOL 68 includes a significant number of features that had not been previously used. Its use of orthogonality, which some may argue was overdone, was nevertheless revolutionary.

ALGOL 68 repeated one of the sins of ALGOL 60, however, and it was an important factor in its limited popularity. The language was described using an elegant and concise but also unknown metalanguage. Before one could read the language-describing document (van Wijngaarden et al., 1969), he or she had to learn the new metalanguage, called van Wijngaarden grammars, which were far more complex than BNF. To make matters worse, the designers invented a collection of words to explain the grammar and the language. For example, keywords were called *indicants*, substring extraction was called *trimming*, and the process of a subprogram execution was called a *coercion of deproceduring*, which might be *meek*, *firm*, or something else.

It is natural to contrast the design of PL/I with that of ALGOL 68, because they appeared only a few years apart. ALGOL 68 achieved writability by the principle of orthogonality: a few primitive concepts and the unrestricted use of a few combining mechanisms. PL/I achieved writability by including a large number of fixed constructs. ALGOL 68 extended the elegant simplicity of ALGOL 60, whereas PL/I simply threw together the features of several languages to attain its goals. Of course, it must be remembered that the goal of PL/I was to provide a unified tool for a broad class of problems, whereas ALGOL 68 was targeted to a single class: scientific applications.

PL/I achieved far greater acceptance than ALGOL 68, due largely to IBM's promotional efforts and the problems of understanding and implementing

ALGOL 68. Implementation was a difficult problem for both, but PL/I had the resources of IBM to apply to building a compiler. ALGOL 68 enjoyed no such benefactor.

2.12 Some Early Descendants of the ALGOLs

All imperative languages owe some of their design to ALGOL 60 and/or ALGOL 68. This section discusses some of the early descendants of these languages.

2.12.1 Simplicity by Design: Pascal

2.12.1.1 Historical Background

Niklaus Wirth (Wirth is pronounced "Virt") was a member of the International Federation of Information Processing (IFIP) Working Group 2.1, which was created to continue the development of ALGOL in the mid-1960s. In August 1965, Wirth and C. A. R. ("Tony") Hoare contributed to that effort by presenting to the group a somewhat modest proposal for additions and modifications to ALGOL 60 (Wirth and Hoare, 1966). The majority of the group rejected the proposal as being too small an improvement over ALGOL 60. Instead, a much more complex revision was developed, which eventually became ALGOL 68. Wirth, along with a few other group members, did not believe that the ALGOL 68 report should have been released, based on the complexity of both the language and the metalanguage used to describe it. This position later proved to have some validity because the ALGOL 68 documents, and therefore the language, were indeed found to be challenging by the computing community.

The Wirth and Hoare version of ALGOL 60 was named ALGOL-W. It was implemented at Stanford University and was used primarily as an instructional vehicle, but only at a few universities. The primary contributions of ALGOL-W were the value-result method of passing parameters and the `case` statement for multiple selection. The value-result method is an alternative to ALGOL 60's pass-by-name method. Both are discussed in Chapter 9. The `case` statement is discussed in Chapter 8.

Wirth's next major design effort, again based on ALGOL 60, was his most successful: Pascal.[10] The original published definition of Pascal appeared in 1971 (Wirth, 1971). This version was modified somewhat in the implementation process and is described in Wirth (1973). The features that are often ascribed to Pascal in fact came from earlier languages. For example, user-defined data types were introduced in ALGOL 68, the `case` statement in ALGOL-W, and Pascal's records are similar to those of COBOL and PL/I.

10. Pascal is named after Blaise Pascal, a seventeenth-century French philosopher and mathematician who invented the first mechanical adding machine in 1642 (among other things).

2.12.1.2 Evaluation

The largest impact of Pascal was on the teaching of programming. In 1970, most students of computer science, engineering, and science were introduced to programming with Fortran, although some universities used PL/I, languages based on PL/I, and ALGOL-W. By the mid-1970s, Pascal had become the most widely used language for this purpose. This was quite natural, because Pascal was designed specifically for teaching programming. It was not until the late 1990s that Pascal was no longer the most commonly used language for teaching programming in colleges and universities.

Because Pascal was designed as a teaching language, it lacks several features that are essential for many kinds of applications. The best example of this is the impossibility of writing a subprogram that takes as a parameter an array of variable length. Another example is the lack of any separate compilation capability. These deficiencies naturally led to many nonstandard dialects, such as Turbo Pascal.

Pascal's popularity, for both teaching programming and other applications, was based primarily on its remarkable combination of simplicity and expressivity. Although there are some insecurities in Pascal, it is still a relatively safe language, particularly when compared with Fortran or C. By the mid-1990s, the popularity of Pascal was on the decline, both in industry and in universities, primarily due to the rise of Modula-2, Ada, and C++, all of which included features not available in Pascal.

The following is an example of a Pascal program:

```
{Pascal Example Program
 Input:  An integer, listlen, where listlen is less than
         100, followed by listlen-integer values
 Output: The number of input values that are greater than
         the average of all input values }
program pasex (input, output);
  type intlisttype = array [1..99] of integer;
  var
    intlist : intlisttype;
    listlen, counter, sum, average, result : integer;
  begin
  result := 0;
  sum := 0;
  readln (listlen);
  if ((listlen > 0) and (listlen < 100)) then
    begin
{ Read input into an array and compute the sum }
    for counter := 1 to listlen do
      begin
      readln (intlist[counter]);
      sum := sum + intlist[counter]
      end;
```

```
{ Compute the average }
    average := sum / listlen;
{ Count the number of input values that are > average }
    for counter := 1 to listlen do
      if (intlist[counter] > average) then
        result := result + 1;
{ Print the result }
    writeln ('The number of values > average is:',
              result)
    end { of the then clause of if (( listlen > 0 ... }
  else
    writeln ('Error—input list length is not legal')
end.
```

2.12.2 A Portable Systems Language: C

Like Pascal, C contributed little to the previously known collection of language features, but it has been widely used over a long period of time. Although originally designed for systems programming, C is well suited for a wide variety of applications.

2.12.2.1 Historical Background

C's ancestors include CPL, BCPL, B, and ALGOL 68. CPL was developed at Cambridge University in the early 1960s. BCPL is a simple systems language, also developed at Cambridge, this time by Martin Richards in 1967 (Richards, 1969).

The first work on the UNIX operating system was done in the late 1960s by Ken Thompson at Bell Laboratories. The first version was written in assembly language. The first high-level language implemented under UNIX was B, which was based on BCPL. B was designed and implemented by Thompson in 1970.

Neither BCPL nor B is a typed language, which is an oddity among high-level languages, although both are much lower-level than a language such as Java. Being untyped means that all data are considered machine words, which, although simple, leads to many complications and insecurities. For example, there is the problem of specifying floating-point rather than integer arithmetic in an expression. In one implementation of BCPL, the variable operands of a floating-point operation were preceded by periods. Variable operands not preceded by periods were considered to be integers. An alternative to this would have been to use different symbols for the floating-point operators.

This problem, along with several others, led to the development of a new typed language based on B. Originally called NB but later named C, it was designed and implemented by Dennis Ritchie at Bell Laboratories in 1972 (Kernighan and Ritchie, 1978). In some cases through BCPL, and in other cases directly, C was influenced by ALGOL 68. This is seen in its **for**

and **switch** statements, in its assigning operators, and in its treatment of pointers.

The only "standard" for C in its first decade and a half was the book by Kernighan and Ritchie (1978).[11] Over that time span, the language slowly evolved, with different implementors adding different features. In 1989, ANSI produced an official description of C (ANSI, 1989), which included many of the features that implementors had already incorporated into the language. This standard was updated in 1999 (ISO, 1999). This later version includes a few significant changes to the language. Among these are a complex data type, a Boolean data type, and C++-style comments (//). We will refer to the 1989 version, which has long been called ANSI C, as C89; we will refer to the 1999 version as C99.

2.12.2.2 Evaluation

C has adequate control statements and data-structuring facilities to allow its use in many application areas. It also has a rich set of operators that provide a high degree of expressiveness.

One of the most important reasons why C is both liked and disliked is its lack of complete type checking. For example, in versions before C99, functions could be written for which parameters were not type checked. Those who like C appreciate the flexibility; those who do not like it find it too insecure. A major reason for its great increase in popularity in the 1980s was that a compiler for it was part of the widely used UNIX operating system. This inclusion in UNIX provided an essentially free and quite good compiler that was available to programmers on many different kinds of computers.

The following is an example of a C program:

```
/* C Example Program
 Input:  An integer, listlen, where listlen is less than
         100, followed by listlen-integer values
 Output: The number of input values that are greater than
         the average of all input values */
int main (){
  int intlist[99], listlen, counter, sum, average, result;
  sum = 0;
  result = 0;
  scanf("%d", &listlen);
  if ((listlen > 0) && (listlen < 100)) {
/* Read input into an array and compute the sum */
    for (counter = 0; counter < listlen; counter++) {
      scanf("%d", &intlist[counter]);
      sum += intlist[counter];
     }
```

11. This language is often referred to as "K & R C."

```
/* Compute the average */
    average = sum / listlen;
/* Count the input values that are > average */
    for (counter = 0; counter < listlen; counter++)
      if (intlist[counter] > average) result++;
/* Print result */
    printf("Number of values > average is:%d\n", result);
   }
  else
    printf("Error—input list length is not legal\n");
 }
```

2.13 Programming Based on Logic: Prolog

Simply put, logic programming is the use of a formal logic notation to communicate computational processes to a computer. Predicate calculus is the notation used in current logic programming languages.

Programming in logic programming languages is nonprocedural. Programs in such languages do not state exactly *how* a result is to be computed but rather describe the necessary form and/or characteristics of the result. What is needed to provide this capability in logic programming languages is a concise means of supplying the computer with both the relevant information and an inferencing process for computing desired results. Predicate calculus supplies the basic form of communication to the computer, and the proof method, named resolution, developed first by Robinson (1965), supplies the inferencing technique.

2.13.1 Design Process

During the early 1970s, Alain Colmerauer and Phillippe Roussel in the Artificial Intelligence Group at the University of Aix-Marseille, together with Robert Kowalski of the Department of Artificial Intelligence at the University of Edinburgh, developed the fundamental design of Prolog. The primary components of Prolog are a method for specifying predicate calculus propositions and an implementation of a restricted form of resolution. Both predicate calculus and resolution are described in Chapter 16. The first Prolog interpreter was developed at Marseille in 1972. The version of the language that was implemented is described in Roussel (1975). The name Prolog is from *pro*gramming *log*ic.

2.13.2 Language Overview

Prolog programs consist of collections of statements. Prolog has only a few kinds of statements, but they can be complex.

One common use of Prolog is as a kind of intelligent database. This application provides a simple framework for discussing the Prolog language.

The database of a Prolog program consists of two kinds of statements: facts and rules. The following are examples of fact statements:

```
mother(joanne, jake).
father(vern, joanne).
```

These state that `joanne` is the `mother` of `jake`, and `vern` is the `father` of `joanne`.

An example of a rule statement is

```
grandparent(X, Z) :- parent(X, Y), parent(Y, Z).
```

This states that it can be deduced that `X` is the `grandparent` of `Z` if it is true that `X` is the `parent` of `Y` and `Y` is the parent of `Z`, for some specific values for the variables `X`, `Y`, and `Z`.

The Prolog database can be interactively queried with goal statements, an example of which is

```
father(bob, darcie).
```

This asks if `bob` is the `father` of `darcie`. When such a query, or goal, is presented to the Prolog system, it uses its resolution process to attempt to determine the truth of the statement. If it can conclude that the goal is true, it displays "true." If it cannot prove it, it displays "false."

2.13.3 Evaluation

In the 1980s, there was a relatively small group of computer scientists who believed that logic programming provided the best hope for escaping from the complexity of imperative languages, and also from the enormous problem of producing the large amount of reliable software that was needed. So far, however, there are two major reasons why logic programming has not become more widely used. First, as with some other nonimperative approaches, programs written in logic languages thus far have proven to be highly inefficient relative to equivalent imperative programs. Second, it has been determined that it is an effective approach for only a few relatively small areas of application: certain kinds of database management systems and some areas of AI.

There is a dialect of Prolog that supports object-oriented programming—Prolog++ (Moss, 1994). Logic programming and Prolog are described in greater detail in Chapter 16.

2.14 History's Largest Design Effort: Ada

The Ada language is the result of the most extensive and expensive language design effort ever undertaken. The following paragraphs briefly describe the evolution of Ada.

2.14.1 Historical Background

The Ada language was developed for the Department of Defense (DoD), so the state of their computing environment was instrumental in determining its form. By 1974, over half of the applications of computers in DoD were embedded systems. An embedded system is one in which the computer hardware is embedded in the device it controls or for which it provides services. Software costs were rising rapidly, primarily because of the increasing complexity of systems. More than 450 different programming languages were in use for DoD projects, and none of them was standardized by DoD. Every defense contractor could define a new and different language for every contract.[12] Because of this language proliferation, application software was rarely reused. Furthermore, no software development tools were created (because they are usually language dependent). A great many languages were in use, but none was actually suitable for embedded systems applications. For these reasons, in 1974, the Army, Navy, and Air Force each independently proposed the development of a single high-level language for embedded systems.

2.14.2 Design Process

Noting this widespread interest, in January 1975, Malcolm Currie, director of Defense Research and Engineering, formed the High-Order Language Working Group (HOLWG), initially headed by Lt. Col. William Whitaker of the Air Force. The HOLWG had representatives from all of the military services and liaisons with Great Britain, France, and what was then West Germany. Its initial charter was to do the following:

- Identify the requirements for a new DoD high-level language.
- Evaluate existing languages to determine whether there was a viable candidate.
- Recommend adoption or implementation of a minimal set of programming languages.

In April 1975, the HOLWG produced the Strawman requirements document for the new language (Department of Defense, 1975a). This was distributed to military branches, federal agencies, selected industrial and university representatives, and interested parties in Europe.

12. This result was largely due to the widespread use of assembly language for embedded systems, along with the fact that most embedded systems used specialized processors.

The Strawman document was followed by Woodenman (Department of Defense, 1975b) in August 1975, Tinman (Department of Defense, 1976) in January 1976, Ironman (Department of Defense, 1977) in January 1977, and finally Steelman (Department of Defense, 1978) in June 1978.

After a tedious process, the many submitted proposals for the language were narrowed down to four finalists, all of which were based on Pascal. In May 1979, the Cii Honeywell/Bull language design proposal was chosen from the four finalists as the design that would be used. The Cii Honeywell/Bull design team in France, the only foreign competitor among the final four, was led by Jean Ichbiah.

In the spring of 1979, Jack Cooper of the Navy Materiel Command recommended the name for the new language, Ada, which was then adopted. The name commemorates Augusta Ada Byron (1815–1851), countess of Lovelace, mathematician, and daughter of poet Lord Byron. She is generally recognized as being the world's first programmer. She worked with Charles Babbage on his mechanical computers, the Difference and Analytical Engines, writing programs for several numerical processes.

The design and the rationale for Ada were published by ACM in its *SIGPLAN Notices* (ACM, 1979) and distributed to a readership of more than 10,000 people. A public test and evaluation conference was held in October 1979 in Boston, with representatives from over 100 organizations from the United States and Europe. By November, more than 500 language reports had been received from 15 different countries. Most of the reports suggested small modifications rather than drastic changes and outright rejections. Based on the language reports, the next version of the requirements specification, the Stoneman document (Department of Defense, 1980a), was released in February 1980.

A revised version of the language design was completed in July 1980 and was accepted as MIL-STD 1815, the standard *Ada Language Reference Manual.* The number 1815 was chosen because it was the year of the birth of Augusta Ada Byron. Another revised version of the *Ada Language Reference Manual* was released in July 1982. In 1983, the American National Standards Institute standardized Ada. This "final" official version is described in Goos and Hartmanis (1983). The Ada language design was then frozen for a minimum of five years.

2.14.3 Language Overview

This subsection briefly describes four of the major contributions of the Ada language.

Packages in the Ada language provide the means for encapsulating data objects, specifications for data types, and procedures. This, in turn, provides the support for the use of data abstraction in program design, as described in Chapter 11.

The Ada language includes extensive facilities for exception handling, which allow the programmer to gain control after any one of a wide variety

of exceptions, or run-time errors, has been detected. Exception handling is discussed in Chapter 14.

Program units can be generic in Ada. For example, it is possible to write a sort procedure that uses an unspecified type for the data to be sorted. Such a generic procedure must be instantiated for a specified type before it can be used, which is done with a statement that causes the compiler to generate a version of the procedure with the given type. The availability of such generic units increases the range of program units that might be reused, rather than duplicated, by programmers. Generics are discussed in Chapters 9 and 11.

The Ada language also provides for concurrent execution of special program units, named tasks, using the rendezvous mechanism. Rendezvous is the name of a method of intertask communication and synchronization. Concurrency is discussed in Chapter 13.

2.14.4 Evaluation

Perhaps the most important aspects of the design of the Ada language to consider are the following:

- Because the design was competitive, there were no limits on participation.
- The Ada language embodies most of the concepts of software engineering and language design of the late 1970s. Although one can question the actual approaches used to incorporate these features, as well as the wisdom of including such a large number of features in a language, most agree that the features are valuable.
- Although most people did not anticipate it, the development of a compiler for the Ada language was a difficult task. Only in 1985, almost four years after the language design was completed, did truly usable Ada compilers begin to appear.

The most serious criticism of Ada in its first few years was that it was too large and too complex. In particular, Hoare (1981) stated that it should not be used for any application where reliability is critical, which is precisely the type of application for which it was designed. On the other hand, others have praised it as the epitome of language design for its time. In fact, even Hoare eventually softened his view of the language.

The following is an example of an Ada program:

```
-- Ada Example Program
-- Input:  An integer, List_Len, where List_Len is less
--         than 100, followed by List_Len-integer values
-- Output: The number of input values that are greater
--         than the average of all input values
with Ada.Text_IO, Ada.Integer.Text_IO;
use Ada.Text_IO, Ada.Integer.Text_IO;
```

```
procedure Ada_Ex is
  type Int_List_Type is array (1..99) of Integer;
  Int_List : Int_List_Type;
  List_Len, Sum, Average, Result : Integer;
begin
  Result:= 0;
  Sum := 0;
  Get (List_Len);
  if (List_Len > 0) and (List_Len < 100) then
-- Read input data into an array and compute the sum
    for Counter := 1 .. List_Len loop
      Get (Int_List(Counter));
      Sum := Sum + Int_List(Counter);
    end loop;
-- Compute the average
    Average := Sum / List_Len;
-- Count the number of values that are > average
    for Counter := 1 .. List_Len loop
      if Int_List(Counter) > Average then
        Result:= Result+ 1;
      end if;
    end loop;
-- Print result
    Put ("The number of values > average is:");
    Put (Result);
    New_Line;
  else
    Put_Line ("Error—input list length is not legal");
  end if;
end Ada_Ex;
```

2.14.5 Ada 95 and Ada 2005

Two of the most important new features of Ada 95 are described briefly in the following paragraphs. In the remainder of the book, we will use the name Ada 83 for the original version and Ada 95 (its actual name) for the later version when it is important to distinguish between the two versions. In discussions of language features common to both versions, we will use the name Ada. The Ada 95 standard language is defined in ARM (1995).

The type derivation mechanism of Ada 83 is extended in Ada 95 to allow adding new components to those inherited from a base class. This provides for inheritance, a key ingredient in object-oriented programming languages. Dynamic binding of subprogram calls to subprogram definitions is accomplished through subprogram dispatching, which is based on the tag value of derived types through classwide types. This feature provides for polymorphism,

another principal feature of object-oriented programming. These features of Ada 95 are discussed in Chapter 12.

The rendezvous mechanism of Ada 83 provided only a cumbersome and inefficient means of sharing data among concurrent processes. It was necessary to introduce a new task to control access to the shared data. The protected objects of Ada 95 offer an attractive alternative to this. The shared data is encapsulated in a syntactic structure that controls all access to the data, either by rendezvous or by subprogram call. The new features of Ada 95 for concurrency and shared data are discussed in Chapter 13.

It is widely believed that the popularity of Ada 95 suffered because the Department of Defense stopped requiring its use in military software systems. There were, of course, other factors that hindered its growth in popularity. Most important among these was the widespread acceptance of C++ for object-oriented programming, which occurred before Ada 95 was released.

There were several additions to Ada 95 to get Ada 2005. Among these were interfaces, similar to those of Java, more control of scheduling algorithms, and synchronized interfaces.

Ada is widely used in both commercial and defense avionics, air traffic control, and rail transportation, as well as in other areas.

2.15 Object-Oriented Programming: Smalltalk

Smalltalk was the first programming language that fully supported object-oriented programming. It is therefore an important part of any discussion of the evolution of programming languages.

2.15.1 Design Process

The concepts that led to the development of Smalltalk originated in the Ph.D. dissertation work of Alan Kay in the late 1960s at the University of Utah (Kay, 1969). Kay had the remarkable foresight to predict the future availability of powerful desktop computers. Recall that the first microcomputer systems were not marketed until the mid-1970s, and they were only remotely related to the machines envisioned by Kay, which were seen to execute a million or more instructions per second and contain several megabytes of memory. Such machines, in the form of workstations, became widely available only in the early 1980s.

Kay believed that desktop computers would be used by nonprogrammers and thus would need very powerful human-interfacing capabilities. The computers of the late 1960s were largely batch oriented and were used exclusively by professional programmers and scientists. For use by nonprogrammers, Kay determined, a computer would have to be highly interactive and use sophisticated graphics in its user interface. Some of the graphics concepts came from

the LOGO experience of Seymour Papert, in which graphics were used to aid children in the use of computers (Papert, 1980).

Kay originally envisioned a system he called Dynabook, which was meant to be a general information processor. It was based in part on the Flex language, which he had helped design. Flex was based primarily on SIMULA 67. Dynabook used the paradigm of the typical desk, on which there are a number of papers, some partially covered. The top sheet is often the focus of attention, with the others temporarily out of focus. The display of Dynabook would model this scene, using screen windows to represent various sheets of paper on the desktop. The user would interact with such a display both through keystrokes and by touching the screen with his or her fingers. After the preliminary design of Dynabook earned him a Ph.D., Kay's goal became to see such a machine constructed.

Kay found his way to the Xerox Palo Alto Research Center (Xerox PARC) and presented his ideas on Dynabook. This led to his employment there and the subsequent birth of the Learning Research Group at Xerox. The first charge of the group was to design a language to support Kay's programming paradigm and implement it on the best personal computer then available. These efforts resulted in an "Interim" Dynabook, consisting of a Xerox Alto workstation and Smalltalk-72 software. Together, they formed a research tool for further development. A number of research projects were conducted with this system, including several experiments to teach programming to children. Along with the experiments came further developments, leading to a sequence of languages that ended with Smalltalk-80. As the language grew, so did the power of the hardware on which it resided. By 1980, both the language and the Xerox hardware nearly matched the early vision of Alan Kay.

2.15.2 Language Overview

The Smalltalk world is populated by nothing but objects, from integer constants to large complex software systems. All computing in Smalltalk is done by the same uniform technique: sending a message to an object to invoke one of its methods. A reply to a message is an object, which either returns the requested information or simply notifies the sender that the requested processing has been completed. The fundamental difference between a message and a subprogram call is this: A message is sent to a data object, specifically to one of the methods defined for the object. The called method is then executed, often modifying the data of the object to which the message was sent; a subprogram call is a message to the code of a subprogram. Usually the data to be processed by the subprogram is sent to it as a parameter.[13]

In Smalltalk, object abstractions are classes, which are very similar to the classes of SIMULA 67. Instances of the class can be created and are then the objects of the program.

The syntax of Smalltalk is unlike that of most other programming language, in large part because of the use of messages, rather than arithmetic and

13. Of course, a method call can also pass data to be processed by the called method.

logic expressions and conventional control statements. One of the Smalltalk control constructs is illustrated in the example in the next subsection.

2.15.3 Evaluation

Smalltalk has done a great deal to promote two separate aspects of computing: graphical user interfaces and object-oriented programming. The windowing systems that are now the dominant method of user interfaces to software systems grew out of Smalltalk. Today, the most significant software design methodologies and programming languages are object oriented. Although the origin of some of the ideas of object-oriented languages came from SIMULA 67, they reached maturation in Smalltalk. It is clear that Smalltalk's impact on the computing world is extensive and will be long-lived.

The following is an example of a Smalltalk class definition:

```
"Smalltalk Example Program"
"The following is a class definition, instantiations
of which can draw equilateral polygons of any number of
sides"
class name                    Polygon
superclass                    Object
instance variable names       ourPen
numSides
sideLength
"Class methods"
  "Create an instance"
  new
     ^ super new getPen

  "Get a pen for drawing polygons"
  getPen
     ourPen <- Pen new defaultNib: 2

  "Instance methods"
  "Draw a polygon"
  draw
     numSides timesRepeat: [ourPen go: sideLength;
                            turn: 360 // numSides]

  "Set length of sides"
  length: len
     sideLength <- len

  "Set number of sides"
  sides: num
     numSides <- num
```

2.16 Combining Imperative and Object-Oriented Features: C++

The origins of C were discussed in Section 2.12; the origins of Simula 67 were discussed in Section 2.10; the origins of Smalltalk were discussed in Section 2.15. C++ builds language facilities, borrowed from Simula 67, on top of C to support much of what Smalltalk pioneered. C++ has evolved from C through a sequence of modifications to improve its imperative features and to add constructs to support object-oriented programming.

2.16.1 Design Process

The first step from C toward C++ was made by Bjarne Stroustrup at Bell Laboratories in 1980. The initial modifications to C included the addition of function parameter type checking and conversion and, more significantly, classes, which are related to those of SIMULA 67 and Smalltalk. Also included were derived classes, public/private access control of inherited components, constructor and destructor methods, and friend classes. During 1981, inline functions, default parameters, and overloading of the assignment operator were added. The resulting language was called C with Classes and is described in Stroustrup (1983).

It is useful to consider some goals of C with Classes. The primary goal was to provide a language in which programs could be organized as they could be organized in SIMULA 67—that is, with classes and inheritance. A second important goal was that there should be little or no performance penalty relative to C. For example, array index range checking was not even considered because a significant performance disadvantage, relative to C, would result. A third goal of C with Classes was that it could be used for every application for which C could be used, so virtually none of the features of C would be removed, not even those considered to be unsafe.

By 1984, this language was extended by the inclusion of virtual methods, which provide dynamic binding of method calls to specific method definitions, method name and operator overloading, and reference types. This version of the language was called C++. It is described in Stroustrup (1984).

In 1985, the first available implementation appeared: a system named Cfront, which translated C++ programs into C programs. This version of Cfront and the version of C++ it implemented were named Release 1.0. It is described in Stroustrup (1986).

Between 1985 and 1989, C++ continued to evolve, based largely on user reactions to the first distributed implementation. This next version was named Release 2.0. Its Cfront implementation was released in June 1989. The most important features added to C++ Release 2.0 were support for multiple inheritance (classes with more than one parent class) and abstract classes, along with some other enhancements. Abstract classes are described in Chapter 12.

Release 3.0 of C++ evolved between 1989 and 1990. It added templates, which provide parameterized types, and exception handling. The current version of C++, which was standardized in 1998, is described in ISO (1998).

In 2002, Microsoft released its .NET computing platform, which included a new version of C++, named Managed C++, or MC++. MC++ extends C++ to provide access to the functionality of the .NET Framework. The additions include properties, delegates, interfaces, and a reference type for garbage-collected objects. Properties are discussed in Chapter 11. Delegates are briefly discussed in the introduction to C# in Section 2.19. Because .NET does not support multiple inheritance, neither does MC++.

2.16.2 Language Overview

Because C++ has both functions and methods, it supports both procedural and object-oriented programming.

Operators in C++ can be overloaded, meaning the user can create operators for existing operators on user-defined types. C++ methods can also be overloaded, meaning the user can define more than one method with the same name, provided either the numbers or types of their parameters are different.

Dynamic binding in C++ is provided by virtual methods. These methods define type-dependent operations, using overloaded methods, within a collection of classes that are related through inheritance. A pointer to an object of class A can also point to objects of classes that have class A as an ancestor. When this pointer points to an overloaded virtual method, the method of the current type is chosen dynamically.

Both methods and classes can be templated, which means that they can be parameterized. For example, a method can be written as a templated method to allow it to have versions for a variety of parameter types. Classes enjoy the same flexibility.

C++ supports multiple inheritance. It also includes exception handling that is significantly different from that of Ada. One difference is that hardware-detectable exceptions cannot be handled. The exception-handling constructs of Ada and C++ are discussed in Chapter 14.

2.16.3 Evaluation

C++ rapidly became and remains a widely used language. One factor in its popularity is the availability of good and inexpensive compilers. Another factor is that it is almost completely backward compatible with C (meaning that C programs can be, with few changes, compiled as C++ programs), and in most implementations it is possible to link C++ code with C code—and thus relatively easy for the many C programmers to learn C++. Finally, at the time C++ first appeared, when object-oriented programming began to receive widespread interest, C++ was the only available language that was suitable for large commercial software projects.

On the negative side, because C++ is a very large and complex language, it clearly suffers drawbacks similar to those of PL/I. It inherited most of the insecurities of C, which make it less safe than languages such as Ada and Java.

2.16.4 A Related Language: Objective-C

Objective-C (Kochan, 2009) is another hybrid language with both imperative and object-oriented features. Objective-C was designed by Brad Cox and Tom Love in the early 1980s. Initially, it consisted of C plus the classes and message passing of Smalltalk. Among the programming languages that were created by adding support for object-oriented programming to an imperative language, Objective-C is the only one to use the Smalltalk syntax for that support.

After Steve Jobs left Apple and founded NeXT, he licensed Objective-C and it was used to write the NeXT computer system software. NeXT also released its Objective-C compiler, along with the NeXTstep development environment and a library of utilities. After the NeXT project failed, Apple bought NeXT and used Objective-C to write MAC OS X. Objective-C is the language of all iPhone software, which explains its rapid rise in popularity after the iPhone appeared.

One characteristic that Objective-C inherited from Smalltalk is the dynamic binding of messages to objects. This means that there is no static checking of messages. If a message is sent to an object and the object cannot respond to the message, it is not known until run time, when an exception is raised.

In 2006, Apple announced Objective-C 2.0, which added a form of garbage collection and new syntax for declaring properties. Unfortunately, garbage collection is not supported by the iPhone run-time system.

Objective-C is a strict superset of C, so all of the insecurities of that language are present in Objective-C.

2.16.5 Another Related Language: Delphi

Delphi (Lischner, 2000) is a hybrid language, similar to C++ and Objetive-C in that it was created by adding object-oriented support, among other things, to an existing imperative language, in this case Pascal. Many of the differences between C++ and Delphi are a result of the predecessor languages and the surrounding programming cultures from which they are derived. Because C is a powerful but potentially unsafe language, C++ also fits that description, at least in the areas of array subscript range checking, pointer arithmetic, and its numerous type coercions. Likewise, because Pascal is more elegant and safer than C, Delphi is more elegant and safer than C++. Delphi is also less complex than C++. For example, Delphi does not allow user-defined operator overloading, generic subprograms, and parameterized classes, all of which are part of C++.

Delphi, like Visual C++, provides a graphical user interface (GUI) to the developer and simple ways to create GUI interfaces to applications written in Delphi. Delphi was designed by Anders Hejlsberg, who had previously developed the Turbo Pascal system. Both of these were marketed and distributed by Borland. Hejlsberg was also the lead designer of C#.

2.16.6 A Loosely Related Language: Go

The Go programming language is not directly related to C++, although it is C-based. It is in this section in part because it does not deserve its own section and it does not fit elsewhere.

Go was designed by Rob Pike, Ken Thompson, and Robert Griesemer at Google. Thompson is the designer of the predecessor of C, B, as well as the codesigner with Dennis Ritchie of UNIX. He and Pike were both formerly employed at Bell Labs. The initial design was begun in 2007 and the first implementation was released in late 2009. One of the initial motivations for Go was the slowness of compilation of large C++ programs at Google. One of the characteristics of the initial compiler for Go is that is it extremely fast. The Go language borrows some of its syntax and constructs from C. Some of the new features of Go include the following: (1) Data declarations are syntactically reversed from the other C-based languages; (2) the variables precede the type name; (3) variable declarations can be given a type by inference if the variable is given an initial value; and (4) functions can return multiple values. Go does not support traditional object-oriented programming, as it has no form of inheritance. However, methods can be defined for any type. It also does not have generics. The control statements of Go are similar to those of other C-based languages, although the switch does not include the implicit fall through to the next segment. Go includes a goto statement, pointers, associative arrays, interfaces (though they are different from those of Java and C#), and support for concurrency using its goroutines.

2.17 An Imperative-Based Object-Oriented Language: Java

Java's designers started with C++, removed some constructs, changed some, and added a few others. The resulting language provides much of the power and flexibility of C++, but in a smaller, simpler, and safer language.

2.17.1 Design Process

Java, like many programming languages, was designed for an application for which there appeared to be no satisfactory existing language. In 1990, Sun Microsystems determined there was a need for a programming language for embedded consumer electronic devices, such as toasters, microwave ovens, and interactive TV systems. Reliability was one of the primary goals for such a language. It may not seem that reliability would be an important factor in the software for a microwave oven. If an oven had malfunctioning software, it probably would not pose a grave danger to anyone and most likely would not lead to large legal settlements. However, if the software in a particular model was found to be erroneous after a million units had been manufactured and sold, their recall would entail significant cost. Therefore, reliability *is* an important characteristic of the software in consumer electronic products.

After considering C and C++, it was decided that neither would be satisfactory for developing software for consumer electronic devices. Although C was relatively small, it did not provide support for object-oriented programming, which they deemed a necessity. C++ supported object-oriented programming, but it was judged to be too large and complex, in part because it also supported procedure-oriented programming. It was also believed that neither C nor C++ provided the necessary level of reliability. So, a new language, later named Java, was designed. Its design was guided by the fundamental goal of providing greater simplicity and reliability than C++ was believed to provide.

Although the initial impetus for Java was consumer electronics, none of the products with which it was used in its early years were ever marketed. Starting in 1993, when the World Wide Web became widely used, and largely because of the new graphical browsers, Java was found to be a useful tool for Web programming. In particular, Java applets, which are relatively small Java programs that are interpreted in Web browsers and whose output can be included in displayed Web documents, quickly became very popular in the middle to late 1990s. In the first few years of Java popularity, the Web was its most common application.

The Java design team was headed by James Gosling, who had previously designed the UNIX emacs editor and the NeWS windowing system.

2.17.2 Language Overview

As we stated previously, Java is based on C++ but it was specifically designed to be smaller, simpler, and more reliable. Like C++, Java has both classes and primitive types. Java arrays are instances of a predefined class, whereas in C++ they are not, although many C++ users build wrapper classes for arrays to add features like index range checking, which is implicit in Java.

Java does not have pointers, but its reference types provide some of the capabilities of pointers. These references are used to point to class instances. All objects are allocated on the heap. References are always implicitly dereferenced, when necessary. So they behave more like ordinary scalar variables.

Java has a primitive Boolean type named **`boolean`**, used mainly for the control expressions of its control statements (such as **`if`** and **`while`**). Unlike C and C++, arithmetic expressions cannot be used for control expressions.

One significant difference between Java and many of its predecessors that support object-oriented programming, including C++, is that it is not possible to write stand-alone subprograms in Java. All Java subprograms are methods and are defined in classes. Furthermore, methods can be called through a class or object only. One consequence of this is that while C++ supports both procedural and object-oriented programming, Java supports object-oriented programming only.

Another important difference between C++ and Java is that C++ supports multiple inheritance directly in its class definitions. Java supports only single

inheritance of classes, although some of the benefits of multiple inheritance can be gained by using its interface construct.

Among the C++ constructs that were not copied into Java are structs and unions.

Java includes a relatively simple form of concurrency control through its **`synchronized`** modifier, which can appear on methods and blocks. In either case, it causes a lock to be attached. The lock ensures mutually exclusive access or execution. In Java, it is relatively easy to create concurrent processes, which in Java are called *threads*.

Java uses implicit storage deallocation for its objects, often called **garbage collection**. This frees the programmer from needing to delete objects explicitly when they are no longer needed. Programs written in languages that do not have garbage collection often suffer from what is sometimes called memory leakage, which means that storage is allocated but never deallocated. This can obviously lead to eventual depletion of all available storage. Object deallocation is discussed in detail in Chapter 6.

Unlike C and C++, Java includes assignment type coercions (implicit type conversions) only if they are widening (from a "smaller" type to a "larger" type). So **`int`** to **`float`** coercions are done across the assignment operator, but **`float`** to **`int`** coercions are not.

2.17.3 Evaluation

The designers of Java did well at trimming the excess and/or unsafe features of C++. For example, the elimination of half of the assignment coercions that are done in C++ was clearly a step toward higher reliability. Index range checking of array accesses also makes the language safer. The addition of concurrency enhances the scope of applications that can be written in the language, as do the class libraries for graphical user interfaces, database access, and networking.

Java's portability, at least in intermediate form, has often been attributed to the design of the language, but it is not. Any language can be translated to an intermediate form and "run" on any platform that has a virtual machine for that intermediate form. The price of this kind of portability is the cost of interpretation, which traditionally has been about an order of magnitude more than execution of machine code. The initial version of the Java interpreter, called the Java Virtual Machine (JVM), indeed was at least 10 times slower than equivalent compiled C programs. However, many Java programs are now translated to machine code before being executed, using Just-in-Time (JIT) compilers. This makes the efficiency of Java programs competitive with that of programs in conventionally compiled languages such as C++.

The use of Java increased faster than that of any other programming language. Initially, this was due to its value in programming dynamic Web documents. Clearly, one of the reasons for Java's rapid rise to prominence is simply that programmers like its design. Some developers thought C++ was simply too

large and complex to be practical and safe. Java offered them an alternative that has much of the power of C++, but in a simpler, safer language. Another reason is that the compiler/interpreter system for Java is free and easily obtained on the Web. Java is now widely used in a variety of different applications areas.

The most recent version of Java, Java 7, appeared in 2011. Since its beginning, many features have been added to the language, including an enumeration class, generics, and a new iteration construct.

The following is an example of a Java program:

```
// Java Example Program
//  Input: An integer, listlen, where listlen is less
//         than 100, followed by length-integer values
// Output: The number of input data that are greater than
//         the average of all input values
import java.io.*;
class IntSort {
public static void main(String args[]) throws IOException {
  DataInputStream in = new DataInputStream(System.in);
  int listlen,
      counter,
      sum = 0,
      average,
      result = 0;
  int[] intlist = new int[99];
  listlen = Integer.parseInt(in.readLine());
  if ((listlen > 0) && (listlen < 100)) {
/* Read input into an array and compute the sum  */
    for (counter = 0; counter < listlen; counter++) {
      intlist[counter] =
             Integer.valueOf(in.readLine()).intValue();
      sum += intlist[counter];
    }
/* Compute the average */
    average = sum / listlen;
/* Count the input values that are > average */
    for (counter = 0; counter < listlen; counter++)
      if (intlist[counter] > average) result++;
/* Print result */
      System.out.println(
          "\nNumber of values > average is:" + result);
    }  //** end of then clause of if ((listlen > 0) ...
    else System.out.println(
              "Error—input list length is not legal\n");
  }  //** end of method main
}  //** end of class IntSort
```

2.18 Scripting Languages

Scripting languages have evolved over the past 25 years. Early scripting languages were used by putting a list of commands, called a **script**, in a file to be interpreted. The first of these languages, named `sh` (for shell), began as a small collection of commands that were interpreted as calls to system subprograms that performed utility functions, such as file management and simple file filtering. To this were added variables, control flow statements, functions, and various other capabilities, and the result is a complete programming language. One of the most powerful and widely known of these is `ksh` (Bolsky and Korn, 1995), which was developed by David Korn at Bell Laboratories.

Another scripting language is `awk`, developed by Al Aho, Brian Kernighan, and Peter Weinberger at Bell Laboratories (Aho et al., 1988). `awk` began as a report-generation language but later became a more general-purpose language.

2.18.1 Origins and Characteristics of Perl

The Perl language, developed by Larry Wall, was originally a combination of `sh` and `awk`. Perl has grown significantly since its beginnings and is now a powerful, although still somewhat primitive, programming language. Although it is still often called a scripting language, it is actually more similar to a typical imperative language, since it is always compiled, at least into an intermediate language, before it is executed. Furthermore, it has all the constructs to make it applicable to a wide variety of areas of computational problems.

Perl has a number of interesting features, only a few of which are mentioned in this chapter and later discussed in the book.

Variables in Perl are statically typed and implicitly declared. There are three distinctive namespaces for variables, denoted by the first character of the variables' names. All scalar variable names begin with dollar signs (`$`), all array names begin with at signs (`@`), and all hash names (hashes are briefly described below) begin with percent signs (`%`). This convention makes variable names in programs more readable than those of any other programming language.

Perl includes a large number of implicit variables. Some of them are used to store Perl parameters, such as the particular form of newline character or characters that are used in the implementation. Implicit variables are commonly used as default parameters to built-in functions and default operands for some operators. The implicit variables have distinctive—although cryptic—names, such as `$!` and `@_`. The implicit variables' names, like the user-defined variable names, use the three namespaces, so `$!` is a scalar.

Perl's arrays have two characteristics that set them apart from the arrays of the common imperative languages. First, they have dynamic length, meaning that they can grow and shrink as needed during execution. Second, arrays can be sparse, meaning that there can be gaps between the elements. These

gaps do not take space in memory, and the iteration statement used for arrays, **foreach**, iterates over the missing elements.

Perl includes associative arrays, which are called **hashes**. These data structures are indexed by strings and are implicitly controlled hash tables. The Perl system supplies the hash function and increases the size of the structure when necessary.

Perl is a powerful, but somewhat dangerous, language. Its scalar type stores both strings and numbers, which are normally stored in double-precision floating-point form. Depending on the context, numbers may be coerced to strings and vice versa. If a string is used in numeric context and the string cannot be converted to a number, zero is used and there is no warning or error message provided for the user. This effect can lead to errors that are not detected by the compiler or run-time system. Array indexing cannot be checked, because there is no set subscript range for any array. References to nonexistent elements return **undef**, which is interpreted as zero in numeric context. So, there is also no error detection in array element access.

Perl's initial use was as a UNIX utility for processing text files. It was and still is widely used as a UNIX system administration tool. When the World Wide Web appeared, Perl achieved widespread use as a Common Gateway Interface language for use with the Web, although it is now rarely used for that purpose. Perl is used as a general-purpose language for a variety of applications, such as computational biology and artificial intelligence.

The following is an example of a Perl program:

```
# Perl Example Program
# Input:  An integer, $listlen, where $listlen is less
#         than 100, followed by $listlen-integer values.
# Output: The number of input values that are greater than
#        the average of all input values.
($sum, $result) = (0, 0);
$listlen = <STDIN>;
if (($listlen > 0) && ($listlen < 100)) {
# Read input into an array and compute the sum
  for ($counter = 0; $counter < $listlen; $counter++) {
    $intlist[$counter] = <STDIN>;
  } #- end of for (counter ...
# Compute the average
  $average = $sum / $listlen;
# Count the input values that are > average
  foreach $num (@intlist) {
    if ($num > $average) { $result++; }
  } #- end of foreach $num ...
# Print result
  print "Number of values > average is: $result \n";
} #- end of if (($listlen ...
```

```
else {
  print "Error--input list length is not legal \n";
}
```

2.18.2 Origins and Characteristics of JavaScript

Use of the Web exploded in the mid-1990s after the first graphical browsers appeared. The need for computation associated with HTML documents, which by themselves are completely static, quickly became critical. Computation on the server side was made possible with the Common Gateway Interface (CGI), which allowed HTML documents to request the execution of programs on the server, with the results of such computations returned to the browser in the form of HTML documents. Computation on the browser end became available with the advent of Java applets. Both of these approaches have now been replaced for the most part by newer technologies, primarily scripting languages.

JavaScript (Flanagan, 2002) was originally developed by Brendan Eich at Netscape. Its original name was Mocha. It was later renamed LiveScript. In late 1995, LiveScript became a joint venture of Netscape and Sun Microsystems and its name was changed to JavaScript. JavaScript has gone through extensive evolution, moving from version 1.0 to version 1.5 by adding many new features and capabilities. A language standard for JavaScript was developed in the late 1990s by the European Computer Manufacturers Association (ECMA) as ECMA-262. This standard has also been approved by the International Standards Organization (ISO) as ISO-16262. Microsoft's version of JavaScript is named JScript .NET.

Although a JavaScript interpreter could be embedded in many different applications, its most common use is embedded in Web browsers. JavaScript code is embedded in HTML documents and interpreted by the browser when the documents are displayed. The primary uses of JavaScript in Web programming are to validate form input data and create dynamic HTML documents. JavaScript also is now used with the Rails Web development framework.

In spite of its name, JavaScript is related to Java only through the use of similar syntax. Java is strongly typed, but JavaScript is dynamically typed (see Chapter 5). JavaScript's character strings and its arrays have dynamic length. Because of this, array indices are not checked for validity, although this is required in Java. Java fully supports object-oriented programming, but JavaScript supports neither inheritance nor dynamic binding of method calls to methods.

One of the most important uses of JavaScript is for dynamically creating and modifying HTML documents. JavaScript defines an object hierarchy that matches a hierarchical model of an HTML document, which is defined by the Document Object Model. Elements of an HTML document are accessed through these objects, providing the basis for dynamic control of the elements of documents.

Following is a JavaScript script for the problem previously solved in several languages in this chapter. Note that it is assumed that this script will be called from an HTML document and interpreted by a Web browser.

```
// example.js
//   Input: An integer, listLen, where listLen is less
//          than 100, followed by listLen-numeric values
//  Output: The number of input values that are greater
//          than the average of all input values

var intList = new Array(99);
var listLen, counter, sum = 0, result = 0;

listLen = prompt (
        "Please type the length of the input list", "");
if ((listLen > 0) && (listLen < 100)) {

// Get the input and compute its sum
   for (counter = 0; counter < listLen; counter++) {
      intList[counter] = prompt (
                      "Please type the next number", "");
      sum += parseInt(intList[counter]);
   }

// Compute the average
   average = sum / listLen;

// Count the input values that are > average
   for (counter = 0; counter < listLen; counter++)
      if (intList[counter] > average) result++;

// Display the results
   document.write("Number of values > average is: ",
                 result, "<br />");
} else
   document.write(
       "Error - input list length is not legal <br />");
```

2.18.3 Origins and Characteristics of PHP

PHP (Converse and Park, 2000) was developed by Rasmus Lerdorf, a member of the Apache Group, in 1994. His initial motivation was to provide a tool to help track visitors to his personal Web site. In 1995, he developed a package called Personal Home Page Tools, which became the first publicly distributed version of PHP. Originally, PHP was an abbreviation for Personal Home Page. Later, its user community began using the recursive name PHP: Hypertext

Preprocessor, which subsequently forced the original name into obscurity. PHP is now developed, distributed, and supported as an open-source product. PHP processors are resident on most Web servers.

PHP is an HTML-embedded server-side scripting language specifically designed for Web applications. PHP code is interpreted on the Web server when an HTML document in which it is embedded has been requested by a browser. PHP code usually produces HTML code as output, which replaces the PHP code in the HTML document. Therefore, a Web browser never sees PHP code.

PHP is similar to JavaScript, in its syntactic appearance, the dynamic nature of its strings and arrays, and its use of dynamic typing. PHP's arrays are a combination of JavaScript's arrays and Perl's hashes.

The original version of PHP did not support object-oriented programming, but that support was added in the second release. However, PHP does not support abstract classes or interfaces, destructors, or access controls for class members.

PHP allows simple access to HTML form data, so form processing is easy with PHP. PHP provides support for many different database management systems. This makes it a useful language for building programs that need Web access to databases.

2.18.4 Origins and Characteristics of Python

Python (Lutz and Ascher, 2004) is a relatively recent object-oriented interpreted scripting language. Its initial design was by Guido van Rossum at Stichting Mathematisch Centrum in the Netherlands in the early 1990s. Its development is now being done by the Python Software Foundation. Python is being used for the same kinds of applications as Perl: system administration, CGI programming, and other relatively small computing tasks. Python is an open-source system and is available for most common computing platforms. The Python implementation is available at `www.python.org`, which also has extensive information regarding Python.

Python's syntax is not based directly on any commonly used language. It is type checked, but dynamically typed. Instead of arrays, Python includes three kinds of data structures: lists; immutable lists, which are called **tuples**; and hashes, which are called **dictionaries**. There is a collection of list methods, such as `append`, `insert`, `remove`, and `sort`, as well as a collection of methods for dictionaries, such as `keys`, `values`, `copy`, and `has_key`. Python also supports list comprehensions, which originated with the Haskell language. List comprehensions are discussed in Section 15.8.

Python is object oriented, includes the pattern-matching capabilities of Perl, and has exception handling. Garbage collection is used to reclaim objects when they are no longer needed.

Support for CGI programming, and form processing in particular, is provided by the `cgi` module. Modules that support cookies, networking, and database access are also available.

Python includes support for concurrency with its threads, as well as support for network programming with its sockets. It also has more support for functional programming than other nonfunctional programming languages.

One of the more interesting features of Python is that it can be easily extended by any user. The modules that support the extensions can be written in any compiled language. Extensions can add functions, variables, and object types. These extensions are implemented as additions to the Python interpreter.

2.18.5 Origins and Characteristics of Ruby

Ruby (Thomas et al., 2005) was designed by Yukihiro Matsumoto (aka Matz) in the early 1990s and released in 1996. Since then it has continually evolved. The motivation for Ruby was dissatisfaction of its designer with Perl and Python. Although both Perl and Python support object-oriented programming,[14] neither is a pure object-oriented language, at least in the sense that each has primitive (nonobject) types and each supports functions.

The primary characterizing feature of Ruby is that it is a pure object-oriented language, just as is Smalltalk. Every data value is an object and all operations are via method calls. The operators in Ruby are only syntactic mechanisms to specify method calls for the corresponding operations. Because they are methods, they can be redefined. All classes, predefined or user defined, can be subclassed.

Both classes and objects in Ruby are dynamic in the sense that methods can be dynamically added to either. This means that both classes and objects can have different sets of methods at different times during execution. So, different instantiations of the same class can behave differently. Collections of methods, data, and constants can be included in the definition of a class.

The syntax of Ruby is related to that of Eiffel and Ada. There is no need to declare variables, because dynamic typing is used. The scope of a variable is specified in its name: A variable whose name begins with a letter has local scope; one that begins with `@` is an instance variable; one that begins with `$` has global scope. A number of features of Perl are present in Ruby, including implicit variables with silly names, such as `$_`.

As is the case with Python, any user can extend and/or modify Ruby. Ruby is culturally interesting because it is the first programming language designed in Japan that has achieved relatively widespread use in the United States.

2.18.6 Origins and Characteristics of Lua

Lua[15] was designed in the early 1990s by Roberto Ierusalimschy, Waldemar Celes, and Luis Henrique de Figueiredo at the Pontifical University of Rio de Janeiro in Brazil. It is a scripting language that supports procedural and

14. Actully, Python's support for object-oriented programming is partial.

15. The name Lua is derived from the Portuguese word for moon.

functional programming with extensibility as one of its primary goals. Among the languages that influenced its design are Scheme, Icon, and Python.

Lua is similar to JavaScript in that it does not support object-oriented programming but it was clearly influenced by it. Both have objects that play the role of both classes and objects and both have prototype inheritance rather than class inheritance. However, in Lua, the language can be extended to support object-oriented programming.

As in Scheme, Lua's functions are first-class values. Also, Lua supports closures. These capabilities allow it to be used for functional programming. Also like Scheme, Lua has only a single data structure, although in Lua's case, it is the table. Lua's tables extend PHP's associate arrays, which subsume the arrays of traditional imperative languages. References to table elements can take the form of references to traditional arrays, associative arrays, or records. Because functions are first-class values, they can be stored in tables, and such tables can serve as namespaces.

Lua uses garbage collection for its objects, which are all heap allocated. It uses dynamic typing, as do most of the other scripting languages.

Lua is a relatively small and simple language, having only 21 reserved words. The design philosophy of the language was to provide the bare essentials and relatively simple ways to extend the language to allow it to fit a variety of application areas. Much of its extensibility derives from its table data structure, which can be customized using Lua's metatable concept.

Lua can conveniently be used as a scripting language extension to other languages. Like early implementations of Java, Lua is translated to an intermediate code and interpreted. It easily can be embedded simply in other systems, in part because of the small size of its interpreter, which is only about 150K bytes.

During 2006 and 2007, the popularity of Lua grew rapidly, in part due to its use in the gaming industry. The sequence of scripting languages that have appeared over the past 20 years has already produced several widely used languages. Lua, the latest arrival among them, is quickly becoming one.

2.19 The Flagship .NET Language: C#

C#, along with the new development platform .NET,[16] was announced by Microsoft in 2000. In January 2002, production versions of both were released.

2.19.1 Design Process

C# is based on C++ and Java but includes some ideas from Delphi and Visual BASIC. Its lead designer, Anders Hejlsberg, also designed Turbo Pascal and Delphi, which explains the Delphi parts of the heritage of C#.

16. The .NET development system is briefly discussed in Chapter 1.

The purpose of C# is to provide a language for component-based software development, specifically for such development in the .NET Framework. In this environment, components from a variety of languages can be easily combined to form systems. All of the .NET languages, which include C#, Visual Basic .NET, Managed C++, F#, and JScript .NET,[17] use the Common Type System (CTS). The CTS provides a common class library. All types in all five .NET languages inherit from a single class root, `System.Object`. Compilers that conform to the CTS specification create objects that can be combined into software systems. All .NET languages are compiled into the same intermediate form, Intermediate Language (IL).[18] Unlike Java, however, the IL is never interpreted. A Just-in-Time compiler is used to translate IL into machine code before it is executed.

2.19.2 Language Overview

Many believe that one of Java's most important advances over C++ lies in the fact that it excludes some of C++'s features. For example, C++ supports multiple inheritance, pointers, structs, **`enum`** types, operator overloading, and a goto statement, but Java includes none of these. The designers of C# obviously disagreed with this wholesale removal of features, because all of these except multiple inheritance have been included in the new language.

To the credit of C#'s designers, however, in several cases, the C# version of a C++ feature has been improved. For example, the **`enum`** types of C# are safer than those of C++, because they are never implicitly converted to integers. This allows them to be more type safe. The **`struct`** type was changed significantly, resulting in a truly useful construct, whereas in C++ it serves virtually no purpose. C#'s structs are discussed in Chapter 12. C# takes a stab at improving the **`switch`** statement that is used in C, C++, and Java. C#'s switch is discussed in Chapter 8.

Although C++ includes function pointers, they share the lack of safety that is inherent in C++'s pointers to variables. C# includes a new type, delegates, which are both object-oriented and type-safe method references to subprograms. Delegates are used for implementing event handlers, controlling the execution of threads, and callbacks.[19] Callbacks are implemented in Java with interfaces; in C++, method pointers are used.

In C#, methods can take a variable number of parameters, as long as they are all the same type. This is specified by the use of a formal parameter of array type, preceded by the **`params`** reserved word.

Both C++ and Java use two distinct typing systems: one for primitives and one for objects. In addition to being confusing, this leads to a frequent need to

---

17. Many other languages have been modified to be .NET languages.

18. Initially, IL was called MSIL (Microsoft Intermediate Language), but apparently many people thought that name was too long.

19. When an object calls a method of another object and needs to be notified when that method has completed its task, the called method calls its caller back. This is known as a callback.

convert values between the two systems—for example, to put a primitive value into a collection that stores objects. C# makes the conversion between values of the two typing systems partially implicit through the implicit boxing and unboxing operations, which are discussed in detail in Chapter 12.[20]

Among the other features of C# are rectangular arrays, which are not supported in most programming languages, and a **foreach** statement, which is an iterator for arrays and collection objects. A similar **foreach** statement is found in Perl, PHP, and Java 5.0. Also, C# includes properties, which are an alternative to public data members. Properties are specified as data members with get and set methods, which are implicitly called when references and assignments are made to the associated data members.

C# has evolved continuously and quickly from its initial release in 2002. The most recent version is C# 2010. C# 2010 adds a form of dynamic typing, implicit typing, and anonymous types (see Chapter 6).

2.19.3 Evaluation

C# was meant to be an improvement over both C++ and Java as a general-purpose programming language. Although it can be argued that some of its features are a step backward, C# clearly includes some constructs that move it beyond its predecessors. Some of its features will surely be adopted by programming languages of the near future. Some already do.

The following is an example of a C# program:

```
// C# Example Program
// Input:  An integer, listlen, where listlen is less than
//         100, followed by listlen-integer values.
// Output: The number of input values that are greater
//         than the average of all input values.
using System;
public class Ch2example {
  static void Main() {
    int[] intlist;
    int listlen,
        counter,
        sum = 0,
        average,
        result = 0;
    intList = new int[99];
    listlen = Int32.Parse(Console.readLine());
    if ((listlen > 0) && (listlen < 100)) {
// Read input into an array and compute the sum
      for (counter = 0; counter < listlen; counter++) {
```

20. This feature was added to Java in Java 5.0.

```
        intList[counter] =
                        Int32.Parse(Console.readLine());
        sum += intList[counter];
      } //- end of for (counter ...
// Compute the average
      average = sum / listlen;
// Count the input values that are > average
      foreach (int num in intList)
        if (num > average) result++;
// Print result
      Console.WriteLine(
         "Number of values > average is:" + result);
    } //- end of if ((listlen ...
    else
      Console.WriteLine(
         "Error--input list length is not legal");
  } //- end of method Main
} //- end of class Ch2example
```

2.20 Markup/Programming Hybrid Languages

A markup/programming hybrid language is a markup language in which some of the elements can specify programming actions, such as control flow and computation. The following subsections introduce two such hybrid languages, XSLT and JSP.

2.20.1 XSLT

eXtensible Markup Language (XML) is a metamarkup language. Such a language is used to define markup languages. XML-derived markup languages are used to define data documents, which are called XML documents. Although XML documents are human readable, they are processed by computers. This processing sometimes consists only of transformations to forms that can be effectively displayed or printed. In many cases, such transformations are to HTML, which can be displayed by a Web browser. In other cases, the data in the document is processed, just as with other forms of data files.

The transformation of XML documents to HTML documents is specified in another markup language, eXtensible Stylesheet Language Transformations (XSLT) (`www.w3.org/TR/XSLT`). XSLT can specify programming-like operations. Therefore, XSLT is a markup/programming hybrid language. XSLT was defined by the World Wide Web Consortium (W3C) in the late 1990s.

An XSLT processor is a program that takes as input an XML data document and an XSLT document (which is also in the form of an XML document). In this processing, the XML data document is transformed to another XML

document,[21] using the transformations described in the XSLT document. The XSLT document specifies transformations by defining templates, which are data patterns that could be found by the XSLT processor in the XML input file. Associated with each template in the XSLT document are its transformation instructions, which specify how the matching data is to be transformed before being put in the output document. So, the templates (and their associated processing) act as subprograms, which are "executed" when the XSLT processor finds a pattern match in the data of the XML document.

XSLT also has programming constructs at a lower level. For example, a looping construct is included, which allows repeated parts of the XML document to be selected. There is also a sort process. These lower-level constructs are specified with XSLT tags, such as `<for-each>`.

2.20.2 JSP

The "core" part of the Java Server Pages Standard Tag Library (JSTL) is another markup/programming hybrid language, although its form and purpose are different from those of XSLT. Before discussing JSTL, it is necessary to introduce the ideas of servlets and Java Server Pages (JSP). A **servlet** is an instance of a Java class that resides on and is executed on a Web server system. The execution of a servlet is requested by a markup document being displayed by a Web browser. The servlet's output, which is in the form of an HTML document, is returned to the requesting browser. A program that runs in the Web server process, called a **servlet container**, controls the execution of servlets. Servlets are commonly used for form processing and for database access.

JSP is a collection of technologies designed to support dynamic Web documents and provide other processing needs of Web documents. When a JSP document, which is often a mixture of HTML and Java, is requested by a browser, the JSP processor program, which resides on a Web server system, converts the document to a servlet. The document's embedded Java code is copied to the servlet. The plain HTML is copied into Java print statements that output it as is. The JSTL markup in the JSP document is processed, as discussed in the following paragraph. The servlet produced by the JSP processor is run by the servlet container.

The JSTL defines a collection of XML action elements that control the processing of the JSP document on the Web server. These elements have the same form as other elements of HTML and XML. One of the most commonly used JSTL control action elements is `if`, which specifies a Boolean expression as an attribute.[22] The content of the `if` element (the text between the opening tag (`<if>`) and its closing tag (`</if>`)) is HTML code that will be included in the output document only if the Boolean expression evaluates to true. The `if` element is related to the C/C++ `#if` preprocessor command. The JSP

21. The output document of the XSLT processor could also be in HTML or plain text.

22. An attribute in HTML, which is embedded in the opening tag of an element, provides further information about that element.

container processes the JSTL parts of JSP documents in a way that is similar to how the C/C++ preprocessor processes C and C++ programs. The preprocessor commands are instructions for the preprocessor to specify how the output file is to be constructed from the input file. Similarly, JSTL control action elements are instructions for the JSP processor to specify how to build the XML output file from the XML input file.

One common use of the `if` element is for the validation of form data submitted by a browser user. Form data is accessible by the JSP processor and can be tested with the `if` element to ensure that it is sensible data. If not, the `if` element can insert an error message for the user in the output document.

For multiple selection control, JSTL has `choose`, `when`, and `otherwise` elements. JSTL also includes a `forEach` element, which iterates over collections, which typically are form values from a client. The `forEach` element can include `begin`, `end`, and `step` attributes to control its iterations.

SUMMARY

We have investigated the development and the development environments of a number of programming languages. This chapter gives the reader a good perspective on current issues in language design. We have set the stage for an in-depth discussion of the important features of contemporary languages.

BIBLIOGRAPHIC NOTES

Perhaps the most important source of historical information about the development of early programming languages is *History of Programming Languages*, edited by Richard Wexelblat (1981). It contains the developmental background and environment of 13 important programming languages, as told by the designers themselves. A similar work resulted from a second "history" conference, published as a special issue of *ACM SIGPLAN Notices* (ACM, 1993a). In this work, the history and evolution of 13 more programming languages are discussed.

The paper "Early Development of Programming Languages" (Knuth and Pardo, 1977), which is part of the *Encyclopedia of Computer Science and Technology*, is an excellent 85-page work that details the development of languages up to and including Fortran. The paper includes example programs to demonstrate the features of many of those languages.

Another book of great interest is *Programming Languages: History and Fundamentals*, by Jean Sammet (1969). It is a 785-page work filled with details of 80 programming languages of the 1950s and 1960s. Sammet has also published several updates to her book, such as *Roster of Programming Languages for 1974–75* (1976).

REVIEW QUESTIONS

1. In what year was Plankalkül designed? In what year was that design published?
2. What two common data structures were included in Plankalkül?
3. How were the pseudocodes of the early 1950s implemented?
4. Speedcoding was invented to overcome two significant shortcomings of the computer hardware of the early 1950s. What were they?
5. Why was the slowness of interpretation of programs acceptable in the early 1950s?
6. What hardware capability that first appeared in the IBM 704 computer strongly affected the evolution of programming languages? Explain why.
7. In what year was the Fortran design project begun?
8. What was the primary application area of computers at the time Fortran was designed?
9. What was the source of all of the control flow statements of Fortran I?
10. What was the most significant feature added to Fortran I to get Fortran II?
11. What control flow statements were added to Fortran IV to get Fortran 77?
12. Which version of Fortran was the first to have any sort of dynamic variables?
13. Which version of Fortran was the first to have character string handling?
14. Why were linguists interested in artificial intelligence in the late 1950s?
15. Where was LISP developed? By whom?
16. In what way are Scheme and Common LISP opposites of each other?
17. What dialect of LISP is used for introductory programming courses at some universities?
18. What two professional organizations together designed ALGOL 60?
19. In what version of ALGOL did block structure appear?
20. What missing language element of ALGOL 60 damaged its chances for widespread use?
21. What language was designed to describe the syntax of ALGOL 60?
22. On what language was COBOL based?
23. In what year did the COBOL design process begin?
24. What data structure that appeared in COBOL originated with Plankalkül?
25. What organization was most responsible for the early success of COBOL (in terms of extent of use)?

26. What user group was the target of the first version of BASIC?
27. Why was BASIC an important language in the early 1980s?
28. PL/I was designed to replace what two languages?
29. For what new line of computers was PL/I designed?
30. What features of SIMULA 67 are now important parts of some object-oriented languages?
31. What innovation of data structuring was introduced in ALGOL 68 but is often credited to Pascal?
32. What design criterion was used extensively in ALGOL 68?
33. What language introduced the `case` statement?
34. What operators in C were modeled on similar operators in ALGOL 68?
35. What are two characteristics of C that make it less safe than Pascal?
36. What is a nonprocedural language?
37. What are the two kinds of statements that populate a Prolog database?
38. What is the primary application area for which Ada was designed?
39. What are the concurrent program units of Ada called?
40. What Ada construct provides support for abstract data types?
41. What populates the Smalltalk world?
42. What three concepts are the basis for object-oriented programming?
43. Why does C++ include the features of C that are known to be unsafe?
44. From what language does Objective-C borrow its syntax for method calls?
45. What programming paradigm that nearly all recently designed languages support is not supported by Go?
46. What is the primary application for Objective-C?
47. What language designer worked on both C and Go?
48. What do the Ada and COBOL languages have in common?
49. What was the first application for Java?
50. What characteristic of Java is most evident in JavaScript?
51. How does the typing system of PHP and JavaScript differ from that of Java?
52. What array structure is included in C# but not in C, C++, or Java?
53. What two languages was the original version of Perl meant to replace?
54. For what application area is JavaScript most widely used?
55. What is the relationship between JavaScript and PHP, in terms of their use?
56. PHP's primary data structure is a combination of what two data structures from other languages?

57. What data structure does Python use in place of arrays?
58. What characteristic does Ruby share with Smalltalk?
59. What characteristic of Ruby's arithmetic operators makes them unique among those of other languages?
60. What data structures are built into Lua?
61. Is Lua normally compiled, purely interpreted, or impurely interpreted?
62. What feature of Delphi's classes is included in C#?
63. What deficiency of the `switch` statement of C is addressed with the changes made by C# to that statement?
64. What is the primary platform on which C# is used?
65. What are the inputs to an XSLT processor?
66. What is the output of an XSLT processor?
67. What element of the JSTL is related to a subprogram?
68. To what is a JSP document converted by a JSP processor?
69. Where are servlets executed?

PROBLEM SET

1. What features of Plankalkül do you think would have had the greatest influence on Fortran 0 if the Fortran designers had been familiar with Plankalkül?
2. Determine the capabilities of Backus's 701 Speedcoding system, and compare them with those of a contemporary programmable hand calculator.
3. Write a short history of the A-0, A-1, and A-2 systems designed by Grace Hopper and her associates.
4. As a research project, compare the facilities of Fortran 0 with those of the Laning and Zierler system.
5. Which of the three original goals of the ALGOL design committee, in your opinion, was most difficult to achieve at that time?
6. Make an educated guess as to the most common syntax error in LISP programs.
7. LISP began as a pure functional language but gradually acquired more and more imperative features. Why?
8. Describe in detail the three most important reasons, in your opinion, why ALGOL 60 did not become a very widely used language.
9. Why, in your opinion, did COBOL allow long identifiers when Fortran and ALGOL did not?

10. Outline the major motivation of IBM in developing PL/I.
11. Was IBM's assumption, on which it based its decision to develop PL/I, correct, given the history of computers and language developments since 1964?
12. Describe, in your own words, the concept of orthogonality in programming language design.
13. What is the primary reason why PL/I became more widely used than ALGOL 68?
14. What are the arguments both for and against the idea of a typeless language?
15. Are there any logic programming languages other than Prolog?
16. What is your opinion of the argument that languages that are too complex are too dangerous to use, and we should therefore keep all languages small and simple?
17. Do you think language design by committee is a good idea? Support your opinion.
18. Languages continually evolve. What sort of restrictions do you think are appropriate for changes in programming languages? Compare your answers with the evolution of Fortran.
19. Build a table identifying all of the major language developments, together with when they occurred, in what language they first appeared, and the identities of the developers.
20. There have been some public interchanges between Microsoft and Sun concerning the design of Microsoft's J++ and C# and Sun's Java. Read some of these documents, which are available on their respective Web sites, and write an analysis of the disagreements concerning the delegates.
21. In recent years data structures have evolved within scripting languages to replace traditional arrays. Explain the chronological sequence of these developments.
22. Explain two reasons why pure interpretation is an acceptable implementation method for several recent scripting languages.
23. Perl 6, when it arrives, will likely be a significantly enlarged language. Make an educated guess as to whether a language like Lua will also grow continuously over its lifetime. Support your answer.
24. Why, in your opinion, do new scripting languages appear more frequently than new compiled languages?
25. Give a brief general description of a markup/programming hybrid language.

PROGRAMMING EXERCISES

1. To understand the value of records in a programming language, write a small program in a C-based language that uses an array of structs that store student information, including name, age, GPA as a float, and grade level as a string (e.g., "freshmen," etc.). Also, write the same program in the same language without using structs.
2. To understand the value of recursion in a programming language, write a program that implements quicksort, first using recursion and then without recursion.
3. To understand the value of counting loops, write a program that implements matrix multiplication using counting loop constructs. Then write the same program using only logical loops—for example, **`while`** loops.

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3

Describing Syntax and Semantics

3.1 Introduction

3.2 The General Problem of Describing Syntax

3.3 Formal Methods of Describing Syntax

3.4 Attribute Grammars

3.5 Describing the Meanings of Programs: Dynamic Semantics

This chapter covers the following topics. First, the terms *syntax* and *semantics* are defined. Then, a detailed discussion of the most common method of describing syntax, context-free grammars (also known as Backus-Naur Form), is presented. Included in this discussion are derivations, parse trees, ambiguity, descriptions of operator precedence and associativity, and extended Backus-Naur Form. Attribute grammars, which can be used to describe both the syntax and static semantics of programming languages, are discussed next. In the last section, three formal methods of describing semantics—operational, axiomatic, and denotational semantics—are introduced. Because of the inherent complexity of the semantics description methods, our discussion of them is brief. One could easily write an entire book on just one of the three (as several authors have).

3.1 Introduction

The task of providing a concise yet understandable description of a programming language is difficult but essential to the language's success. ALGOL 60 and ALGOL 68 were first presented using concise formal descriptions; in both cases, however, the descriptions were not easily understandable, partly because each used a new notation. The levels of acceptance of both languages suffered as a result. On the other hand, some languages have suffered the problem of having many slightly different dialects, a result of a simple but informal and imprecise definition.

One of the problems in describing a language is the diversity of the people who must understand the description. Among these are initial evaluators, implementors, and users. Most new programming languages are subjected to a period of scrutiny by potential users, often people within the organization that employs the language's designer, before their designs are completed. These are the initial evaluators. The success of this feedback cycle depends heavily on the clarity of the description.

Programming language implementors obviously must be able to determine how the expressions, statements, and program units of a language are formed, and also their intended effect when executed. The difficulty of the implementors' job is, in part, determined by the completeness and precision of the language description.

Finally, language users must be able to determine how to encode software solutions by referring to a language reference manual. Textbooks and courses enter into this process, but language manuals are usually the only authoritative printed information source about a language.

The study of programming languages, like the study of natural languages, can be divided into examinations of syntax and semantics. The **`syntax`** of a programming language is the form of its expressions, statements, and program units. Its **semantics** is the meaning of those expressions, statements, and program units. For example, the syntax of a Java **`while`** statement is

`while` (boolean_expr) statement

The semantics of this statement form is that when the current value of the Boolean expression is true, the embedded statement is executed. Otherwise, control continues after the **`while`** construct. Then control implicitly returns to the Boolean expression to repeat the process.

Although they are often separated for discussion purposes, syntax and semantics are closely related. In a well-designed programming language, semantics should follow directly from syntax; that is, the appearance of a statement should strongly suggest what the statement is meant to accomplish.

Describing syntax is easier than describing semantics, partly because a concise and universally accepted notation is available for syntax description, but none has yet been developed for semantics.

3.2 The General Problem of Describing Syntax

A language, whether natural (such as English) or artificial (such as Java), is a set of strings of characters from some alphabet. The strings of a language are called **sentences** or statements. The syntax rules of a language specify which strings of characters from the language's alphabet are in the language. English, for example, has a large and complex collection of rules for specifying the syntax of its sentences. By comparison, even the largest and most complex programming languages are syntactically very simple.

Formal descriptions of the syntax of programming languages, for simplicity's sake, often do not include descriptions of the lowest-level syntactic units. These small units are called **lexemes**. The description of lexemes can be given by a lexical specification, which is usually separate from the syntactic description of the language. The lexemes of a programming language include its numeric literals, operators, and special words, among others. One can think of programs as strings of lexemes rather than of characters.

Lexemes are partitioned into groups—for example, the names of variables, methods, classes, and so forth in a programming language form a group called *identifiers*. Each lexeme group is represented by a name, or token. So, a **token** of a language is a category of its lexemes. For example, an identifier is a token that can have lexemes, or instances, such as `sum` and `total`. In some cases, a token has only a single possible lexeme. For example, the token for the arithmetic operator symbol + has just one possible lexeme. Consider the following Java statement:

```
index = 2 * count + 17;
```

The lexemes and tokens of this statement are

Lexemes	*Tokens*
`index`	identifier
`=`	equal_sign
`2`	int_literal

`*`	mult_op
`count`	identifier
`+`	plus_op
`17`	int_literal
`;`	semicolon

The example language descriptions in this chapter are very simple, and most include lexeme descriptions.

3.2.1 Language Recognizers

In general, languages can be formally defined in two distinct ways: by **recognition** and by **generation** (although neither provides a definition that is practical by itself for people trying to learn or use a programming language). Suppose we have a language L that uses an alphabet Σ of characters. To define L formally using the recognition method, we would need to construct a mechanism R, called a recognition device, capable of reading strings of characters from the alphabet Σ. R would indicate whether a given input string was or was not in L. In effect, R would either accept or reject the given string. Such devices are like filters, separating legal sentences from those that are incorrectly formed. If R, when fed any string of characters over Σ, accepts it only if it is in L, then R is a description of L. Because most useful languages are, for all practical purposes, infinite, this might seem like a lengthy and ineffective process. Recognition devices, however, are not used to enumerate all of the sentences of a language—they have a different purpose.

The syntax analysis part of a compiler is a recognizer for the language the compiler translates. In this role, the recognizer need not test all possible strings of characters from some set to determine whether each is in the language. Rather, it need only determine whether given programs are in the language. In effect then, the syntax analyzer determines whether the given programs are syntactically correct. The structure of syntax analyzers, also known as parsers, is discussed in Chapter 4.

3.2.2 Language Generators

A language generator is a device that can be used to generate the sentences of a language. We can think of the generator as having a button that produces a sentence of the language every time it is pushed. Because the particular sentence that is produced by a generator when its button is pushed is unpredictable, a generator seems to be a device of limited usefulness as a language descriptor. However, people prefer certain forms of generators over recognizers because they can more easily read and understand them. By contrast, the syntax-checking portion of a compiler (a language recognizer) is not as useful a language description for a programmer because it can be used only in trial-and-error mode. For example, to determine the correct syntax of a particular statement using a compiler, the programmer can only submit a speculated version and note whether

the compiler accepts it. On the other hand, it is often possible to determine whether the syntax of a particular statement is correct by comparing it with the structure of the generator.

There is a close connection between formal generation and recognition devices for the same language. This was one of the seminal discoveries in computer science, and it led to much of what is now known about formal languages and compiler design theory. We return to the relationship of generators and recognizers in the next section.

3.3 Formal Methods of Describing Syntax

This section discusses the formal language-generation mechanisms, usually called **grammars**, that are commonly used to describe the syntax of programming languages.

3.3.1 Backus-Naur Form and Context-Free Grammars

In the middle to late 1950s, two men, Noam Chomsky and John Backus, in unrelated research efforts, developed the same syntax description formalism, which subsequently became the most widely used method for programming language syntax.

3.3.1.1 Context-Free Grammars

In the mid-1950s, Chomsky, a noted linguist (among other things), described four classes of generative devices or grammars that define four classes of languages (Chomsky, 1956, 1959). Two of these grammar classes, named *context-free* and *regular*, turned out to be useful for describing the syntax of programming languages. The forms of the tokens of programming languages can be described by regular grammars. The syntax of whole programming languages, with minor exceptions, can be described by context-free grammars. Because Chomsky was a linguist, his primary interest was the theoretical nature of natural languages. He had no interest at the time in the artificial languages used to communicate with computers. So it was not until later that his work was applied to programming languages.

3.3.1.2 Origins of Backus-Naur Form

Shortly after Chomsky's work on language classes, the ACM-GAMM group began designing ALGOL 58. A landmark paper describing ALGOL 58 was presented by John Backus, a prominent member of the ACM-GAMM group, at an international conference in 1959 (Backus, 1959). This paper introduced a new formal notation for specifying programming language syntax. The new notation was later modified slightly by Peter Naur for the description of

ALGOL 60 (Naur, 1960). This revised method of syntax description became known as **Backus-Naur Form**, or simply **BNF**.

BNF is a natural notation for describing syntax. In fact, something similar to BNF was used by Panini to describe the syntax of Sanskrit several hundred years before Christ (Ingerman, 1967).

Although the use of BNF in the ALGOL 60 report was not immediately accepted by computer users, it soon became and is still the most popular method of concisely describing programming language syntax.

It is remarkable that BNF is nearly identical to Chomsky's generative devices for context-free languages, called **context-free grammars**. In the remainder of the chapter, we refer to context-free grammars simply as grammars. Furthermore, the terms BNF and grammar are used interchangeably.

3.3.1.3 Fundamentals

A **metalanguage** is a language that is used to describe another language. BNF is a metalanguage for programming languages.

BNF uses abstractions for syntactic structures. A simple Java assignment statement, for example, might be represented by the abstraction <assign> (pointed brackets are often used to delimit names of abstractions). The actual definition of <assign> can be given by

<assign> → <var> = <expression>

The text on the left side of the arrow, which is aptly called the **left-hand side** (LHS), is the abstraction being defined. The text to the right of the arrow is the definition of the LHS. It is called the **right-hand side** (RHS) and consists of some mixture of tokens, lexemes, and references to other abstractions. (Actually, tokens are also abstractions.) Altogether, the definition is called a **rule**, or **production**. In the example rule just given, the abstractions <var> and <expression> obviously must be defined for the <assign> definition to be useful.

This particular rule specifies that the abstraction <assign> is defined as an instance of the abstraction <var>, followed by the lexeme =, followed by an instance of the abstraction <expression>. One example sentence whose syntactic structure is described by the rule is

```
total = subtotal1 + subtotal2
```

The abstractions in a BNF description, or grammar, are often called **nonterminal symbols**, or simply **nonterminals**, and the lexemes and tokens of the rules are called **terminal symbols**, or simply **terminals**. A BNF description, or **grammar**, is a collection of rules.

Nonterminal symbols can have two or more distinct definitions, representing two or more possible syntactic forms in the language. Multiple definitions can be written as a single rule, with the different definitions separated by

the symbol |, meaning logical OR. For example, a Java **`if`** statement can be described with the rules

<if_stmt> → **`if`** (<logic_expr>) <stmt>
<if_stmt> → **`if`** (<logic_expr>) <stmt> **`else`** <stmt>

or with the rule

<if_stmt> → **`if`** (<logic_expr>) <stmt>
| **`if`** (<logic_expr>) <stmt> **`else`** <stmt>

In these rules, <stmt> represents either a single statement or a compound statement.

Although BNF is simple, it is sufficiently powerful to describe nearly all of the syntax of programming languages. In particular, it can describe lists of similar constructs, the order in which different constructs must appear, and nested structures to any depth, and even imply operator precedence and operator associativity.

3.3.1.4 Describing Lists

Variable-length lists in mathematics are often written using an ellipsis (. . .); 1, 2, . . . is an example. BNF does not include the ellipsis, so an alternative method is required for describing lists of syntactic elements in programming languages (for example, a list of identifiers appearing on a data declaration statement). For BNF, the alternative is recursion. A rule is **recursive** if its LHS appears in its RHS. The following rules illustrate how recursion is used to describe lists:

<ident_list> → identifier
| identifier, <ident_list>

This defines <ident_list> as either a single token (identifier) or an identifier followed by a comma and another instance of <ident_list>. Recursion is used to describe lists in many of the example grammars in the remainder of this chapter.

3.3.1.5 Grammars and Derivations

A grammar is a generative device for defining languages. The sentences of the language are generated through a sequence of applications of the rules, beginning with a special nonterminal of the grammar called the **start symbol**. This sequence of rule applications is called a **derivation**. In a grammar for a complete programming language, the start symbol represents a complete program and is often named <program>. The simple grammar shown in Example 3.1 is used to illustrate derivations.

EXAMPLE 3.1 A Grammar for a Small Language

```
<program> → begin <stmt_list> end

<stmt_list> → <stmt>
            | <stmt> ; <stmt_list>
<stmt> → <var> = <expression>
<var> → A | B | C
<expression> → <var> + <var>
             | <var> – <var>
             | <var>
```

The language described by the grammar of Example 3.1 has only one statement form: assignment. A program consists of the special word **begin**, followed by a list of statements separated by semicolons, followed by the special word **end**. An expression is either a single variable or two variables separated by either a + or - operator. The only variable names in this language are A, B, and C.

A derivation of a program in this language follows:

```
<program> => begin <stmt_list> end
          => begin <stmt> ; <stmt_list> end
          => begin <var> = <expression> ; <stmt_list> end
          => begin A = <expression> ; <stmt_list> end
          => begin A = <var> + <var> ; <stmt_list> end
          => begin A = B + <var> ; <stmt_list> end
          => begin A = B + C ; <stmt_list> end
          => begin A = B + C ; <stmt> end
          => begin A = B + C ; <var> = <expression> end
          => begin A = B + C ; B = <expression> end
          => begin A = B + C ; B = <var> end
          => begin A = B + C ; B = C end
```

This derivation, like all derivations, begins with the start symbol, in this case <program>. The symbol => is read "derives." Each successive string in the sequence is derived from the previous string by replacing one of the nonterminals with one of that nonterminal's definitions. Each of the strings in the derivation, including <program>, is called a **sentential form**.

In this derivation, the replaced nonterminal is always the leftmost nonterminal in the previous sentential form. Derivations that use this order of replacement are called **leftmost derivations**. The derivation continues until the sentential form contains no nonterminals. That sentential form, consisting of only terminals, or lexemes, is the generated sentence.

In addition to leftmost, a derivation may be rightmost or in an order that is neither leftmost nor rightmost. Derivation order has no effect on the language generated by a grammar.

By choosing alternative RHSs of rules with which to replace nonterminals in the derivation, different sentences in the language can be generated. By exhaustively choosing all combinations of choices, the entire language can be generated. This language, like most others, is infinite, so one cannot generate *all* the sentences in the language in finite time.

Example 3.2 is another example of a grammar for part of a typical programming language.

EXAMPLE 3.2 A Grammar for Simple Assignment Statements

```
<assign> → <id> = <expr>
<id> → A | B | C
<expr> → <id> + <expr>
       | <id> * <expr>
       | ( <expr> )
       | <id>
```

The grammar of Example 3.2 describes assignment statements whose right sides are arithmetic expressions with multiplication and addition operators and parentheses. For example, the statement

```
A = B * ( A + C )
```

is generated by the leftmost derivation:

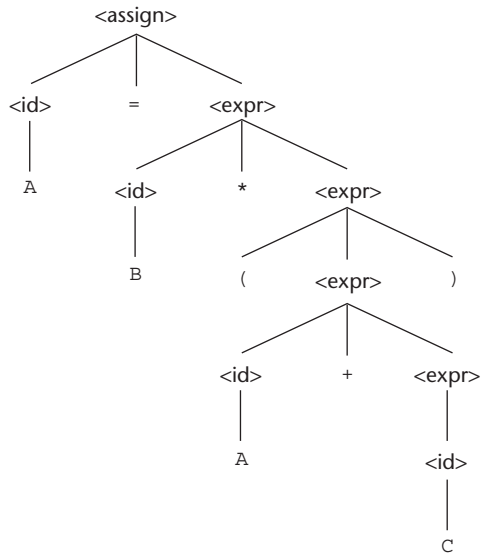
```
<assign> => <id> = <expr>
         => A = <expr>
         => A = <id> * <expr>
         => A = B * <expr>
         => A = B * ( <expr> )
         => A = B * ( <id> + <expr> )
         => A = B * ( A + <expr> )
         => A = B * ( A + <id> )
         => A = B * ( A + C )
```

3.3.1.6 Parse Trees

One of the most attractive features of grammars is that they naturally describe the hierarchical syntactic structure of the sentences of the languages they define. These hierarchical structures are called **parse trees**. For example, the parse tree in Figure 3.1 shows the structure of the assignment statement derived previously.

Figure 3.1

A parse tree for the simple statement `A = B * (A + C)`



Every internal node of a parse tree is labeled with a nonterminal symbol; every leaf is labeled with a terminal symbol. Every subtree of a parse tree describes one instance of an abstraction in the sentence.

3.3.1.7 Ambiguity

A grammar that generates a sentential form for which there are two or more distinct parse trees is said to be **ambiguous**. Consider the grammar shown in Example 3.3, which is a minor variation of the grammar shown in Example 3.2.

EXAMPLE 3.3 An Ambiguous Grammar for Simple Assignment Statements

```
<assign> → <id> = <expr>
<id> → A | B | C
<expr> → <expr> + <expr>
         | <expr> * <expr>
         | ( <expr> )
         | <id>
```

The grammar of Example 3.3 is ambiguous because the sentence

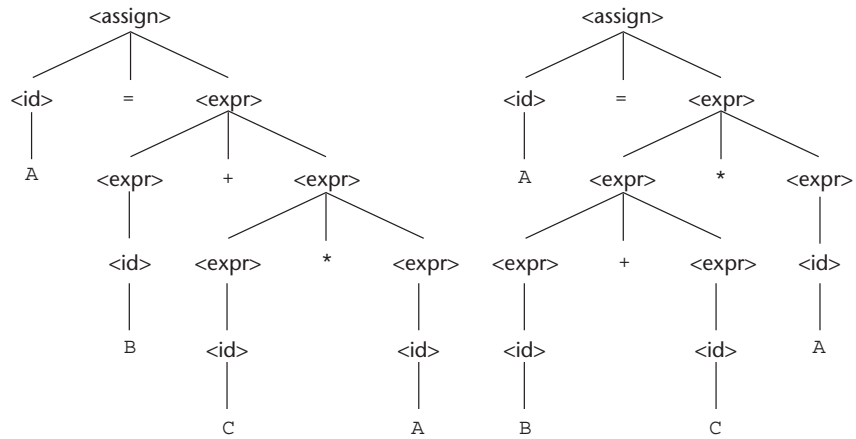
```
A = B + C * A
```

has two distinct parse trees, as shown in Figure 3.2. The ambiguity occurs because the grammar specifies slightly less syntactic structure than does the grammar of

Example 3.2. Rather than allowing the parse tree of an expression to grow only on the right, this grammar allows growth on both the left and the right.

Figure 3.2

Two distinct parse trees for the same sentence, `A = B + C * A`



Syntactic ambiguity of language structures is a problem because compilers often base the semantics of those structures on their syntactic form. Specifically, the compiler chooses the code to be generated for a statement by examining its parse tree. If a language structure has more than one parse tree, then the meaning of the structure cannot be determined uniquely. This problem is discussed in two specific examples in the following subsections.

There are several other characteristics of a grammar that are sometimes useful in determining whether a grammar is ambiguous.[1] They include the following: (1) if the grammar generates a sentence with more than one leftmost derivation and (2) if the grammar generates a sentence with more than one rightmost derivation.

Some parsing algorithms can be based on ambiguous grammars. When such a parser encounters an ambiguous construct, it uses nongrammatical information provided by the designer to construct the correct parse tree. In many cases, an ambiguous grammar can be rewritten to be unambiguous but still generate the desired language.

3.3.1.8 Operator Precedence

When an expression includes two different operators, for example, `x + y * z`, one obvious semantic issue is the order of evaluation of the two operators (for example, in this expression is it add and then multiply, or vice versa?). This semantic question can be answered by assigning different precedence levels to operators. For example, if `*` has been assigned higher precedence than `+` (by the language

1. Note that it is mathematically impossible to determine whether an arbitrary grammar is ambiguous.

designer), multiplication will be done first, regardless of the order of appearance of the two operators in the expression.

As stated previously, a grammar can describe a certain syntactic structure so that part of the meaning of the structure can be determined from its parse tree. In particular, the fact that an operator in an arithmetic expression is generated lower in the parse tree (and therefore must be evaluated first) can be used to indicate that it has precedence over an operator produced higher up in the tree. In the first parse tree of Figure 3.2, for example, the multiplication operator is generated lower in the tree, which could indicate that it has precedence over the addition operator in the expression. The second parse tree, however, indicates just the opposite. It appears, therefore, that the two parse trees indicate conflicting precedence information.

Notice that although the grammar of Example 3.2 is not ambiguous, the precedence order of its operators is not the usual one. In this grammar, a parse tree of a sentence with multiple operators, regardless of the particular operators involved, has the rightmost operator in the expression at the lowest point in the parse tree, with the other operators in the tree moving progressively higher as one moves to the left in the expression. For example, in the expression `A + B * C`, `*` is the lowest in the tree, indicating it is to be done first. However, in the expression `A * B + C`, `+` is the lowest, indicating it is to be done first.

A grammar can be written for the simple expressions we have been discussing that is both unambiguous and specifies a consistent precedence of the `+` and `*` operators, regardless of the order in which the operators appear in an expression. The correct ordering is specified by using separate nonterminal symbols to represent the operands of the operators that have different precedence. This requires additional nonterminals and some new rules. Instead of using <expr> for both operands of both `+` and `*`, we could use three nonterminals to represent operands, which allows the grammar to force different operators to different levels in the parse tree. If <expr> is the root symbol for expressions, `+` can be forced to the top of the parse tree by having <expr> directly generate only `+` operators, using the new nonterminal, <term>, as the right operand of `+`. Next, we can define <term> to generate `*` operators, using <term> as the left operand and a new nonterminal, <factor>, as its right operand. Now, `*` will always be lower in the parse tree, simply because it is farther from the start symbol than `+` in every derivation. The grammar of Example 3.4 is such a grammar.

EXAMPLE 3.4 An Unambiguous Grammar for Expressions

```
<assign> → <id> = <expr>
<id> → A | B | C
<expr> → <expr> + <term>
       | <term>
<term> → <term> * <factor>
       | <factor>
<factor> → ( <expr> )
         | <id>
```

The grammar in Example 3.4 generates the same language as the grammars of Examples 3.2 and 3.3, but it is unambiguous and it specifies the usual precedence order of multiplication and addition operators. The following derivation of the sentence `A = B + C * A` uses the grammar of Example 3.4:

```
<assign> => <id> = <expr>
         => A = <expr>
         => A = <expr> + <term>
         => A = <term> + <term>
         => A = <factor> + <term>
         => A = <id> + <term>
         => A = B + <term>
         => A = B + <term> * <factor>
         => A = B + <factor> * <factor>
         => A = B + <id> * <factor>
         => A = B + C * <factor>
         => A = B + C * <id>
         => A = B + C * A
```

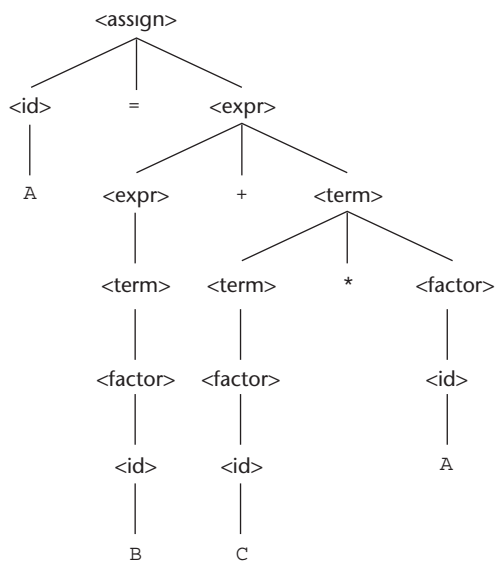
The unique parse tree for this sentence, using the grammar of Example 3.4, is shown in Figure 3.3.

The connection between parse trees and derivations is very close: Either can easily be constructed from the other. Every derivation with an unambiguous grammar has a unique parse tree, although that tree can be represented by different derivations. For example, the following derivation of the sentence `A = B + C * A` is different from the derivation of the same sentence given previously. This is a rightmost derivation, whereas the previous one is leftmost. Both of these derivations, however, are represented by the same parse tree.

```
<assign> => <id> = <expr>
         => <id> = <expr> + <term>
         => <id> = <expr> + <term> * <factor>
         => <id> = <expr> + <term> * <id>
         => <id> = <expr> + <term> * A
         => <id> = <expr> + <factor> * A
         => <id> = <expr> + <id> * A
         => <id> = <expr> + C * A
         => <id> = <term> + C * A
         => <id> = <factor> + C * A
         => <id> = <id> + C * A
         => <id> = B + C * A
         => A = B + C * A
```

Figure 3.3

The unique parse tree for `A = B + C * A` using an unambiguous grammar



3.3.1.9 Associativity of Operators

When an expression includes two operators that have the same precedence (as `*` and `/` usually have)—for example, `A / B * C`—a semantic rule is required to specify which should have precedence.[2] This rule is named *associativity*.

2. An expression with two occurrences of the same operator has the same issue; for example, `A / B / C`.

As was the case with precedence, a grammar for expressions may correctly imply operator associativity. Consider the following example of an assignment statement:

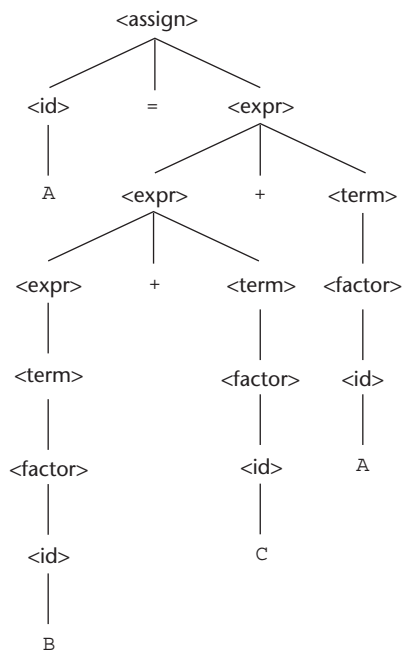
```
A = B + C + A
```

The parse tree for this sentence, as defined with the grammar of Example 3.4, is shown in Figure 3.4.

The parse tree of Figure 3.4 shows the left addition operator lower than the right addition operator. This is the correct order if addition is meant to be left associative, which is typical. In most cases, the associativity of addition in a computer is irrelevant. In mathematics, addition is associative, which means that left and right associative orders of evaluation mean the same thing. That is, `(A + B) + C = A + (B + C)`. Floating-point addition in a computer, however, is not necessarily associative. For example, suppose floating-point values store seven digits of accuracy. Consider the problem of adding 11 numbers together, where one of the numbers is 10^7 and the other ten are 1. If the small numbers (the 1's) are each added to the large number, one at a time, there is no effect on that number, because the small numbers occur in the eighth digit of the large number. However, if the small numbers are first added together and the result is added to the large number, the result in seven-digit accuracy is $1.000001 * 10^7$. Subtraction and division are not associative, whether in mathematics or in a computer. Therefore, correct associativity may be essential for an expression that contains either of them.

Figure 3.4

A parse tree for `A = B + C + A` illustrating the associativity of addition



When a grammar rule has its LHS also appearing at the beginning of its RHS, the rule is said to be **left recursive**. This left recursion specifies left associativity. For example, the left recursion of the rules of the grammar of Example 3.4 causes it to make both addition and multiplication left associative. Unfortunately, left recursion disallows the use of some important syntax analysis algorithms. When such algorithms are to be used, the grammar must be modified to remove the left recursion. This, in turn, disallows the grammar from precisely specifying that certain operators are left associative. Fortunately, left associativity can be enforced by the compiler, even though the grammar does not dictate it.

In most languages that provide it, the exponentiation operator is right associative. To indicate right associativity, right recursion can be used. A grammar rule is **right recursive** if the LHS appears at the right end of the RHS. Rules such as

```
<factor> → <exp> ** <factor>
         | <exp>
<exp> → ( <expr> )
      | id
```

could be used to describe exponentiation as a right-associative operator.

3.3.1.10 An Unambiguous Grammar for `if-then-else`

The BNF rules for an Ada **`if-then-else`** statement are as follows:

```
<if_stmt> → if <logic_expr> then <stmt>
          | if <logic_expr> then <stmt> else <stmt>
```

If we also have <stmt> → <if_stmt>, this grammar is ambiguous. The simplest sentential form that illustrates this ambiguity is

```
if <logic_expr> then if <logic_expr> then <stmt> else <stmt>
```

The two parse trees in Figure 3.5 show the ambiguity of this sentential form. Consider the following example of this construct:

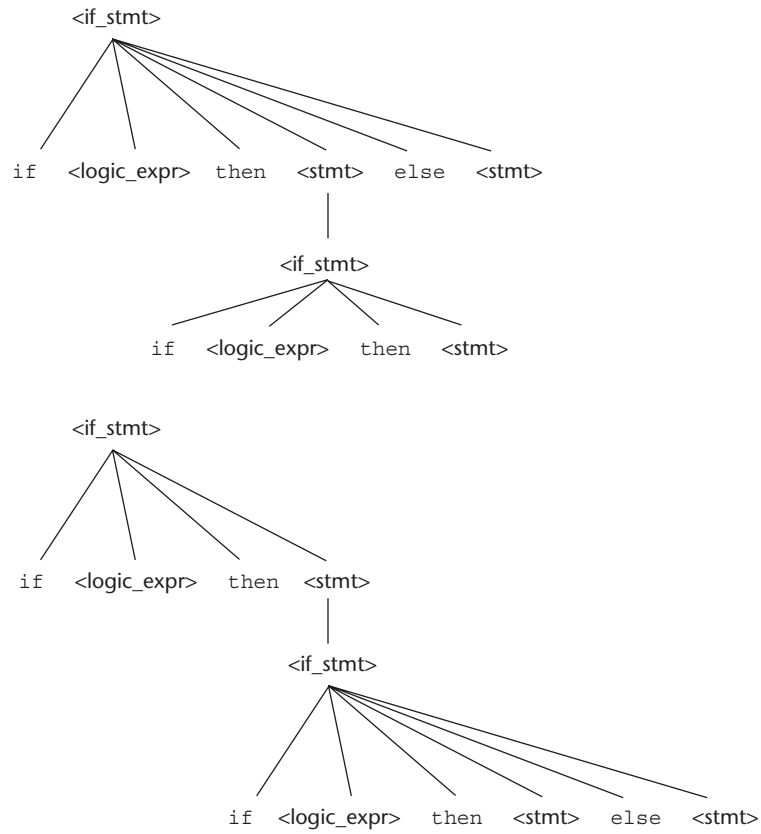
```
if done == true
  then if denom == 0
    then quotient = 0;
    else quotient = num / denom;
```

The problem is that if the upper parse tree in Figure 3.5 is used as the basis for translation, the else clause would be executed when `done` is not true, which probably is not what was intended by the author of the construct. We will examine the practical problems associated with this else-association problem in Chapter 8.

We will now develop an unambiguous grammar that describes this **`if`** statement. The rule for **`if`** constructs in many languages is that an else clause, when present, is matched with the nearest previous unmatched **`then`**.

Figure 3.5

Two distinct parse trees for the same sentential form



Therefore, there cannot be an **`if`** statement without an **`else`** between a **`then`** and its matching **`else`**. So, for this situation, statements must be distinguished between those that are matched and those that are unmatched, where unmatched statements are **`else`**-less **`if`**s and all other statements are matched. The problem with the earlier grammar is that it treats all statements as if they had equal syntactic significance—that is, as if they were all matched.

To reflect the different categories of statements, different abstractions, or nonterminals, must be used. The unambiguous grammar based on these ideas follows:

<stmt> → <matched> | <unmatched>

<matched> → **`if`** <logic_expr> **`then`** <matched> **`else`** <matched>
| any non-if statement

<unmatched> → **`if`** <logic_expr> **`then`** <stmt>
| **`if`** <logic_expr> **`then`** <matched> **`else`** <unmatched>

There is just one possible parse tree, using this grammar, for the following sentential form:

`if` <logic_expr> **`then`** **`if`** <logic_expr> **`then`** <stmt> **`else`** <stmt>

3.3.2 Extended BNF

Because of a few minor inconveniences in BNF, it has been extended in several ways. Most extended versions are called Extended BNF, or simply EBNF, even though they are not all exactly the same. The extensions do not enhance the descriptive power of BNF; they only increase its readability and writability.

Three extensions are commonly included in the various versions of EBNF. The first of these denotes an optional part of an RHS, which is delimited by brackets. For example, a C **`if-else`** statement can be described as

```
<if_stmt> → if (<expression>) <statement> [else <statement>]
```

Without the use of the brackets, the syntactic description of this statement would require the following two rules:

```
<if_stmt> → if (<expression>) <statement>
          | if (<expression>) <statement> else <statement>
```

The second extension is the use of braces in an RHS to indicate that the enclosed part can be repeated indefinitely or left out altogether. This extension allows lists to be built with a single rule, instead of using recursion and two rules. For example, lists of identifiers separated by commas can be described by the following rule:

```
<ident_list> → <identifier> {, <identifier>}
```

This is a replacement of the recursion by a form of implied iteration; the part enclosed within braces can be iterated any number of times.

The third common extension deals with multiple-choice options. When a single element must be chosen from a group, the options are placed in parentheses and separated by the OR operator, | . For example,

```
<term> → <term> (* | / | %) <factor>
```

In BNF, a description of this <term> would require the following three rules:

```
<term> → <term> * <factor>
       | <term> / <factor>
       | <term> % <factor>
```

The brackets, braces, and parentheses in the EBNF extensions are **metasymbols**, which means they are notational tools and not terminal symbols in the syntactic entities they help describe. In cases where these metasymbols are also terminal symbols in the language being described, the instances that are terminal symbols can be underlined or quoted. Example 3.5 illustrates the use of braces and multiple choices in an EBNF grammar.

EXAMPLE 3.5 BNF and EBNF Versions of an Expression Grammar

BNF:

```
<expr> → <expr> + <term>
       | <expr> - <term>
       | <term>
<term> → <term> * <factor>
       | <term> / <factor>
       | <factor>
<factor> → <exp> ** <factor>
         <exp>
<exp> → (<expr>)
      | id
```

EBNF:

```
<expr> → <term> {(+ | -) <term>}
<term> → <factor> {(* | /) <factor>}
<factor> → <exp> { ** <exp>}
<exp> → (<expr>)
      | id
```

The BNF rule

<expr> → <expr> + <term>

clearly specifies—in fact forces—the + operator to be left associative. However, the EBNF version,

<expr> → <term> {+ <term>}

does not imply the direction of associativity. This problem is overcome in a syntax analyzer based on an EBNF grammar for expressions by designing the syntax analysis process to enforce the correct associativity. This is further discussed in Chapter 4.

Some versions of EBNF allow a numeric superscript to be attached to the right brace to indicate an upper limit to the number of times the enclosed part can be repeated. Also, some versions use a plus (+) superscript to indicate one or more repetitions. For example,

<compound> → **begin** <stmt> {<stmt>} **end**

and

<compound> → **begin** {<stmt>}$^{+}$ **end**

are equivalent.

In recent years, some variations on BNF and EBNF have appeared. Among these are the following:

- In place of the arrow, a colon is used and the RHS is placed on the next line.
- Instead of a vertical bar to separate alternative RHSs, they are simply placed on separate lines.
- In place of square brackets to indicate something being optional, the subscript opt is used. For example,

  Constructor Declarator → SimpleName (FormalParameterList$_{opt}$)
- Rather than using the | symbol in a parenthesized list of elements to indicate a choice, the words "one of" are used. For example,

```
AssignmentOperator → one of   =   *=   /=   %=   +=   -=
                              <<=  >>=  &=   ^=   |=
```

There is a standard for EBNF, ISO/IEC 14977:1996(1996), but it is rarely used. The standard uses the equal sign (=) instead of an arrow in rules, terminates each RHS with a semicolon, and requires quotes on all terminal symbols. It also specifies a host of other notational rules.

3.3.3 Grammars and Recognizers

Earlier in this chapter, we suggested that there is a close relationship between generation and recognition devices for a given language. In fact, given a context-free grammar, a recognizer for the language generated by the grammar can be algorithmically constructed. A number of software systems have been developed that perform this construction. Such systems allow the quick creation of the syntax analysis part of a compiler for a new language and are therefore quite valuable. One of the first of these syntax analyzer generators is named yacc[3] (Johnson, 1975). There are now many such systems available.

3.4 Attribute Grammars

An **attribute grammar** is a device used to describe more of the structure of a programming language than can be described with a context-free grammar. An attribute grammar is an extension to a context-free grammar. The extension

3. The term yacc is an acronym for "yet another compiler compiler."

history note

Attribute grammars have been used in a wide variety of applications. They have been used to provide complete descriptions of the syntax and static semantics of programming languages (Watt, 1979); they have been used as the formal definition of a language that can be input to a compiler generation system (Farrow, 1982); and they have been used as the basis of several syntax-directed editing systems (Teitelbaum and Reps, 1981; Fischer et al., 1984). In addition, attribute grammars have been used in natural-language processing systems (Correa, 1992).

allows certain language rules to be conveniently described, such as type compatibility. Before we formally define the form of attribute grammars, we must clarify the concept of static semantics.

3.4.1 Static Semantics

There are some characteristics of the structure of programming languages that are difficult to describe with BNF, and some that are impossible. As an example of a syntax rule that is difficult to specify with BNF, consider type compatibility rules. In Java, for example, a floating-point value cannot be assigned to an integer type variable, although the opposite is legal. Although this restriction can be specified in BNF, it requires additional nonterminal symbols and rules. If all of the typing rules of Java were specified in BNF, the grammar would become too large to be useful, because the size of the grammar determines the size of the syntax analyzer.

As an example of a syntax rule that cannot be specified in BNF, consider the common rule that all variables must be declared before they are referenced. It has been proven that this rule cannot be specified in BNF.

These problems exemplify the categories of language rules called static semantics rules. The **static semantics** of a language is only indirectly related to the meaning of programs during execution; rather, it has to do with the legal forms of programs (syntax rather than semantics). Many static semantic rules of a language state its type constraints. Static semantics is so named because the analysis required to check these specifications can be done at compile time.

Because of the problems of describing static semantics with BNF, a variety of more powerful mechanisms has been devised for that task. One such mechanism, attribute grammars, was designed by Knuth (1968a) to describe both the syntax and the static semantics of programs.

Attribute grammars are a formal approach both to describing and checking the correctness of the static semantics rules of a program. Although they are not always used in a formal way in compiler design, the basic concepts of attribute grammars are at least informally used in every compiler (see Aho et al., 1986).

Dynamic semantics, which is the meaning of expressions, statements, and program units, is discussed in Section 3.5.

3.4.2 Basic Concepts

Attribute grammars are context-free grammars to which have been added attributes, attribute computation functions, and predicate functions. **Attributes**, which are associated with grammar symbols (the terminal and nonterminal symbols), are similar to variables in the sense that they can have values assigned to them. **Attribute computation functions**, sometimes called semantic

functions, are associated with grammar rules. They are used to specify how attribute values are computed. **Predicate functions**, which state the static semantic rules of the language, are associated with grammar rules.

These concepts will become clearer after we formally define attribute grammars and provide an example.

3.4.3 Attribute Grammars Defined

An attribute grammar is a grammar with the following additional features:

- Associated with each grammar symbol X is a set of attributes A(X). The set A(X) consists of two disjoint sets S(X) and I(X), called synthesized and inherited attributes, respectively. **Synthesized attributes** are used to pass semantic information up a parse tree, while **inherited attributes** pass semantic information down and across a tree.
- Associated with each grammar rule is a set of semantic functions and a possibly empty set of predicate functions over the attributes of the symbols in the grammar rule. For a rule $X_0 \rightarrow X_1 \ldots X_n$, the synthesized attributes of X_0 are computed with semantic functions of the form $S(X_0) = f(A(X_1), \ldots, A(X_n))$. So the value of a synthesized attribute on a parse tree node depends only on the values of the attributes on that node's children nodes. Inherited attributes of symbols X_j, $1 \leq j \leq n$ (in the rule above), are computed with a semantic function of the form $I(X_j) = f(A(X_0), \ldots, A(X_n))$. So the value of an inherited attribute on a parse tree node depends on the attribute values of that node's parent node and those of its sibling nodes. Note that, to avoid circularity, inherited attributes are often restricted to functions of the form $I(X_j) = f(A(X_0), \ldots, A(X_{(j-1)}))$. This form prevents an inherited attribute from depending on itself or on attributes to the right in the parse tree.
- A predicate function has the form of a Boolean expression on the union of the attribute set $\{A(X_0), \ldots, A(X_n)\}$ and a set of literal attribute values. The only derivations allowed with an attribute grammar are those in which every predicate associated with every nonterminal is true. A false predicate function value indicates a violation of the syntax or static semantics rules of the language.

A parse tree of an attribute grammar is the parse tree based on its underlying BNF grammar, with a possibly empty set of attribute values attached to each node. If all the attribute values in a parse tree have been computed, the tree is said to be **fully attributed**. Although in practice it is not always done this way, it is convenient to think of attribute values as being computed after the complete unattributed parse tree has been constructed by the compiler.

3.4.4 Intrinsic Attributes

Intrinsic attributes are synthesized attributes of leaf nodes whose values are determined outside the parse tree. For example, the type of an instance of a variable in a program could come from the symbol table, which is used to store variable names

and their types. The contents of the symbol table are set based on earlier declaration statements. Initially, assuming that an unattributed parse tree has been constructed and that attribute values are needed, the only attributes with values are the intrinsic attributes of the leaf nodes. Given the intrinsic attribute values on a parse tree, the semantic functions can be used to compute the remaining attribute values.

3.4.5 Examples of Attribute Grammars

As a very simple example of how attribute grammars can be used to describe static semantics, consider the following fragment of an attribute grammar that describes the rule that the name on the **end** of an Ada procedure must match the procedure's name. (This rule cannot be stated in BNF.) The string attribute of <proc_name>, denoted by <proc_name>.string, is the actual string of characters that were found immediately following the reserved word **procedure** by the compiler. Notice that when there is more than one occurrence of a nonterminal in a syntax rule in an attribute grammar, the nonterminals are subscripted with brackets to distinguish them. Neither the subscripts nor the brackets are part of the described language.

Syntax rule: <proc_def> → **procedure** <proc_name> [1]
<proc_body> **end** <proc_name> [2] ;
Predicate: <proc_name> [1] string == <proc_name> [2] .string

In this example, the predicate rule states that the name string attribute of the <proc_name> nonterminal in the subprogram header must match the name string attribute of the <proc_name> nonterminal following the end of the subprogram.

Next, we consider a larger example of an attribute grammar. In this case, the example illustrates how an attribute grammar can be used to check the type rules of a simple assignment statement. The syntax and static semantics of this assignment statement are as follows: The only variable names are A, B, and C. The right side of the assignments can be either a variable or an expression in the form of a variable added to another variable. The variables can be one of two types: int or real. When there are two variables on the right side of an assignment, they need not be the same type. The type of the expression when the operand types are not the same is always real. When they are the same, the expression type is that of the operands. The type of the left side of the assignment must match the type of the right side. So the types of operands in the right side can be mixed, but the assignment is valid only if the target and the value resulting from evaluating the right side have the same type. The attribute grammar specifies these static semantic rules.

The syntax portion of our example attribute grammar is

<assign> → <var> = <expr>
<expr> → <var> + <var>
| <var>
<var> → A | B | C

The attributes for the nonterminals in the example attribute grammar are described in the following paragraphs:

- *actual_type*—A synthesized attribute associated with the nonterminals <var> and <expr>. It is used to store the actual type, int or real, of a variable or expression. In the case of a variable, the actual type is intrinsic. In the case of an expression, it is determined from the actual types of the child node or children nodes of the <expr> nonterminal.
- *expected_type*—An inherited attribute associated with the nonterminal <expr>. It is used to store the type, either int or real, that is expected for the expression, as determined by the type of the variable on the left side of the assignment statement.

The complete attribute grammar follows in Example 3.6.

EXAMPLE 3.6 An Attribute Grammar for Simple Assignment Statements

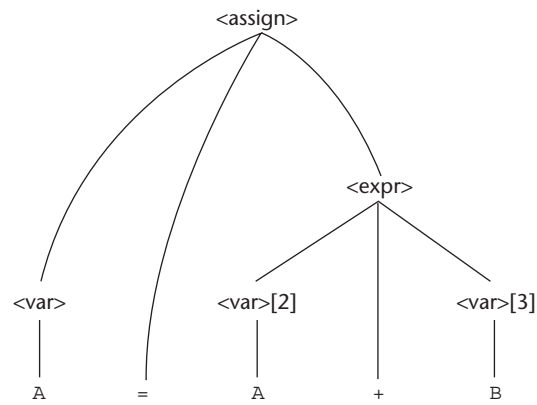
1. Syntax rule: <assign> → <var> = <expr>
   Semantic rule: <expr>.expected_type ← <var>.actual_type
2. Syntax rule: <expr> → <var>[2] + <var>[3]
   Semantic rule: <expr>.actual_type ←
   if (<var>[2].actual_type = int) and
   (<var>[3].actual_type = int)
   then int
   else real
   end if
   Predicate: <expr>.actual_type == <expr>.expected_type
3. Syntax rule: <expr> → <var>
   Semantic rule: <expr>.actual_type ← <var>.actual_type
   Predicate: <expr>.actual_type == <expr>.expected_type
4. Syntax rule: <var> → `A` | `B` | `C`
   Semantic rule: <var>.actual_type ← look-up (<var>.string)

The look-up function looks up a given variable name in the symbol table and returns the variable's type.

A parse tree of the sentence `A = A + B` generated by the grammar in Example 3.6 is shown in Figure 3.6. As in the grammar, bracketed numbers are added after the repeated node labels in the tree so they can be referenced unambiguously.

Figure 3.6

A parse tree for
`A = A + B`



3.4.6 Computing Attribute Values

Now, consider the process of computing the attribute values of a parse tree, which is sometimes called **decorating** the parse tree. If all attributes were inherited, this could proceed in a completely top-down order, from the root to the leaves. Alternatively, it could proceed in a completely bottom-up order, from the leaves to the root, if all the attributes were synthesized. Because our grammar has both synthesized and inherited attributes, the evaluation process cannot be in any single direction. The following is an evaluation of the attributes, in an order in which it is possible to compute them:

1. <var>.actual_type ← look-up(`A`) (Rule 4)
2. <expr>.expected_type ← <var>.actual_type (Rule 1)
3. <var>[2].actual_type ← look-up(`A`) (Rule 4)
   <var>[3].actual_type ← look-up(`B`) (Rule 4)
4. <expr>.actual_type ← either int or real (Rule 2)
5. <expr>.expected_type == <expr>.actual_type is either TRUE or FALSE (Rule 2)

The tree in Figure 3.7 shows the flow of attribute values in the example of Figure 3.6. Solid lines are used for the parse tree; dashed lines show attribute flow in the tree.

The tree in Figure 3.8 shows the final attribute values on the nodes. In this example, `A` is defined as a real and `B` is defined as an int.

Determining attribute evaluation order for the general case of an attribute grammar is a complex problem, requiring the construction of a dependency graph to show all attribute dependencies.

Figure 3.7

The flow of attributes in the tree

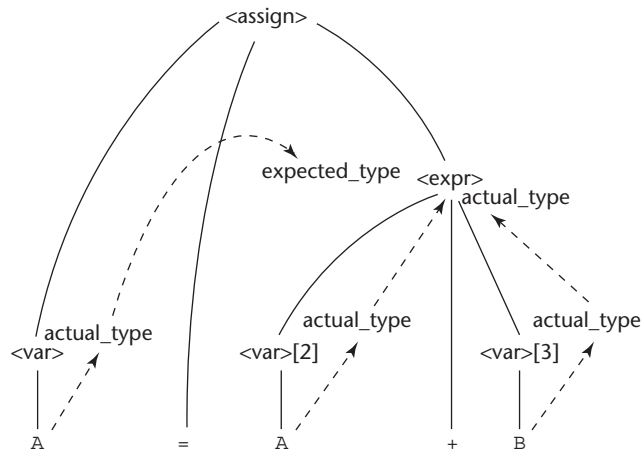
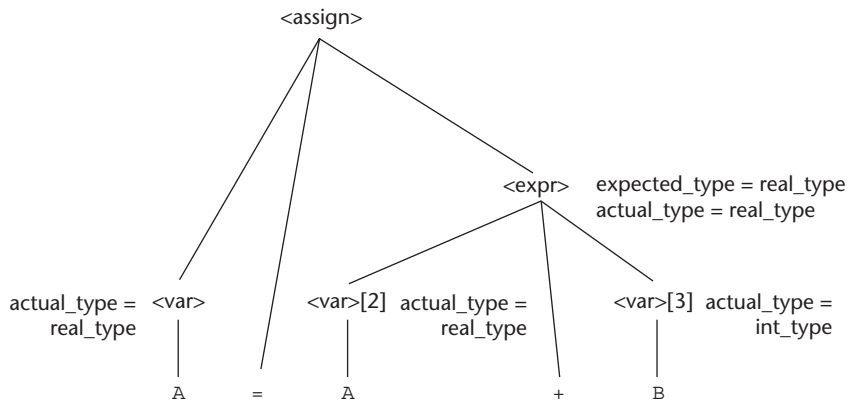


Figure 3.8

A fully attributed parse tree



3.4.7 Evaluation

Checking the static semantic rules of a language is an essential part of all compilers. Even if a compiler writer has never heard of an attribute grammar, he or she would need to use their fundamental ideas to design the checks of static semantics rules for his or her compiler.

One of the main difficulties in using an attribute grammar to describe all of the syntax and static semantics of a real contemporary programming language is the size and complexity of the attribute grammar. The large number of attributes and semantic rules required for a complete programming language make such grammars difficult to write and read. Furthermore, the attribute values on a large parse tree are costly to evaluate. On the other hand, less formal attribute

grammars are a powerful and commonly used tool for compiler writers, who are more interested in the process of producing a compiler than they are in formalism.

3.5 Describing the Meanings of Programs: Dynamic Semantics

We now turn to the difficult task of describing the **dynamic semantics**, or meaning, of the expressions, statements, and program units of a programming language. Because of the power and naturalness of the available notation, describing syntax is a relatively simple matter. On the other hand, no universally accepted notation or approach has been devised for dynamic semantics. In this section, we briefly describe several of the methods that have been developed. For the remainder of this section, when we use the term *semantics*, we mean dynamic semantics.

There are several different reasons underlying the need for a methodology and notation for describing semantics. Programmers obviously need to know precisely what the statements of a language do before they can use them effectively in their programs. Compiler writers must know exactly what language constructs mean to design implementations for them correctly. If there were a precise semantics specification of a programming language, programs written in the language potentially could be proven correct without testing. Also, compilers could be shown to produce programs that exhibited exactly the behavior given in the language definition; that is, their correctness could be verified. A complete specification of the syntax and semantics of a programming language could be used by a tool to generate a compiler for the language automatically. Finally, language designers, who would develop the semantic descriptions of their languages, could in the process discover ambiguities and inconsistencies in their designs.

Software developers and compiler designers typically determine the semantics of programming languages by reading English explanations in language manuals. Because such explanations are often imprecise and incomplete, this approach is clearly unsatisfactory. Due to the lack of complete semantics specifications of programming languages, programs are rarely proven correct without testing, and commercial compilers are never generated automatically from language descriptions.

Scheme, a functional language described in Chapter 15, is one of only a few programming languages whose definition includes a formal semantics description. However, the method used is not one described in this chapter, as this chapter is focused on approaches that are suitable for imperative languages.

3.5.1 Operational Semantics

The idea behind **operational semantics** is to describe the meaning of a statement or program by specifying the effects of running it on a machine. The effects on the machine are viewed as the sequence of changes in its

state, where the machine's state is the collection of the values in its storage. An obvious operational semantics description, then, is given by executing a compiled version of the program on a computer. Most programmers have, on at least one occasion, written a small test program to determine the meaning of some programming language construct, often while learning the language. Essentially, what such a programmer is doing is using operational semantics to determine the meaning of the construct.

There are several problems with using this approach for complete formal semantics descriptions. First, the individual steps in the execution of machine language and the resulting changes to the state of the machine are too small and too numerous. Second, the storage of a real computer is too large and complex. There are usually several levels of memory devices, as well as connections to enumerable other computers and memory devices through networks. Therefore, machine languages and real computers are not used for formal operational semantics. Rather, intermediate-level languages and interpreters for idealized computers are designed specifically for the process.

There are different levels of uses of operational semantics. At the highest level, the interest is in the final result of the execution of a complete program. This is sometimes called **natural operational semantics**. At the lowest level, operational semantics can be used to determine the precise meaning of a program through an examination of the complete sequence of state changes that occur when the program is executed. This use is sometimes called **structural operational semantics**.

3.5.1.1 The Basic Process

The first step in creating an operational semantics description of a language is to design an appropriate intermediate language, where the primary characteristic of the language is clarity. Every construct of the intermediate language must have an obvious and unambiguous meaning. This language is at the intermediate level, because machine language is too low-level to be easily understood and another high-level language is obviously not suitable. If the semantics description is to be used for natural operational semantics, a virtual machine (an interpreter) must be constructed for the intermediate language. The virtual machine can be used to execute either single statements, code segments, or whole programs. The semantics description can be used without a virtual machine if the meaning of a single statement is all that is required. In this use, which is structural operational semantics, the intermediate code can be visually inspected.

The basic process of operational semantics is not unusual. In fact, the concept is frequently used in programming textbooks and programming language reference manuals. For example, the semantics of the C `for` construct can be described in terms of simpler statements, as in

C Statement	*Meaning*
`for (expr1; expr2; expr3) {`	`expr1;`
`...`	`loop: if expr2 == 0 goto out`
`}`	`...`
	`expr3;`
	`goto loop`
	`out: ...`

The human reader of such a description is the virtual computer and is assumed to be able to "execute" the instructions in the definition correctly and recognize the effects of the "execution."

The intermediate language and its associated virtual machine used for formal operational semantics descriptions are often highly abstract. The intermediate language is meant to be convenient for the virtual machine, rather than for human readers. For our purposes, however, a more human-oriented intermediate language could be used. As such an example, consider the following list of statements, which would be adequate for describing the semantics of the simple control statements of a typical programming language:

```
ident = var
ident = ident + 1
ident = ident - 1
goto label
if var relop var goto label
```

In these statements, relop is one of the relational operators from the set {`=, <>, >, <, >=, <=`}, ident is an identifier, and var is either an identifier or a constant. These statements are all simple and therefore easy to understand and implement.

A slight generalization of these three assignment statements allows more general arithmetic expressions and assignment statements to be described. The new statements are

```
ident = var bin_op var
ident = un_op var
```

where bin_op is a binary arithmetic operator and un_op is a unary operator. Multiple arithmetic data types and automatic type conversions, of course, complicate this generalization. Adding just a few more relatively simple instructions would allow the semantics of arrays, records, pointers, and subprograms to be described.

In Chapter 8, the semantics of various control statements are described using this intermediate language.

3.5.1.2 Evaluation

The first and most significant use of formal operational semantics was to describe the semantics of PL/I (Wegner, 1972). That particular abstract machine and the translation rules for PL/I were together named the Vienna Definition Language (VDL), after the city where IBM designed it.

Operational semantics provides an effective means of describing semantics for language users and language implementors, as long as the descriptions are kept simple and informal. The VDL description of PL/I, unfortunately, is so complex that it serves no practical purpose.

Operational semantics depends on programming languages of lower levels, not mathematics. The statements of one programming language are described in terms of the statements of a lower-level programming language. This approach can lead to circularities, in which concepts are indirectly defined in terms of themselves. The methods described in the following two sections are much more formal, in the sense that they are based on mathematics and logic, not programming languages.

3.5.2 Denotational Semantics

Denotational semantics is the most rigorous and most widely known formal method for describing the meaning of programs. It is solidly based on recursive function theory. A thorough discussion of the use of denotational semantics to describe the semantics of programming languages is necessarily long and complex. It is our intent to provide the reader with an introduction to the central concepts of denotational semantics, along with a few simple examples that are relevant to programming language specifications.

The process of constructing a denotational semantics specification for a programming language requires one to define for each language entity both a mathematical object and a function that maps instances of that language entity onto instances of the mathematical object. Because the objects are rigorously defined, they model the exact meaning of their corresponding entities. The idea is based on the fact that there are rigorous ways of manipulating mathematical objects but not programming language constructs. The difficulty with this method lies in creating the objects and the mapping functions. The method is named *denotational* because the mathematical objects denote the meaning of their corresponding syntactic entities.

The mapping functions of a denotational semantics programming language specification, like all functions in mathematics, have a domain and a range. The domain is the collection of values that are legitimate parameters to the function; the range is the collection of objects to which the parameters are mapped. In denotational semantics, the domain is called the **syntactic domain**, because it is syntactic structures that are mapped. The range is called the **semantic domain**.

Denotational semantics is related to operational semantics. In operational semantics, programming language constructs are translated into simpler programming language constructs, which become the basis of the meaning of the

construct. In denotational semantics, programming language constructs are mapped to mathematical objects, either sets or, more often, functions. However, unlike operational semantics, denotational semantics does not model the step-by-step computational processing of programs.

3.5.2.1 Two Simple Examples

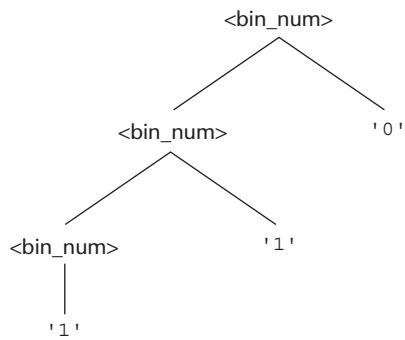
We use a very simple language construct, character string representations of binary numbers, to introduce the denotational method. The syntax of such binary numbers can be described by the following grammar rules:

```
<bin_num> → '0'
          | '1'
          | <bin_num> '0'
          | <bin_num> '1'
```

A parse tree for the example binary number, `110`, is shown in Figure 3.9. Notice that we put apostrophes around the syntactic digits to show they are not mathematical digits. This is similar to the relationship between ASCII coded digits and mathematical digits. When a program reads a number as a string, it must be converted to a mathematical number before it can be used as a value in the program.

Figure 3.9

A parse tree of the binary number `110`



The syntactic domain of the mapping function for binary numbers is the set of all character string representations of binary numbers. The semantic domain is the set of nonnegative decimal numbers, symbolized by N.

To describe the meaning of binary numbers using denotational semantics, we associate the actual meaning (a decimal number) with each rule that has a single terminal symbol as its RHS.

In our example, decimal numbers must be associated with the first two grammar rules. The other two grammar rules are, in a sense, computational rules, because they combine a terminal symbol, to which an object can be associated, with a nonterminal, which can be expected to represent some construct. Presuming an evaluation that progresses upward in the parse tree,

the nonterminal in the right side would already have its meaning attached. So, a syntax rule with a nonterminal as its RHS would require a function that computed the meaning of the LHS, which represents the meaning of the complete RHS.

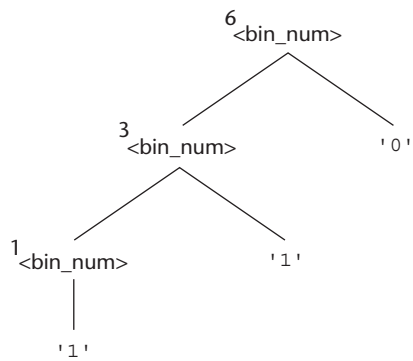
The semantic function, named M_{bin}, maps the syntactic objects, as described in the previous grammar rules, to the objects in N, the set of non-negative decimal numbers. The function M_{bin} is defined as follows:

M_{bin}('0') = 0
M_{bin}('1') = 1
M_{bin}(<bin_num> '0') = 2 * M_{bin}(<bin_num>)
M_{bin}(<bin_num> '1') = 2 * M_{bin}(<bin_num>) + 1

The meanings, or denoted objects (which in this case are decimal numbers), can be attached to the nodes of the parse tree shown on the previous page, yielding the tree in Figure 3.10. This is syntax-directed semantics. Syntactic entities are mapped to mathematical objects with concrete meaning.

Figure 3.10

A parse tree with denoted objects for 110



In part because we need it later, we now show a similar example for describing the meaning of syntactic decimal literals. In this case, the syntactic domain is the set of character string representations of decimal numbers. The semantic domain is once again the set N.

<dec_num> → '0'|'1'|'2'|'3'|'4'|'5'|'6'|'7''8'|'9'
|<dec_num> ('0'|'1'|'2'|'3'|'4'|'5'|'6'|'7'|'8'|'9')

The denotational mappings for these syntax rules are

M_{dec}('0') = 0, M_{dec}('1') = 1, M_{dec}('2') = 2, . . ., M_{dec}('9') = 9
M_{dec}(<dec_num> '0') = 10 * M_{dec}(<dec_num>)
M_{dec}(<dec_num> '1') = 10 * M_{dec}(<dec_num>) + 1
. . .
M_{dec}(<dec_num> '9') = 10 * M_{dec}(<dec_num>) + 9

In the following sections, we present the denotational semantics descriptions of a few simple constructs. The most important simplifying assumption made here is that both the syntax and static semantics of the constructs are correct. In addition, we assume that only two scalar types are included: integer and Boolean.

3.5.2.2 The State of a Program

The denotational semantics of a program could be defined in terms of state changes in an ideal computer. Operational semantics are defined in this way, and denotational semantics are defined in nearly the same way. In a further simplification, however, denotational semantics is defined in terms of only the values of all of the program's variables. So, denotational semantics uses the state of the program to describe meaning, whereas operational semantics uses the state of a machine. The key difference between operational semantics and denotational semantics is that state changes in operational semantics are defined by coded algorithms, written in some programming language, whereas in denotational semantics, state changes are defined by mathematical functions.

Let the state s of a program be represented as a set of ordered pairs, as follows:

$$s = \{<i_1, v_1>, <i_2, v_2>, \ldots, <i_n, v_n>\}$$

Each i is the name of a variable, and the associated v's are the current values of those variables. Any of the v's can have the special value **undef**, which indicates that its associated variable is currently undefined. Let VARMAP be a function of two parameters: a variable name and the program state. The value of VARMAP (i_j, s) is v_j (the value paired with i_j in state s). Most semantics mapping functions for programs and program constructs map states to states. These state changes are used to define the meanings of programs and program constructs. Some language constructs—for example, expressions—are mapped to values, not states.

3.5.2.3 Expressions

Expressions are fundamental to most programming languages. We assume here that expressions have no side effects. Furthermore, we deal with only very simple expressions: The only operators are + and *, and an expression can have at most one operator; the only operands are scalar integer variables and integer literals; there are no parentheses; and the value of an expression is an integer. Following is the BNF description of these expressions:

<expr> → <dec_num> | <var> | <binary_expr>
<binary_expr> → <left_expr> <operator> <right_expr>
<left_expr> → <dec_num> | <var>
<right_expr> → <dec_num> | <var>
<operator> → + | *

The only error we consider in expressions is a variable having an undefined value. Obviously, other errors can occur, but most of them are machine-dependent. Let Z be the set of integers, and let **error** be the error value. Then Z ∪ {**error**} is the semantic domain for the denotational specification for our expressions.

The mapping function for a given expression E and state s follows. To distinguish between mathematical function definitions and the assignment statements of programming languages, we use the symbol Δ= to define mathematical functions. The implication symbol, =>, used in this definition connects the form of an operand with its associated case (or switch) construct. Dot notation is used to refer to the child nodes of a node. For example, <binary_expr>.<left_expr> refers to the left child node of <binary_expr>.

```
Me (<expr>, s) Δ= case <expr> of
                  <dec_num>=>Mdec (<dec_num>, s)
                  <var> =>if VARMAP (<var>, s) == undef
                              then error
                              else VARMAP (<var>, s)
                  <binary_expr> =>
                   if(Me (<binary_expr>.<left_expr>,s) == undef OR
                      Me (<binary_expr>.<right_expr>, s) == undef)
                   then error
                   else if (<binary_expr>.<operator> == '+')
                           then Me (<binary_expr>.<left_expr>, s) +
                                Me (<binary_expr>.<right_expr>, s)
                           else Me (<binary_expr>.<left_expr>, s) *
                                Me (<binary_expr>.<right_expr>, s)
```

3.5.2.4 Assignment Statements

An assignment statement is an expression evaluation plus the setting of the target variable to the expression's value. In this case, the meaning function maps a state to a state. This function can be described with the following:

```
Ma (x = E, s) Δ= if Me (E, s) == error
                 then error
                 else s′ = {<i1, v1′>, <i2, v2′>, . . . , <in, vn′>}, where
                        for j = 1, 2, . . . , n
                          if ij == x
                            then vj′ = Me (E, s)
                            else vj′ = VARMAP (ij, s)
```

Note that the comparison in the third last line above, i_j == x, is of names, not values.

3.5.2.5 Logical Pretest Loops

The denotational semantics of a logical pretest loop is deceptively simple. To expedite the discussion, we assume that there are two other existing mapping functions, M_{sl} and M_b, that map statement lists and states to states and Boolean expressions to Boolean values (or **error**), respectively. The function is

```
Ml(while B do L, s) Δ= if Mb(B, s) == undef
                        then error
                        else if Mb(B, s) == false
                               then s
                               else if Msl(L, s) == error
                                      then error
                                      else Ml(while B do L, Msl(L, s))
```

The meaning of the loop is simply the value of the program variables after the statements in the loop have been executed the prescribed number of times, assuming there have been no errors. In essence, the loop has been converted from iteration to recursion, where the recursion control is mathematically defined by other recursive state mapping functions. Recursion is easier to describe with mathematical rigor than iteration.

One significant observation at this point is that this definition, like actual program loops, may compute nothing because of nontermination.

3.5.2.6 Evaluation

Objects and functions, such as those used in the earlier constructs, can be defined for the other syntactic entities of programming languages. When a complete system has been defined for a given language, it can be used to determine the meaning of complete programs in that language. This provides a framework for thinking about programming in a highly rigorous way.

As stated previously, denotational semantics can be used as an aid to language design. For example, statements for which the denotational semantic description is complex and difficult may indicate to the designer that such statements may also be difficult for language users to understand and that an alternative design may be in order.

Because of the complexity of denotational descriptions, they are of little use to language users. On the other hand, they provide an excellent way to describe a language concisely.

Although the use of denotational semantics is normally attributed to Scott and Strachey (1971), the general denotational approach to language description can be traced to the nineteenth century (Frege, 1892).

3.5.3 Axiomatic Semantics

Axiomatic semantics, thus named because it is based on mathematical logic, is the most abstract approach to semantics specification discussed in this chapter. Rather than directly specifying the meaning of a program, axiomatic semantics specifies what can be proven about the program. Recall that one of the possible uses of semantic specifications is to prove the correctness of programs.

In axiomatic semantics, there is no model of the state of a machine or program or model of state changes that take place when the program is executed. The meaning of a program is based on relationships among program variables and constants, which are the same for every execution of the program.

Axiomatic semantics has two distinct applications: program verification and program semantics specification. This section focuses on program verification in its description of axiomatic semantics.

Axiomatic semantics was defined in conjunction with the development of an approach to proving the correctness of programs. Such correctness proofs, when they can be constructed, show that a program performs the computation described by its specification. In a proof, each statement of a program is both preceded and followed by a logical expression that specifies constraints on program variables. These, rather than the entire state of an abstract machine (as with operational semantics), are used to specify the meaning of the statement. The notation used to describe constraints—indeed, the language of axiomatic semantics—is predicate calculus. Although simple Boolean expressions are often adequate to express constraints, in some cases they are not.

When axiomatic semantics is used to specify formally the meaning of a statement, the meaning is defined by the statement's effect on assertions about the data affected by the statement.

3.5.3.1 Assertions

The logical expressions used in axiomatic semantics are called predicates, or **assertions**. An assertion immediately preceding a program statement describes the constraints on the program variables at that point in the program. An assertion immediately following a statement describes the new constraints on those variables (and possibly others) after execution of the statement. These assertions are called the **precondition** and **postcondition**, respectively, of the statement. For two adjacent statements, the postcondition of the first serves as the precondition of the second. Developing an axiomatic description or proof of a given program requires that every statement in the program has both a precondition and a postcondition.

In the following sections, we examine assertions from the point of view that preconditions for statements are computed from given postconditions, although it is possible to consider these in the opposite sense. We assume all variables are integer type. As a simple example, consider the following assignment statement and postcondition:

```
sum = 2 * x + 1 {sum > 1}
```

Precondition and postcondition assertions are presented in braces to distinguish them from parts of program statements. One possible precondition for this statement is {`x > 10`}.

In axiomatic semantics, the meaning of a specific statement is defined by its precondition and its postcondition. In effect, the two assertions specify precisely the effect of executing the statement.

In the following subsections, we focus on correctness proofs of statements and programs, which is a common use of axiomatic semantics. The more general concept of axiomatic semantics is to state precisely the meaning of statements and programs in terms of logic expressions. Program verification is one application of axiomatic descriptions of languages.

3.5.3.2 Weakest Preconditions

The **weakest precondition** is the least restrictive precondition that will guarantee the validity of the associated postcondition. For example, in the statement and postcondition given in Section 3.5.3.1, {`x > 10`}, {`x > 50`}, and {`x > 1000`} are all valid preconditions. The weakest of all preconditions in this case is {`x > 0`}.

If the weakest precondition can be computed from the most general postcondition for each of the statement types of a language, then the processes used to compute these preconditions provide a concise description of the semantics of that language. Furthermore, correctness proofs can be constructed for programs in that language. A program proof is begun by using the characteristics of the results of the program's execution as the postcondition of the last statement of the program. This postcondition, along with the last statement, is used to compute the weakest precondition for the last statement. This precondition is then used as the postcondition for the second last statement. This process continues until the beginning of the program is reached. At that point, the precondition of the first statement states the conditions under which the program will compute the desired results. If these conditions are implied by the input specification of the program, the program has been verified to be correct.

An **inference rule** is a method of inferring the truth of one assertion on the basis of the values of other assertions. The general form of an inference rule is as follows:

$$\frac{S1, S2, \ldots, Sn}{S}$$

This rule states that if S1, S2, . . . , and S*n* are true, then the truth of S can be inferred. The top part of an inference rule is called its **antecedent**; the bottom part is called its **consequent**.

An **axiom** is a logical statement that is assumed to be true. Therefore, an axiom is an inference rule without an antecedent.

For some program statements, the computation of a weakest precondition from the statement and a postcondition is simple and can be specified by an

axiom. In most cases, however, the weakest precondition can be specified only by an inference rule.

To use axiomatic semantics with a given programming language, whether for correctness proofs or for formal semantics specifications, either an axiom or an inference rule must exist for each kind of statement in the language. In the following subsections, we present an axiom for assignment statements and inference rules for statement sequences, selection statements, and logical pretest loop statements. Note that we assume that neither arithmetic nor Boolean expressions have side effects.

3.5.3.3 Assignment Statements

The precondition and postcondition of an assignment statement together define precisely its meaning. To define the meaning of an assignment statement, given a postcondition, there must be a way to compute its precondition from that postcondition.

Let x = E be a general assignment statement and Q be its postcondition. Then, its precondition, P, is defined by the axiom

$$P = Q_{x \rightarrow E}$$

which means that P is computed as Q with all instances of x replaced by E. For example, if we have the assignment statement and postcondition

```
a = b / 2 - 1 {a < 10}
```

the weakest precondition is computed by substituting `b / 2 - 1` for `a` in the postcondition {`a < 10`}, as follows:

```
b / 2 - 1 < 10
b < 22
```

Thus, the weakest precondition for the given assignment statement and postcondition is {`b < 22`}. Remember that the assignment axiom is guaranteed to be correct only in the absence of side effects. An assignment statement has a side effect if it changes some variable other than its target.

The usual notation for specifying the axiomatic semantics of a given statement form is

$$\{P\}S\{Q\}$$

where P is the precondition, Q is the postcondition, and S is the statement form. In the case of the assignment statement, the notation is

$$\{Q_{x \rightarrow E}\}\ x = E\{Q\}$$

As another example of computing a precondition for an assignment statement, consider the following:

```
x = 2 * y - 3 {x > 25}
```

The precondition is computed as follows:

```
2 * y - 3 > 25
y > 14
```

So {`y > 14`} is the weakest precondition for this assignment statement and postcondition.

Note that the appearance of the left side of the assignment statement in its right side does not affect the process of computing the weakest precondition. For example, for

```
x = x + y - 3 {x > 10}
```

the weakest precondition is

```
x + y - 3 > 10
y > 13 - x
```

Recall that axiomatic semantics was developed to prove the correctness of programs. In light of that, it is natural at this point to wonder how the axiom for assignment statements can be used to prove anything. Here is how: A given assignment statement with both a precondition and a postcondition can be considered a logical statement, or theorem. If the assignment axiom, when applied to the postcondition and the assignment statement, produces the given precondition, the theorem is proved. For example, consider the logical statement

```
{x > 3} x = x - 3 {x > 0}
```

Using the assignment axiom on

```
x = x - 3 {x > 0}
```

produces {`x > 3`}, which is the given precondition. Therefore, we have proven the example logical statement.

Next, consider the logical statement

```
{x > 5} x = x - 3 {x > 0}
```

In this case, the given precondition, {`x > 5`}, is not the same as the assertion produced by the axiom. However, it is obvious that {`x > 5`} implies {`x > 3`}.

To use this in a proof, an inference rule, named the **rule of consequence**, is needed. The form of the rule of consequence is

$$\frac{\{P\}\ S\ \{Q\},\ P' \Rightarrow P,\ Q \Rightarrow Q'}{\{P'\}\ S\ \{Q'\}}$$

The => symbol means "implies," and S can be any program statement. The rule can be stated as follows: If the logical statement {P} S {Q} is true, the assertion P′ implies the assertion P, and the assertion Q implies the assertion Q′, then it can be inferred that {P′} S {Q′}. In other words, the rule of consequence says that a postcondition can always be weakened and a precondition can always be strengthened. This is quite useful in program proofs. For example, it allows the completion of the proof of the last logical statement example above. If we let P be {x > 3}, Q and Q′ be {x > 0}, and P′ be {x > 5}, we have

$$\frac{\{x>3\}x = x-3\{x>0\},\ (x>5) \Rightarrow \{x>3\},\ (x>0) \Rightarrow (x>0)}{\{x>5\}x = x-3\{x>0\}}$$

The first term of the antecedent ({x > 3} x = x − 3 {x > 0}) was proven with the assignment axiom. The second and third terms are obvious. Therefore, by the rule of consequence, the consequent is true.

3.5.3.4 Sequences

The weakest precondition for a sequence of statements cannot be described by an axiom, because the precondition depends on the particular kinds of statements in the sequence. In this case, the precondition can only be described with an inference rule. Let S1 and S2 be adjacent program statements. If S1 and S2 have the following pre- and postconditions

{P1} S1 {P2}
{P2} S2 {P3}

the inference rule for such a two-statement sequence is

$$\frac{\{P1\}\ S1\ \{P2\},\ \{P2\}\ S2\ \{P3\}}{\{P1\}\ S1,\ S2\ \{P3\}}$$

So, for our example, {P1} S1; S2 {P3} describes the axiomatic semantics of the sequence S1; S2. The inference rule states that to get the sequence precondition, the precondition of the second statement is computed. This new assertion is then used as the postcondition of the first statement, which can then be used to compute the precondition of the first statement, which is also the precondition of the whole sequence. If S1 and S2 are the assignment statements

x1 = E1

and

x2 = E2

then we have

$\{P3_{x2 \rightarrow E2}\}$ x2 = E2 {P3}
$\{(P3_{x2 \rightarrow E2})_{x1 \rightarrow E1}\}$ x1 = E1 $\{P3_{x2 \rightarrow E2}\}$

Therefore, the weakest precondition for the sequence x1 = E1; x2 = E2 with postcondition P3 is $\{(P3_{x2 \rightarrow E2})_{x1 \rightarrow E1}\}$.

For example, consider the following sequence and postcondition:

```
y = 3 * x + 1;
x = y + 3;
{x < 10}
```

The precondition for the second assignment statement is

```
y < 7
```

which is used as the postcondition for the first statement. The precondition for the first assignment statement can now be computed:

```
3 * x + 1 < 7
x < 2
```

So, `{x < 2}` is the precondition of both the first statement and the two-statement sequence.

3.5.3.5 Selection

We next consider the inference rule for selection statements, the general form of which is

`if` B **`then`** S1 **`else`** S2

We consider only selections that include **`else`** clauses. The inference rule is

$$\frac{\{B \text{ and } P\}\ S1\ \{Q\},\ \{(\text{not } B) \text{ and } P\}\ S2\{Q\}}{\{P\}\ \textbf{if}\ B\ \textbf{then}\ S1\ \textbf{else}\ S2\ \{Q\}}$$

This rule indicates that selection statements must be proven both when the Boolean control expression is true and when it is false. The first logical statement above the line represents the **`then`** clause; the second represents the **`else`**

clause. According to the inference rule, we need a precondition P that can be used in the precondition of both the **then** and **else** clauses.

Consider the following example of the computation of the precondition using the selection inference rule. The example selection statement is

```
if x > 0 then
  y = y - 1
else
  y = y + 1
```

Suppose the postcondition, Q, for this selection statement is {`y > 0`}. We can use the axiom for assignment on the **then** clause

```
y = y - 1 {y > 0}
```

This produces {`y - 1 > 0`} or {`y > 1`}. It can be used as the P part of the precondition for the **then** clause. Now we apply the same axiom to the **else** clause

```
y = y + 1 {y > 0}
```

which produces the precondition {`y + 1 > 0`} or {`y > -1`}. Because {`y > 1`} => {`y > -1`}, the rule of consequence allows us to use {`y > 1`} for the precondition of the whole selection statement.

history note

A significant amount of work has been done on the possibility of using denotational language descriptions to generate compilers automatically (Jones, 1980; Milos et al., 1984; Bodwin et al., 1982). These efforts have shown that the method is feasible, but the work has never progressed to the point where it can be used to generate useful compilers.

3.5.3.6 Logical Pretest Loops

Another essential construct of imperative programming languages is the logical pretest, or **while** loop. Computing the weakest precondition for a **while** loop is inherently more difficult than for a sequence, because the number of iterations cannot always be predetermined. In a case where the number of iterations is known, the loop can be unrolled and treated as a sequence.

The problem of computing the weakest precondition for loops is similar to the problem of proving a theorem about all positive integers. In the latter case, induction is normally used, and the same inductive method can be used for some loops. The principal step in induction is finding an inductive hypothesis. The corresponding step in the axiomatic semantics of a **while** loop is finding an assertion called a **loop invariant**, which is crucial to finding the weakest precondition.

The inference rule for computing the precondition for a **while** loop is

$$\frac{\{I \text{ and } B\}\ S\ \{I\}}{\{I\}\ \textbf{while}\ B\ \textbf{do}\ S\ \textbf{end}\ \{I \text{ and (not } B)\}}$$

where I is the loop invariant. This seems simple, but it is not. The complexity lies in finding an appropriate loop invariant.

The axiomatic description of a **while** loop is written as

{P} **while** B **do** S **end** {Q}

The loop invariant must satisfy a number of requirements to be useful. First, the weakest precondition for the **while** loop must guarantee the truth of the loop invariant. In turn, the loop invariant must guarantee the truth of the postcondition upon loop termination. These constraints move us from the inference rule to the axiomatic description. During execution of the loop, the truth of the loop invariant must be unaffected by the evaluation of the loop-controlling Boolean expression and the loop body statements. Hence, the name *invariant*.

Another complicating factor for **while** loops is the question of loop termination. A loop that does not terminate cannot be correct, and in fact computes nothing. If Q is the postcondition that holds immediately after loop exit, then a precondition P for the loop is one that guarantees Q at loop exit and also guarantees that the loop terminates.

The complete axiomatic description of a **while** construct requires all of the following to be true, in which I is the loop invariant:

P => I
{I and B} S {I}
(I and (not B)) => Q
the loop terminates

If a loop computes a sequence of numeric values, it may be possible to find a loop invariant using an approach that is used for determining the inductive hypothesis when mathematical induction is used to prove a statement about a mathematical sequence. The relationship between the number of iterations and the precondition for the loop body is computed for a few cases, with the hope that a pattern emerges that will apply to the general case. It is helpful to treat the process of producing a weakest precondition as a function, wp. In general

wp(statement, postcondition) = precondition

A wp function is often called a **predicate transformer**, because it takes a predicate, or assertion, as a parameter and returns another predicate.

To find I, the loop postcondition Q is used to compute preconditions for several different numbers of iterations of the loop body, starting with none. If the loop body contains a single assignment statement, the axiom for assignment statements can be used to compute these cases. Consider the example loop:

```
while y <> x do y = y + 1 end {y = x}
```

Remember that the equal sign is being used for two different purposes here. In assertions, it means mathematical equality; outside assertions, it means the assignment operator.

For zero iterations, the weakest precondition is, obviously,

```
{y = x}
```

For one iteration, it is

wp`(y = y + 1, {y = x}) = {y + 1 = x}`, or `{y = x - 1}`

For two iterations, it is

wp`(y = y + 1, {y = x - 1})={y + 1 = x - 1}`, or `{y = x - 2}`

For three iterations, it is

wp`(y = y + 1, {y = x - 2})={y + 1 = x - 2}`, or `{y = x – 3}`

It is now obvious that `{y < x}` will suffice for cases of one or more iterations. Combining this with `{y = x}` for the zero iterations case, we get `{y <= x}`, which can be used for the loop invariant. A precondition for the **`while`** statement can be determined from the loop invariant. In fact, I can be used as the precondition, P.

We must ensure that our choice satisfies the four criteria for I for our example loop. First, because P = I, P => I. The second requirement is that it must be true that

{I and B} S {I}

In our example, we have

```
{y <= x and y <> x} y = y + 1 {y <= x}
```

Applying the assignment axiom to

```
y = y + 1 {y <= x}
```

we get `{y + 1 <= x}`, which is equivalent to `{y < x}`, which is implied by `{y <= x` and `y <> x}`. So, the earlier statement is proven.

Next, we must have

{I and (not B)} => Q

In our example, we have

```
{ (y <= x) and not (y <> x) } => {y = x}
{ (y <= x) and (y = x) } => {y = x}
{y = x} => {y = x}
```

So, this is obviously true. Next, loop termination must be considered. In this example, the question is whether the loop

```
{y <= x} while y <> x do y = y + 1 end {y = x}
```

terminates. Recalling that x and y are assumed to be integer variables, it is easy to see that this loop does terminate. The precondition guarantees that y initially is not larger than x. The loop body increments y with each iteration, until y is equal to x. No matter how much smaller y is than x initially, it will eventually become equal to x. So the loop will terminate. Because our choice of I satisfies all four criteria, it is a satisfactory loop invariant and loop precondition.

The previous process used to compute the invariant for a loop does not always produce an assertion that is the weakest precondition (although it does in the example).

As another example of finding a loop invariant using the approach used in mathematical induction, consider the following loop statement:

```
while s > 1 do s = s / 2 end {s = 1}
```

As before, we use the assignment axiom to try to find a loop invariant and a precondition for the loop. For zero iterations, the weakest precondition is {s = 1}. For one iteration, it is

wp(s = s / 2, {s = 1}) = {s / 2 = 1}, or {s = 2}

For two iterations, it is

wp(s = s / 2, {s = 2}) = {s / 2 = 2}, or {s = 4}

For three iterations, it is

wp(s = s / 2, {s = 4}) = {s / 2 = 4}, or {s = 8}

From these cases, we can see clearly that the invariant is

{s is a nonnegative power of 2}

Once again, the computed I can serve as P, and I passes the four requirements. Unlike our earlier example of finding a loop precondition, this one clearly is not a weakest precondition. Consider using the precondition {s > 1}. The logical statement

```
{s > 1} while s > 1 do s = s / 2 end {s = 1}
```

can easily be proven, and this precondition is significantly broader than the one computed earlier. The loop and precondition are satisfied for any positive value for s, not just powers of 2, as the process indicates. Because of the rule of consequence, using a precondition that is stronger than the weakest precondition does not invalidate a proof.

Finding loop invariants is not always easy. It is helpful to understand the nature of these invariants. First, a loop invariant is a weakened version of the loop postcondition and also a precondition for the loop. So, I must be weak enough to be satisfied prior to the beginning of loop execution, but when combined with the loop exit condition, it must be strong enough to force the truth of the postcondition.

Because of the difficulty of proving loop termination, that requirement is often ignored. If loop termination can be shown, the axiomatic description of the loop is called **total correctness**. If the other conditions can be met but termination is not guaranteed, it is called **partial correctness**.

In more complex loops, finding a suitable loop invariant, even for partial correctness, requires a good deal of ingenuity. Because computing the precondition for a **`while`** loop depends on finding a loop invariant, proving the correctness of programs with **`while`** loops using axiomatic semantics can be difficult.

3.5.3.7 Program Proofs

This section provides validations for two simple programs. The first example of a correctness proof is for a very short program, consisting of a sequence of three assignment statements that interchange the values of two variables.

```
{x = A AND y = B}
t = x;
x = y;
y = t;
{x = B AND y = A}
```

Because the program consists entirely of assignment statements in a sequence, the assignment axiom and the inference rule for sequences can be used to prove its correctness. The first step is to use the assignment axiom on the last statement and the postcondition for the whole program. This yields the precondition

```
{x = B AND t = A}
```

Next, we use this new precondition as a postcondition on the middle statement and compute its precondition, which is

```
{y = B AND t = A}
```

Next, we use this new assertion as the postcondition on the first statement and apply the assignment axiom, which yields

```
{y = B AND x = A}
```

which is the same as the precondition on the program, except for the order of operands on the `AND` operator. Because `AND` is a symmetric operator, our proof is complete.

The following example is a proof of correctness of a pseudocode program that computes the factorial function.

```
{n >= 0}
count = n;
fact = 1;
while count <> 0 do
    fact = fact * count;
    count = count - 1;
end
{fact = n!}
```

The method described earlier for finding the loop invariant does not work for the loop in this example. Some ingenuity is required here, which can be aided by a brief study of the code. The loop computes the factorial function in order of the last multiplication first; that is, `(n - 1) * n` is done first, assuming `n` is greater than `1`. So, part of the invariant can be

```
fact = (count + 1) * (count + 2) *...* (n - 1) * n
```

But we must also ensure that `count` is always nonnegative, which we can do by adding that to the assertion above, to get

```
I = (fact = (count + 1) *...* n) AND (count >= 0)
```

Next, we must confirm that this I meets the requirements for invariants. Once again we let I also be used for P, so P clearly implies I. The next question is

{I and B} S {I}

I and B is

```
((fact = (count + 1) *...* n) AND (count >= 0)) AND
   (count <> 0)
```

which reduces to

```
(fact = (count + 1) *...* n) AND (count > 0)
```

In our case, we must compute the precondition of the body of the loop, using the invariant for the postcondition. For

{P} `count = count - 1` {`I`}

we compute P to be

```
{(fact = count * (count + 1) *...* n) AND
    (count >= 1)}
```

Using this as the postcondition for the first assignment in the loop body,

```
{P} fact = fact * count {(fact = count * (count + 1)
                          *...* n) AND (count >= 1)}
```

In this case, P is

```
{(fact = (count + 1) *...* n) AND (count >= 1)}
```

It is clear that I and B implies this P, so by the rule of consequence,

{I AND B} S {I}

is true. Finally, the last test of I is

I AND (NOT B) => Q

For our example, this is

```
((fact = (count + 1) *...* n) AND (count >= 0)) AND
   (count = 0)) => fact = n!
```

This is clearly true, for when `count = 0`, the first part is precisely the definition of factorial. So, our choice of I meets the requirements for a loop invariant. Now we can use our P (which is the same as I) from the **`while`** as the postcondition on the second assignment of the program

```
{P} fact = 1 {(fact = (count + 1) *...* n) AND
    (count >= 0)}
```

which yields for P

```
(1 = (count + 1) *...* n) AND (count >= 0))
```

Using this as the postcondition for the first assignment in the code

```
{P} count = n {(1 = (count + 1) *...* n) AND
    (count >= 0))}
```

produces for P

```
{(n + 1) *...* n = 1) AND (n >= 0)}
```

The left operand of the AND operator is true (because `1 = 1`) and the right operand is exactly the precondition of the whole code segment, {`n >= 0`}. Therefore, the program has been proven to be correct.

3.5.3.8 Evaluation

As stated previously, to define the semantics of a complete programming language using the axiomatic method, there must be an axiom or an inference rule for each statement type in the language. Defining axioms or inference rules for

some of the statements of programming languages has proven to be a difficult task. An obvious solution to this problem is to design the language with the axiomatic method in mind, so that only statements for which axioms or inference rules can be written are included. Unfortunately, such a language would necessarily leave out some useful and powerful parts.

Axiomatic semantics is a powerful tool for research into program correctness proofs, and it provides an excellent framework in which to reason about programs, both during their construction and later. Its usefulness in describing the meaning of programming languages to language users and compiler writers is, however, highly limited.

SUMMARY

Backus-Naur Form and context-free grammars are equivalent metalanguages that are well suited for the task of describing the syntax of programming languages. Not only are they concise descriptive tools, but also the parse trees that can be associated with their generative actions give graphical evidence of the underlying syntactic structures. Furthermore, they are naturally related to recognition devices for the languages they generate, which leads to the relatively easy construction of syntax analyzers for compilers for these languages.

An attribute grammar is a descriptive formalism that can describe both the syntax and static semantics of a language. Attribute grammars are extensions to context-free grammars. An attribute grammar consists of a grammar, a set of attributes, a set of attribute computation functions, and a set of predicates, which together describe static semantics rules.

This chapter provides a brief introduction to three methods of semantic description: operational, denotational, and axiomatic. Operational semantics is a method of describing the meaning of language constructs in terms of their effects on an ideal machine. In denotational semantics, mathematical objects are used to represent the meanings of language constructs. Language entities are converted to these mathematical objects with recursive functions. Axiomatic semantics, which is based on formal logic, was devised as a tool for proving the correctness of programs.

BIBLIOGRAPHIC NOTES

Syntax description using context-free grammars and BNF are thoroughly discussed in Cleaveland and Uzgalis (1976).

Research in axiomatic semantics was begun by Floyd (1967) and further developed by Hoare (1969). The semantics of a large part of Pascal was described by Hoare and Wirth (1973) using this method. The parts they did not complete involved functional side effects and goto statements. These were found to be the most difficult to describe.

The technique of using preconditions and postconditions during the development of programs is described (and advocated) by Dijkstra (1976) and also discussed in detail in Gries (1981).

Good introductions to denotational semantics can be found in Gordon (1979) and Stoy (1977). Introductions to all of the semantics description methods discussed in this chapter can be found in Marcotty et al. (1976). Another good reference for much of the chapter material is Pagan (1981). The form of the denotational semantic functions in this chapter is similar to that found in Meyer (1990).

REVIEW QUESTIONS

1. Define *syntax* and *semantics*.
2. Who are language descriptions for?
3. Describe the operation of a general language generator.
4. Describe the operation of a general language recognizer.
5. What is the difference between a sentence and a sentential form?
6. Define a left-recursive grammar rule.
7. What three extensions are common to most EBNFs?
8. Distinguish between static and dynamic semantics.
9. What purpose do predicates serve in an attribute grammar?
10. What is the difference between a synthesized and an inherited attribute?
11. How is the order of evaluation of attributes determined for the trees of a given attribute grammar?
12. What is the primary use of attribute grammars?
13. Explain the primary uses of a methodology and notation for describing the semantics of programming languages.
14. Why can machine languages not be used to define statements in operational semantics?
15. Describe the two levels of uses of operational semantics.
16. In denotational semantics, what are the syntactic and semantic domains?
17. What is stored in the state of a program for denotational semantics?
18. Which semantics approach is most widely known?
19. What two things must be defined for each language entity in order to construct a denotational description of the language?
20. Which part of an inference rule is the antecedent?
21. What is a predicate transformer function?
22. What does partial correctness mean for a loop construct?
23. On what branch of mathematics is axiomatic semantics based?
24. On what branch of mathematics is denotational semantics based?

25. What is the problem with using a software pure interpreter for operational semantics?
26. Explain what the preconditions and postconditions of a given statement mean in axiomatic semantics.
27. Describe the approach of using axiomatic semantics to prove the correctness of a given program.
28. Describe the basic concept of denotational semantics.
29. In what fundamental way do operational semantics and denotational semantics differ?

PROBLEM SET

1. The two mathematical models of language description are generation and recognition. Describe how each can define the syntax of a programming language.
2. Write EBNF descriptions for the following:
   a. A Java class definition header statement
   b. A Java method call statement
   c. A C **`switch`** statement
   d. A C **`union`** definition
   e. C **`float`** literals
3. Rewrite the BNF of Example 3.4 to give `+` precedence over `*` and force `+` to be right associative.
4. Rewrite the BNF of Example 3.4 to add the `++` and `--` unary operators of Java.
5. Write a BNF description of the Boolean expressions of Java, including the three operators `&&`, `||`, and `!` and the relational expressions.
6. Using the grammar in Example 3.2, show a parse tree and a leftmost derivation for each of the following statements:
   a. `A = A * (B + (C * A))`
   b. `B = C * (A * C + B)`
   c. `A = A * (B + (C))`
7. Using the grammar in Example 3.4, show a parse tree and a leftmost derivation for each of the following statements:
   a. `A = ( A + B ) * C`
   b. `A = B + C + A`
   c. `A = A * (B + C)`
   d. `A = B * (C * (A + B))`

8. Prove that the following grammar is ambiguous:

   <S> → <A>

   <A> → <A> + <A> | <id>

   <id> → a | b | c

9. Modify the grammar of Example 3.4 to add a unary minus operator that has higher precedence than either + or * .

10. Describe, in English, the language defined by the following grammar:

    <S> → <A> <B> <C>

    <A> → a <A> | a

    <B> → b <B> | b

    <C> → c <C> | c

11. Consider the following grammar:

    <S> → <A> a <B> b

    <A> → <A> b | b

    <B> → a <B> | a

    Which of the following sentences are in the language generated by this grammar?

    a. baab
    b. bbbab
    c. bbaaaaa
    d. bbaab

12. Consider the following grammar:

    <S> → a <S> c <B> | <A> | b

    <A> → c <A> | c

    <B> → d | <A>

    Which of the following sentences are in the language generated by this grammar?

    a. abcd
    b. acccbd
    c. acccbcc
    d. acd
    e. accc

13. Write a grammar for the language consisting of strings that have n copies of the letter a followed by the same number of copies of the letter b, where $n > 0$. For example, the strings ab, aaaabbbb, and aaaaaaaabbbbbbbb are in the language but a, abb, ba, and aaabb are not.

14. Draw parse trees for the sentences aabb and aaaabbbb, as derived from the grammar of Problem 13.

15. Convert the BNF of Example 3.1 to EBNF.
16. Convert the BNF of Example 3.3 to EBNF.
17. Convert the following EBNF to BNF:

    $S \rightarrow A\{bA\}$

    $A \rightarrow a[b]A$
18. What is the difference between an intrinsic attribute and a nonintrinsic synthesized attribute?
19. Write an attribute grammar whose BNF basis is that of Example 3.6 in Section 3.4.5 but whose language rules are as follows: Data types cannot be mixed in expressions, but assignment statements need not have the same types on both sides of the assignment operator.
20. Write an attribute grammar whose base BNF is that of Example 3.2 and whose type rules are the same as for the assignment statement example of Section 3.4.5.
21. Using the virtual machine instructions given in Section 3.5.1.1, give an operational semantic definition of the following:
    a. Java **`do-while`**
    b. Ada **`for`**
    c. C++ **`if-then-else`**
    d. C **`for`**
    e. C **`switch`**
22. Write a denotational semantics mapping function for the following statements:
    a. Ada **`for`**
    b. Java **`do-while`**
    c. Java Boolean expressions
    d. Java **`for`**
    e. C **`switch`**
23. Compute the weakest precondition for each of the following assignment statements and postconditions:
    a. `a = 2 * (b - 1) - 1 {a > 0}`
    b. `b = (c + 10) / 3 {b > 6}`
    c. `a = a + 2 * b - 1 {a > 1}`
    d. `x = 2 * y + x - 1 {x > 11}`
24. Compute the weakest precondition for each of the following sequences of assignment statements and their postconditions:
    a.
    ```
    a = 2 * b + 1;
    b = a - 3
    {b < 0}
    ```

b.
```
a = 3 * (2 * b + a);
b = 2 * a - 1
{b > 5}
```

25. Compute the weakest precondition for each of the following selection constructs and their postconditions:

a.
```
if (a == b)
  b = 2 * a + 1
else
  b = 2 * a;
{b > 1}
```

b.
```
if (x < y)
  x = x + 1
else
  x = 3 * x
{x < 0}
```

c.
```
if (x > y)
  y = 2 * x + 1
else
  y = 3 * x - 1;
{y > 3}
```

26. Explain the four criteria for proving the correctness of a logical pretest loop construct of the form **while** B **do** S **end**

27. Prove that (n + 1) * . . . * n = 1

28. Prove the following program is correct:

```
{n > 0}
count = n;
sum = 0;
while count <> 0 do
  sum = sum + count;
  count = count - 1;
end
{sum = 1 + 2 + . . . + n}
```

4

Lexical and Syntax Analysis

4.1 Introduction
4.2 Lexical Analysis
4.3 The Parsing Problem
4.4 Recursive-Descent Parsing
4.5 Bottom-Up Parsing

A serious investigation of compiler design requires at least a semester of intensive study, including the design and implementation of a compiler for a small but realistic programming language. The first part of such a course is devoted to lexical and syntax analyses. The syntax analyzer is the heart of a compiler, because several other important components, including the semantic analyzer and the intermediate code generator, are driven by the actions of the syntax analyzer.

Some readers may wonder why a chapter on any part of a compiler would be included in a book on programming languages. There are at least two reasons to include a discussion of lexical and syntax analyses in this book: First, syntax analyzers are based directly on the grammars discussed in Chapter 3, so it is natural to discuss them as an application of grammars. Second, lexical and syntax analyzers are needed in numerous situations outside compiler design. Many applications, among them program listing formatters, programs that compute the complexity of programs, and programs that must analyze and react to the contents of a configuration file, all need to do lexical and syntax analyses. Therefore, lexical and syntax analyses are important topics for software developers, even if they never need to write a compiler. Furthermore, some computer science programs no longer require students to take a compiler design course, which leaves students with no instruction in lexical or syntax analysis. In those cases, this chapter can be covered in the programming language course. In degree programs that require a compiler design course, this chapter can be skipped.

This chapter begins with an introduction to lexical analysis, along with a simple example. Next, the general parsing problem is discussed, including the two primary approaches to parsing and the complexity of parsing. Then, we introduce the recursive-descent implementation technique for top-down parsers, including examples of parts of a recursive-descent parser and a trace of a parse using one. The last section discusses bottom-up parsing and the LR parsing algorithm. This section includes an example of a small LR parsing table and the parse of a string using the LR parsing process.

4.1 Introduction

Three different approaches to implementing programming languages are introduced in Chapter 1: compilation, pure interpretation, and hybrid implementation. The compilation approach uses a program called a compiler, which translates programs written in a high-level programming language into machine code. Compilation is typically used to implement programming languages that are used for large applications, often written in languages such as C++ and COBOL. Pure interpretation systems perform no translation; rather, programs are interpreted in their original form by a software interpreter. Pure interpretation is usually used for smaller systems in which execution efficiency is not critical, such as scripts embedded in HTML documents, written in languages such as JavaScript. Hybrid implementation systems translate programs written in high-level languages into intermediate forms, which are interpreted. These systems are now more widely used than ever, thanks in large part to the popularity of scripting languages. Traditionally, hybrid systems have resulted in much slower program execution than compiler systems. However, in recent

years the use of Just-in-Time (JIT) compilers has become widespread, particularly for Java programs and programs written for the Microsoft .NET system. A JIT compiler, which translates intermediate code to machine code, is used on methods at the time they are first called. In effect, a JIT compiler transforms a hybrid system to a delayed compiler system.

All three of the implementation approaches just discussed use both lexical and syntax analyzers.

Syntax analyzers, or parsers, are nearly always based on a formal description of the syntax of programs. The most commonly used syntax-description formalism is context-free grammars, or BNF, which is introduced in Chapter 3. Using BNF, as opposed to using some informal syntax description, has at least three compelling advantages. First, BNF descriptions of the syntax of programs are clear and concise, both for humans and for software systems that use them. Second, the BNF description can be used as the direct basis for the syntax analyzer. Third, implementations based on BNF are relatively easy to maintain because of their modularity.

Nearly all compilers separate the task of analyzing syntax into two distinct parts, named lexical analysis and syntax analysis, although this terminology is confusing. The lexical analyzer deals with small-scale language constructs, such as names and numeric literals. The syntax analyzer deals with the large-scale constructs, such as expressions, statements, and program units. Section 4.2 introduces lexical analyzers. Sections 4.3, 4.4, and 4.5 discuss syntax analyzers.

There are three reasons why lexical analysis is separated from syntax analysis:

1. Simplicity—Techniques for lexical analysis are less complex than those required for syntax analysis, so the lexical-analysis process can be simpler if it is separate. Also, removing the low-level details of lexical analysis from the syntax analyzer makes the syntax analyzer both smaller and less complex.
2. Efficiency—Although it pays to optimize the lexical analyzer, because lexical analysis requires a significant portion of total compilation time, it is not fruitful to optimize the syntax analyzer. Separation facilitates this selective optimization.
3. Portability—Because the lexical analyzer reads input program files and often includes buffering of that input, it is somewhat platform dependent. However, the syntax analyzer can be platform independent. It is always good to isolate machine-dependent parts of any software system.

4.2 Lexical Analysis

A lexical analyzer is essentially a pattern matcher. A pattern matcher attempts to find a substring of a given string of characters that matches a given character pattern. Pattern matching is a traditional part of computing. One of the earliest uses

of pattern matching was with text editors, such as the `ed` line editor, which was introduced in an early version of UNIX. Since then, pattern matching has found its way into some programming languages—for example, Perl and JavaScript. It is also available through the standard class libraries of Java, C++, and C#.

A lexical analyzer serves as the front end of a syntax analyzer. Technically, lexical analysis is a part of syntax analysis. A lexical analyzer performs syntax analysis at the lowest level of program structure. An input program appears to a compiler as a single string of characters. The lexical analyzer collects characters into logical groupings and assigns internal codes to the groupings according to their structure. In Chapter 3, these logical groupings are named **lexemes**, and the internal codes for categories of these groupings are named **tokens**. Lexemes are recognized by matching the input character string against character string patterns. Although tokens are usually represented as integer values, for the sake of readability of lexical and syntax analyzers, they are often referenced through named constants.

Consider the following example of an assignment statement:

```
result = oldsum - value / 100;
```

Following are the tokens and lexemes of this statement:

Token	*Lexeme*
IDENT	result
ASSIGN_OP	=
IDENT	oldsum
SUB_OP	-
IDENT	value
DIV_OP	/
INT_LIT	100
SEMICOLON	;

Lexical analyzers extract lexemes from a given input string and produce the corresponding tokens. In the early days of compilers, lexical analyzers often processed an entire source program file and produced a file of tokens and lexemes. Now, however, most lexical analyzers are subprograms that locate the next lexeme in the input, determine its associated token code, and return them to the caller, which is the syntax analyzer. So, each call to the lexical analyzer returns a single lexeme and its token. The only view of the input program seen by the syntax analyzer is the output of the lexical analyzer, one token at a time.

The lexical-analysis process includes skipping comments and white space outside lexemes, as they are not relevant to the meaning of the program. Also, the lexical analyzer inserts lexemes for user-defined names into the symbol table, which is used by later phases of the compiler. Finally, lexical analyzers detect syntactic errors in tokens, such as ill-formed floating-point literals, and report such errors to the user.

There are three approaches to building a lexical analyzer:

1. Write a formal description of the token patterns of the language using a descriptive language related to regular expressions.[1] These descriptions are used as input to a software tool that automatically generates a lexical analyzer. There are many such tools available for this. The oldest of these, named lex, is commonly included as part of UNIX systems.
2. Design a state transition diagram that describes the token patterns of the language and write a program that implements the diagram.
3. Design a state transition diagram that describes the token patterns of the language and hand-construct a table-driven implementation of the state diagram.

A state transition diagram, or just **state diagram**, is a directed graph. The nodes of a state diagram are labeled with state names. The arcs are labeled with the input characters that cause the transitions among the states. An arc may also include actions the lexical analyzer must perform when the transition is taken.

State diagrams of the form used for lexical analyzers are representations of a class of mathematical machines called **finite automata**. Finite automata can be designed to recognize members of a class of languages called **regular languages**. Regular grammars are generative devices for regular languages. The tokens of a programming language are a regular language, and a lexical analyzer is a finite automaton.

We now illustrate lexical-analyzer construction with a state diagram and the code that implements it. The state diagram could simply include states and transitions for each and every token pattern. However, that approach results in a very large and complex diagram, because every node in the state diagram would need a transition for every character in the character set of the language being analyzed. We therefore consider ways to simplify it.

Suppose we need a lexical analyzer that recognizes only arithmetic expressions, including variable names and integer literals as operands. Assume that the variable names consist of strings of uppercase letters, lowercase letters, and digits but must begin with a letter. Names have no length limitation. The first thing to observe is that there are 52 different characters (any uppercase or lowercase letter) that can begin a name, which would require 52 transitions from the transition diagram's initial state. However, a lexical analyzer is interested only in determining that it is a name and is not concerned with which specific name it happens to be. Therefore, we define a character class named LETTER for all 52 letters and use a single transition on the first letter of any name.

Another opportunity for simplifying the transition diagram is with the integer literal tokens. There are 10 different characters that could begin an integer literal lexeme. This would require 10 transitions from the start state of the state diagram. Because specific digits are not a concern of the lexical analyzer, we can build a much more compact state diagram if we define a character

1. These regular expressions are the basis for the pattern-matching facilities now part of many programming languages, either directly or through a class library.

class named DIGIT for digits and use a single transition on any character in this character class to a state that collects integer literals.

Because our names can include digits, the transition from the node following the first character of a name can use a single transition on LETTER or DIGIT to continue collecting the characters of a name.

Next, we define some utility subprograms for the common tasks inside the lexical analyzer. First, we need a subprogram, which we can name `getChar`, that has several duties. When called, `getChar` gets the next character of input from the input program and puts it in the global variable `nextChar`. `getChar` must also determine the character class of the input character and put it in the global variable `charClass`. The lexeme being built by the lexical analyzer, which could be implemented as a character string or an array, will be named `lexeme`.

We implement the process of putting the character in `nextChar` into the string array `lexeme` in a subprogram named `addChar`. This subprogram must be explicitly called because programs include some characters that need not be put in `lexeme`, for example the white-space characters between lexemes. In a more realistic lexical analyzer, comments also would not be placed in `lexeme`.

When the lexical analyzer is called, it is convenient if the next character of input is the first character of the next lexeme. Because of this, a function named `getNonBlank` is used to skip white space every time the analyzer is called.

Finally, a subprogram named `lookup` is needed to compute the token code for the single-character tokens. In our example, these are parentheses and the arithmetic operators. Token codes are numbers arbitrarily assigned to tokens by the compiler writer.

The state diagram in Figure 4.1 describes the patterns for our tokens. It includes the actions required on each transition of the state diagram.

The following is a C implementation of a lexical analyzer specified in the state diagram of Figure 4.1, including a main driver function for testing purposes:

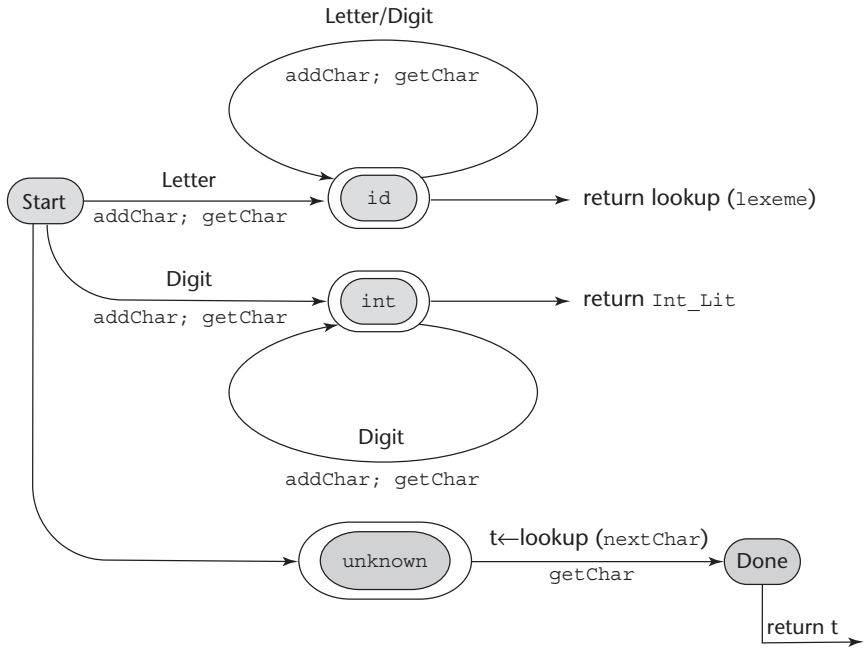
```
/* front.c - a lexical analyzer system for simple
             arithmetic expressions */

#include <stdio.h>
#include <ctype.h>

/* Global declarations */
/* Variables */
int charClass;
char lexeme [100];
char nextChar;
int lexLen;
int token;
int nextToken;
FILE *in_fp, *fopen();
```

Figure 4.1

A state diagram to recognize names, parentheses, and arithmetic operators




```
/* Function declarations */
void addChar();
void getChar();
void getNonBlank();
int lex();

/* Character classes */
#define LETTER 0
#define DIGIT 1
#define UNKNOWN 99

/* Token codes */
#define INT_LIT 10
#define IDENT 11
#define ASSIGN_OP 20
#define ADD_OP 21
#define SUB_OP 22
#define MULT_OP 23
#define DIV_OP 24
#define LEFT_PAREN 25
#define RIGHT_PAREN 26
```

```
/*****************************************************/
/* main driver */
main() {

/* Open the input data file and process its contents */
  if ((in_fp = fopen("front.in", "r")) == NULL)
    printf("ERROR - cannot open front.in \n");
  else {
    getChar();
    do {
      lex();
    } while (nextToken != EOF);
  }
}



/*****************************************************/
/* lookup - a function to lookup operators and parentheses
            and return the token */
int lookup(char ch) {
  switch (ch) {
    case '(':
      addChar();
      nextToken = LEFT_PAREN;
      break;

    case ')':
      addChar();
      nextToken = RIGHT_PAREN;
      break;

    case '+':
      addChar();
      nextToken = ADD_OP;
      break;

    case '-':
      addChar();
      nextToken = SUB_OP;
      break;

    case '*':
      addChar();
      nextToken = MULT_OP;
      break;
```

```
    case '/':
      addChar();
      nextToken = DIV_OP;
      break;

    default:
      addChar();
      nextToken = EOF;
      break;
  }
  return nextToken;
}

/*****************************************************/
/* addChar - a function to add nextChar to lexeme */
void addChar() {
  if (lexLen <= 98) {
    lexeme[lexLen++] = nextChar;
    lexeme[lexLen] = 0;
  }
  else
    printf("Error - lexeme is too long \n");
}

/*****************************************************/
/* getChar - a function to get the next character of
             input and determine its character class */
void getChar() {
  if ((nextChar = getc(in_fp)) != EOF) {
    if (isalpha(nextChar))
      charClass = LETTER;
    else if (isdigit(nextChar))
           charClass = DIGIT;
         else charClass = UNKNOWN;
   }
   else
     charClass = EOF;
}

/*****************************************************/
/* getNonBlank - a function to call getChar until it
                 returns a non-whitespace character */
void getNonBlank() {
  while (isspace(nextChar))
    getChar();
}
```

```
/
*****************************************************/
/* lex - a simple lexical analyzer for arithmetic
         expressions */
int lex() {
  lexLen = 0;
  getNonBlank();
  switch (charClass) {

/* Parse identifiers */
    case LETTER:
      addChar();
      getChar();
      while (charClass == LETTER || charClass == DIGIT) {
        addChar();
        getChar();
      }
    nextToken = IDENT;
    break;

/* Parse integer literals */
    case DIGIT:
      addChar();
      getChar();
      while (charClass == DIGIT) {
        addChar();
        getChar();
      }
      nextToken = INT_LIT;
      break;

/* Parentheses and operators */
    case UNKNOWN:
      lookup(nextChar);
      getChar();
      break;

/* EOF */
    case EOF:
      nextToken = EOF;
      lexeme[0] = 'E';
      lexeme[1] = 'O';
      lexeme[2] = 'F';
      lexeme[3] = 0;
      break;
  } /* End of switch */
```

```
  printf("Next token is: %d, Next lexeme is %s\n",
          nextToken, lexeme);
  return nextToken;
}  /* End of function lex */
```

This code illustrates the relative simplicity of lexical analyzers. Of course, we have left out input buffering, as well as some other important details. Furthermore, we have dealt with a very small and simple input language.

Consider the following expression:

```
(sum + 47) / total
```

Following is the output of the lexical analyzer of `front.c` when used on this expression:

```
Next token is: 25 Next lexeme is (
Next token is: 11 Next lexeme is sum
Next token is: 21 Next lexeme is +
Next token is: 10 Next lexeme is 47
Next token is: 26 Next lexeme is )
Next token is: 24 Next lexeme is /
Next token is: 11 Next lexeme is total
Next token is: -1 Next lexeme is EOF
```

Names and reserved words in programs have similar patterns. Although it is possible to build a state diagram to recognize every specific reserved word of a programming language, that would result in a prohibitively large state diagram. It is much simpler and faster to have the lexical analyzer recognize names and reserved words with the same pattern and use a lookup in a table of reserved words to determine which names are reserved words. Using this approach considers reserved words to be exceptions in the names token category.

A lexical analyzer often is responsible for the initial construction of the symbol table, which acts as a database of names for the compiler. The entries in the symbol table store information about user-defined names, as well as the attributes of the names. For example, if the name is that of a variable, the variable's type is one of its attributes that will be stored in the symbol table. Names are usually placed in the symbol table by the lexical analyzer. The attributes of a name are usually put in the symbol table by some part of the compiler that is subsequent to the actions of the lexical analyzer.

4.3 The Parsing Problem

The part of the process of analyzing syntax that is referred to as *syntax analysis* is often called **parsing**. We will use these two interchangeably.

This section discusses the general parsing problem and introduces the two main categories of parsing algorithms, top-down and bottom-up, as well as the complexity of the parsing process.

4.3.1 Introduction to Parsing

Parsers for programming languages construct parse trees for given programs. In some cases, the parse tree is only implicitly constructed, meaning that perhaps only a traversal of the tree is generated. But in all cases, the information required to build the parse tree is created during the parse. Both parse trees and derivations include all of the syntactic information needed by a language processor.

There are two distinct goals of syntax analysis: First, the syntax analyzer must check the input program to determine whether it is syntactically correct. When an error is found, the analyzer must produce a diagnostic message and recover. In this case, recovery means it must get back to a normal state and continue its analysis of the input program. This step is required so that the compiler finds as many errors as possible during a single analysis of the input program. If it is not done well, error recovery may create more errors, or at least more error messages. The second goal of syntax analysis is to produce a complete parse tree, or at least trace the structure of the complete parse tree, for syntactically correct input. The parse tree (or its trace) is used as the basis for translation.

Parsers are categorized according to the direction in which they build parse trees. The two broad classes of parsers are **top-down**, in which the tree is built from the root downward to the leaves, and **bottom-up**, in which the parse tree is built from the leaves upward to the root.

In this chapter, we use a small set of notational conventions for grammar symbols and strings to make the discussion less cluttered. For formal languages, they are as follows:

1. Terminal symbols—lowercase letters at the beginning of the alphabet (a, b, . . .)
2. Nonterminal symbols—uppercase letters at the beginning of the alphabet (A, B, . . .)
3. Terminals or nonterminals—uppercase letters at the end of the alphabet (W, X, Y, Z)
4. Strings of terminals—lowercase letters at the end of the alphabet (w, x, y, z)
5. Mixed strings (terminals and/or nonterminals)—lowercase Greek letters (α, β, δ, γ)

For programming languages, terminal symbols are the small-scale syntactic constructs of the language, what we have referred to as lexemes. The nonterminal symbols of programming languages are usually connotative names or abbreviations, surrounded by pointed brackets—for example, <while_statement>, <expr>, and <function_def>. The sentences of a language (programs, in the case of a programming language) are strings of terminals. Mixed strings describe right-hand sides (RHSs) of grammar rules and are used in parsing algorithms.

4.3.2 Top-Down Parsers

A top-down parser traces or builds a parse tree in preorder. A preorder traversal of a parse tree begins with the root. Each node is visited before its branches are followed. Branches from a particular node are followed in left-to-right order. This corresponds to a leftmost derivation.

In terms of the derivation, a top-down parser can be described as follows: Given a sentential form that is part of a leftmost derivation, the parser's task is to find the next sentential form in that leftmost derivation. The general form of a left sentential form is $xA\alpha$, whereby our notational conventions x is a string of terminal symbols, A is a nonterminal, and α is a mixed string. Because x contains only terminals, A is the leftmost nonterminal in the sentential form, so it is the one that must be expanded to get the next sentential form in a leftmost derivation. Determining the next sentential form is a matter of choosing the correct grammar rule that has A as its LHS. For example, if the current sentential form is

$$xA\alpha$$

and the A-rules are $A \rightarrow bB$, $A \rightarrow cBb$, and $A \rightarrow a$, a top-down parser must choose among these three rules to get the next sentential form, which could be $xbB\alpha$, $xcBb\alpha$, or $xa\alpha$. This is the parsing decision problem for top-down parsers.

Different top-down parsing algorithms use different information to make parsing decisions. The most common top-down parsers choose the correct RHS for the leftmost nonterminal in the current sentential form by comparing the next token of input with the first symbols that can be generated by the RHSs of those rules. Whichever RHS has that token at the left end of the string it generates is the correct one. So, in the sentential form $xA\alpha$, the parser would use whatever token would be the first generated by A to determine which A-rule should be used to get the next sentential form. In the example above, the three RHSs of the A-rules all begin with different terminal symbols. The parser can easily choose the correct RHS based on the next token of input, which must be a, b, or c in this example. In general, choosing the correct RHS is not so straightforward, because some of the RHSs of the leftmost nonterminal in the current sentential form may begin with a nonterminal.

The most common top-down parsing algorithms are closely related. A **recursive-descent parser** is a coded version of a syntax analyzer based directly on the BNF description of the syntax of language. The most common alternative to recursive descent is to use a parsing table, rather than code, to implement the BNF rules. Both of these, which are called **LL algorithms**, are equally powerful, meaning they work on the same subset of all context-free grammars. The first L in LL specifies a left-to-right scan of the input; the second L specifies that a leftmost derivation is generated. Section 4.4 introduces the recursive-descent approach to implementing an LL parser.

4.3.3 Bottom-Up Parsers

A bottom-up parser constructs a parse tree by beginning at the leaves and progressing toward the root. This parse order corresponds to the reverse of a rightmost derivation. That is, the sentential forms of the derivation are produced in order of last to first. In terms of the derivation, a bottom-up parser can be described as follows: Given a right sentential form α,[2] the parser must determine what substring of α is the RHS of the rule in the grammar that must be reduced to its LHS to produce the previous sentential form in the rightmost derivation. For example, the first step for a bottom-up parser is to determine which substring of the initial given sentence is the RHS to be reduced to its corresponding LHS to get the second last sentential form in the derivation. The process of finding the correct RHS to reduce is complicated by the fact that a given right sentential form may include more than one RHS from the grammar of the language being parsed. The correct RHS is called the **handle**.

Consider the following grammar and derivation:

$S \rightarrow aAc$
$A \rightarrow aA \mid b$

S => aAc => aaAc => aabc

A bottom-up parser of this sentence, aabc, starts with the sentence and must find the handle in it. In this example, this is an easy task, for the string contains only one RHS, b. When the parser replaces b with its LHS, A, it gets the second to last sentential form in the derivation, aaAc. In the general case, as stated previously, finding the handle is much more difficult, because a sentential form may include several different RHSs.

A bottom-up parser finds the handle of a given right sentential form by examining the symbols on one or both sides of a possible handle. Symbols to the right of the possible handle are usually tokens in the input that have not yet been analyzed.

The most common bottom-up parsing algorithms are in the LR family, where the L specifies a left-to-right scan of the input and the R specifies that a rightmost derivation is generated.

4.3.4 The Complexity of Parsing

Parsing algorithms that work for any unambiguous grammar are complicated and inefficient. In fact, the complexity of such algorithms is $O(n^3)$, which means the amount of time they take is on the order of the cube of the length of the string to be parsed. This relatively large amount of time is required because these algorithms frequently must back up and reparse part of the sentence being analyzed. Reparsing is required when the parser has made a mistake in

2. A right sentential form is a sentential form that appears in a rightmost derivation.

the parsing process. Backing up the parser also requires that part of the parse tree being constructed (or its trace) must be dismantled and rebuilt. $O(n^3)$ algorithms are normally not useful for practical processes, such as syntax analysis for a compiler, because they are far too slow. In situations such as this, computer scientists often search for algorithms that are faster, though less general. Generality is traded for efficiency. In terms of parsing, faster algorithms have been found that work for only a subset of the set of all possible grammars. These algorithms are acceptable as long as the subset includes grammars that describe programming languages. (Actually, as discussed in Chapter 3, the whole class of context-free grammars is not adequate to describe all of the syntax of most programming languages.)

All algorithms used for the syntax analyzers of commercial compilers have complexity $O(n)$, which means the time they take is linearly related to the length of the string to be parsed. This is vastly more efficient than $O(n^3)$ algorithms.

4.4 Recursive-Descent Parsing

This section introduces the recursive-descent top-down parser implementation process.

4.4.1 The Recursive-Descent Parsing Process

A recursive-descent parser is so named because it consists of a collection of subprograms, many of which are recursive, and it produces a parse tree in top-down order. This recursion is a reflection of the nature of programming languages, which include several different kinds of nested structures. For example, statements are often nested in other statements. Also, parentheses in expressions must be properly nested. The syntax of these structures is naturally described with recursive grammar rules.

EBNF is ideally suited for recursive-descent parsers. Recall from Chapter 3 that the primary EBNF extensions are braces, which specify that what they enclose can appear zero or more times, and brackets, which specify that what they enclose can appear once or not at all. Note that in both cases, the enclosed symbols are optional. Consider the following examples:

<if_statement> → **`if`** <logic_expr> <statement> [**`else`** <statement>]
<ident_list> → ident {, ident}

In the first rule, the **`else`** clause of an **`if`** statement is optional. In the second, an <ident_list> is an identifier, followed by zero or more repetitions of a comma and an identifier.

A recursive-descent parser has a subprogram for each nonterminal in its associated grammar. The responsibility of the subprogram associated with a particular nonterminal is as follows: When given an input string, it traces

out the parse tree that can be rooted at that nonterminal and whose leaves match the input string. In effect, a recursive-descent parsing subprogram is a parser for the language (set of strings) that is generated by its associated nonterminal.

Consider the following EBNF description of simple arithmetic expressions:

<expr> → <term> {(+ | -) <term>}
<term> → <factor> {(* | /) <factor>}
<factor> → id | int_constant | (<expr>)

Recall from Chapter 3 that an EBNF grammar for arithmetic expressions, such as this one, does not force any associativity rule. Therefore, when using such a grammar as the basis for a compiler, one must take care to ensure that the code generation process, which is normally driven by syntax analysis, produces code that adheres to the associativity rules of the language. This can easily be done when recursive-descent parsing is used.

In the following recursive-descent function, `expr`, the lexical analyzer is the function that is implemented in Section 4.2. It gets the next lexeme and puts its token code in the global variable `nextToken`. The token codes are defined as named constants, as in Section 4.2.

A recursive-descent subprogram for a rule with a single RHS is relatively simple. For each terminal symbol in the RHS, that terminal symbol is compared with `nextToken`. If they do not match, it is a syntax error. If they match, the lexical analyzer is called to get the next input token. For each nonterminal, the parsing subprogram for that nonterminal is called.

The recursive-descent subprogram for the first rule in the previous example grammar, written in C, is

```
/* expr
   Parses strings in the language generated by the rule:
   <expr> -> <term> {(+ | -) <term>}
   */
void expr() {
  printf("Enter <expr>\n");

/* Parse the first term */
  term();

/* As long as the next token is + or -, get
   the next token and parse the next term */
  while (nextToken == ADD_OP || nextToken == SUB_OP) {
    lex();
    term();
  }
  printf("Exit <expr>\n");
}  /* End of function expr */
```

Notice that the `expr` function includes tracing output statements, which are included to produce the example output shown later in this section.

Recursive-descent parsing subprograms are written with the convention that each one leaves the next token of input in `nextToken`. So, whenever a parsing function begins, it assumes that `nextToken` has the code for the leftmost token of the input that has not yet been used in the parsing process.

The part of the language that the `expr` function parses consists of one or more terms, separated by either plus or minus operators. This is the language generated by the nonterminal <expr>. Therefore, first it calls the function that parses terms (`term`). Then it continues to call that function as long as it finds `ADD_OP` or `SUB_OP` tokens (which it passes over by calling `lex`). This recursive-descent function is simpler than most, because its associated rule has only one RHS. Furthermore, it does not include any code for syntax error detection or recovery, because there are no detectable errors associated with the grammar rule.

A recursive-descent parsing subprogram for a nonterminal whose rule has more than one RHS begins with code to determine which RHS is to be parsed. Each RHS is examined (at compiler construction time) to determine the set of terminal symbols that can appear at the beginning of sentences it can generate. By matching these sets against the next token of input, the parser can choose the correct RHS.

The parsing subprogram for <term> is similar to that for <expr>:

```
/* term
   Parses strings in the language generated by the rule:
   <term> -> <factor> {(* | /) <factor>)
   */
void term() {
  printf("Enter <term>\n");

/* Parse the first factor */
  factor();

/* As long as the next token is * or /, get the
   next token and parse the next factor */
  while (nextToken == MULT_OP || nextToken == DIV_OP) {
    lex();
    factor();
  }
  printf("Exit <term>\n");
}  /* End of function term */
```

The function for the <factor> nonterminal of our arithmetic expression grammar must choose between its two RHSs. It also includes error detection. In the function for <factor>, the reaction to detecting a syntax error is simply to call the `error` function. In a real parser, a diagnostic message must be

produced when an error is detected. Furthermore, parsers must recover from the error so that the parsing process can continue.

```
/* factor
   Parses strings in the language generated by the rule:
   <factor> -> id | int_constant | ( <expr )
   */
void factor() {
  printf("Enter <factor>\n");

/* Determine which RHS */
  if (nextToken == IDENT || nextToken == INT_LIT)

/* Get the next token */
    lex();

/* If the RHS is ( <expr>), call lex to pass over the
   left parenthesis, call expr, and check for the right
   parenthesis */
  else {
    if (nextToken == LEFT_PAREN) {
      lex();
      expr();
      if (nextToken == RIGHT_PAREN)
        lex();
      else
        error();
    }  /* End of if (nextToken == ... */

/* It was not an id, an integer literal, or a left
   parenthesis */
    else
      error();
  }  /* End of else */

  printf("Exit <factor>\n");;
}  /* End of function factor */
```

Following is the trace of the parse of the example expression `(sum + 47) / total`, using the parsing functions `expr`, `term`, and `factor`, and the function `lex` from Section 4.2. Note that the parse begins by calling `lex` and the start symbol routine, in this case, `expr`.

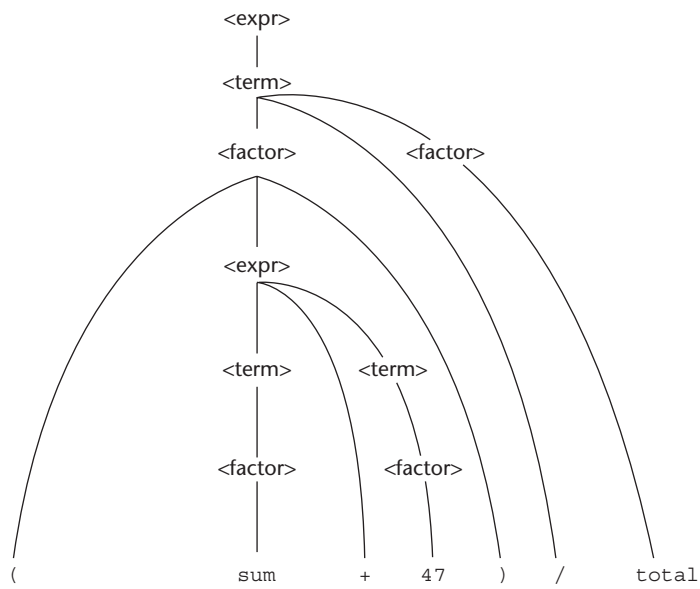
```
Next token is: 25 Next lexeme is (
Enter <expr>
Enter <term>
Enter <factor>
```

```
Next token is: 11 Next lexeme is sum
Enter <expr>
Enter <term>
Enter <factor>
Next token is: 21 Next lexeme is +
Exit <factor>
Exit <term>
Next token is: 10 Next lexeme is 47
Enter <term>
Enter <factor>
Next token is: 26 Next lexeme is )
Exit <factor>
Exit <term>
Exit <expr>
Next token is: 24 Next lexeme is /
Exit <factor>
Next token is: 11 Next lexeme is total
Enter <factor>
Next token is: -1 Next lexeme is EOF
Exit <factor>
Exit <term>
Exit <expr>
```

The parse tree traced by the parser for the preceding expression is shown in Figure 4.2.

Figure 4.2

Parse tree for (sum + 47) / total



One more example grammar rule and parsing function should help solidify the reader's understanding of recursive-descent parsing. Following is a grammatical description of the Java **if** statement:

<ifstmt> → **if** (<boolexpr>) <statement> [**else** <statement>]

The recursive-descent subprogram for this rule follows:

```
/* Function ifstmt
    Parses strings in the language generated by the rule:
    <ifstmt> -> if (<boolexpr>) <statement>
                   [else <statement>]
    */
void ifstmt() {
/* Be sure the first token is 'if' */
  if (nextToken != IF_CODE)
    error();
  else {
/* Call lex to get to the next token */
    lex();
/* Check for the left parenthesis */
    if (nextToken != LEFT_PAREN)
      error();
    else {
/* Call boolexpr to parse the Boolean expression */
      boolexpr();
/* Check for the right parenthesis */
      if (nextToken != RIGHT_PAREN)
        error();
      else {
/* Call statement to parse the then clause */
        statement();
/* If an else is next, parse the else clause */
        if (nextToken == ELSE_CODE) {
/* Call lex to get over the else */
          lex();
          statement();
        } /* end of if (nextToken == ELSE_CODE ... */
      } /* end of else of if (nextToken != RIGHT ... */
    } /* end of else of if (nextToken != LEFT ... */
  } /* end of else of if (nextToken != IF_CODE ... */
} /* end of ifstmt */
```

Notice that this function uses parser functions for statements and Boolean expressions, which are not given in this section.

The objective of these examples is to convince you that a recursive-descent parser can be easily written if an appropriate grammar is available for the

language. The characteristics of a grammar that allows a recursive-descent parser to be built are discussed in the following subsection.

4.4.2 The LL Grammar Class

Before choosing to use recursive descent as a parsing strategy for a compiler or other program analysis tool, one must consider the limitations of the approach, in terms of grammar restrictions. This section discusses these restrictions and their possible solutions.

One simple grammar characteristic that causes a catastrophic problem for LL parsers is left recursion. For example, consider the following rule:

$A \rightarrow A + B$

A recursive-descent parser subprogram for A immediately calls itself to parse the first symbol in its RHS. That activation of the A parser subprogram then immediately calls itself again, and again, and so forth. It is easy to see that this leads nowhere (except to a stack overflow).

The left recursion in the rule $A \rightarrow A + B$ is called **direct left recursion**, because it occurs in one rule. Direct left recursion can be eliminated from a grammar by the following process:

For each nonterminal, A,

1. Group the A-rules as $A \rightarrow A\alpha_1, | \ldots |A\alpha_m | \beta_1 | \beta_2 | \ldots | \beta_n$ where none of the β's begins with A
2. Replace the original A-rules with

$$A \rightarrow \beta_1 A' \mid \beta_2 A' \mid \ldots \mid \beta_n A'$$
$$A' \rightarrow \alpha_1 A' \mid \alpha_2 A' \mid \alpha_m A' \mid \varepsilon$$

Note that ε specifies the empty string. A rule that has ε as its RHS is called an *erasure rule*, because its use in a derivation effectively erases its LHS from the sentential form.

Consider the following example grammar and the application of the above process:

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid id$

For the E-rules, we have $\alpha_1 = +T$ and $\beta = T$, so we replace the E-rules with

$E \rightarrow T E'$

$E' \rightarrow + T E' \mid \varepsilon$

For the T-rules, we have $\alpha_1 = * F$ and $\beta = F$, so we replace the T-rules with

$T \rightarrow F T'$

$T' \rightarrow * F T' \mid \varepsilon$

Because there is no left recursion in the F-rules, they remain the same, so the complete replacement grammar is

$E \rightarrow T\ E'$
$E' \rightarrow +\ T\ E' \mid \varepsilon$
$T \rightarrow F\ T'$
$T' \rightarrow *\ F\ T' \mid \varepsilon$
$F \rightarrow (E) \mid id$

This grammar generates the same language as the original grammar but is not left recursive.

As was the case with the expression grammar written using EBNF in Section 4.4.1, this grammar does not specify left associativity of operators. However, it is relatively easy to design the code generation based on this grammar so that the addition and multiplication operators will have left associativity.

Indirect left recursion poses the same problem as direct left recursion. For example, suppose we have

$A \rightarrow B\ a\ A$
$B \rightarrow A\ b$

A recursive-descent parser for these rules would have the A subprogram immediately call the subprogram for B, which immediately calls the A subprogram. So, the problem is the same as for direct left recursion. The problem of left recursion is not confined to the recursive-descent approach to building top-down parsers. It is a problem for all top-down parsing algorithms. Fortunately, left recursion is not a problem for bottom-up parsing algorithms.

There is an algorithm to modify a given grammar to remove indirect left recursion (Aho et al., 2006), but it is not covered here. When writing a grammar for a programming language, one can usually avoid including left recursion, both direct and indirect.

Left recursion is not the only grammar trait that disallows top-down parsing. Another is whether the parser can always choose the correct RHS on the basis of the next token of input, using only the first token generated by the leftmost nonterminal in the current sentential form. There is a relatively simple test of a non–left recursive grammar that indicates whether this can be done, called the **pairwise disjointness test**. This test requires the ability to compute a set based on the RHSs of a given nonterminal symbol in a grammar. These sets, which are called FIRST, are defined as

$FIRST(\alpha) = \{a \mid \alpha \Rightarrow^* a\beta\}$ (If $\alpha \Rightarrow^* \varepsilon$, ε is in $FIRST(\alpha)$)

in which $\Rightarrow^*$ means 0 or more derivation steps.

An algorithm to compute FIRST for any mixed string α can be found in Aho et al. (2006). For our purposes, FIRST can usually be computed by inspection of the grammar.

The pairwise disjointness test is as follows:

For each nonterminal, A, in the grammar that has more than one RHS, for each pair of rules, $A \rightarrow \alpha_i$ and $A \rightarrow \alpha_j$, it must be true that

$$\text{FIRST}(\alpha_i) \cap \text{FIRST}(\alpha_j) = \phi$$

(The intersection of the two sets, $\text{FIRST}(\alpha_i)$ and $\text{FIRST}(\alpha_j)$, must be empty.)

In other words, if a nonterminal A has more than one RHS, the first terminal symbol that can be generated in a derivation for each of them must be unique to that RHS. Consider the following rules:

A → aB | bAb | Bb

B → cB | d

The FIRST sets for the RHSs of the A-rules are {a}, {b}, and {c, d}, which are clearly disjoint. Therefore, these rules pass the pairwise disjointness test. What this means, in terms of a recursive-descent parser, is that the code of the subprogram for parsing the nonterminal A can choose which RHS it is dealing with by seeing only the first terminal symbol of input (token) that is generated by the nonterminal. Now consider the rules

A → aB | BAb

B → aB | b

The FIRST sets for the RHSs in the A-rules are {a} and {a, b}, which are clearly not disjoint. So, these rules fail the pairwise disjointness test. In terms of the parser, the subprogram for A could not determine which RHS was being parsed by looking at the next symbol of input, because if it were an a, it could be either RHS. This issue is of course more complex if one or more of the RHSs begin with nonterminals.

In many cases, a grammar that fails the pairwise disjointness test can be modified so that it will pass the test. For example, consider the rule

<variable> → identifier | identifier [<expression>]

This states that a <variable> is either an identifier or an identifier followed by an expression in brackets (a subscript). These rules clearly do not pass the pairwise disjointness test, because both RHSs begin with the same terminal, identifier. This problem can be alleviated through a process called **left factoring**.

We now take an informal look at left factoring. Consider our rules for <variable>. Both RHSs begin with identifier. The parts that follow identifier in the two RHSs are ε (the empty string) and [<expression>]. The two rules can be replaced by the following two rules:

<variable> → identifier <new>

<new> → ε | [<expression>]

It is not difficult to see that together, these two rules generate the same language as the two rules with which we began. However, these two pass the pairwise disjointness test.

If the grammar is being used as the basis for a recursive-descent parser, an alternative to left factoring is available. With an EBNF extension, the problem disappears in a way that is very similar to the left factoring solution. Consider the original rules above for <variable>. The subscript can be made optional by placing it in square brackets, as in

<variable> → identifier [[<expression]]

In this rule, the outer brackets are metasymbols that indicate that what is inside is optional. The inner brackets are terminal symbols of the programming language being described. The point is that we replaced two rules with a single rule that generates the same language but passes the pairwise disjointness test.

A formal algorithm for left factoring can be found in Aho et al. (2006). Left factoring cannot solve all pairwise disjointness problems of grammars. In some cases, rules must be rewritten in other ways to eliminate the problem.

4.5 Bottom-Up Parsing

This section introduces the general process of bottom-up parsing and includes a description of the LR parsing algorithm.

4.5.1 The Parsing Problem for Bottom-Up Parsers

Consider the following grammar for arithmetic expressions:

E → E + T | T
T → T * F | F
F → (E) | id

Notice that this grammar generates the same arithmetic expressions as the example in Section 4.4. The difference is that this grammar is left recursive, which is acceptable to bottom-up parsers. Also note that grammars for bottom-up parsers normally do not include metasymbols such as those used to specify extensions to BNF. The following rightmost derivation illustrates this grammar:

E => <u>E + T</u>
=> E + <u>T * F</u>
=> E + T * <u>id</u>
=> E + <u>F</u> * id
=> E + <u>id</u> * id
=> <u>T</u> + id * id
=> <u>F</u> + id * id
=> <u>id</u> + id * id

The underlined part of each sentential form in this derivation is the RHS that is rewritten as its corresponding LHS to get the previous sentential form. The process of bottom-up parsing produces the reverse of a rightmost derivation. So, in the example derivation, a bottom-up parser starts with the last sentential form (the input sentence) and produces the sequence of sentential forms from there until all that remains is the start symbol, which in this grammar is E. In each step, the task of the bottom-up parser is to find the specific RHS, the handle, in the sentential form that must be rewritten to get the next (previous) sentential form. As mentioned earlier, a right sentential form may include more than one RHS. For example, the right sentential form

E + T * id

includes three RHSs, E + T, T, and id. Only one of these is the handle. For example, if the RHS E + T were chosen to be rewritten in this sentential form, the resulting sentential form would be E * id, but E * id is not a legal right sentential form for the given grammar.

The handle of a right sentential form is unique. The task of a bottom-up parser is to find the handle of any given right sentential form that can be generated by its associated grammar. Formally, handle is defined as follows:

> Definition: β is the **handle** of the right sentential form $\gamma = \alpha\beta w$ if and only if $S \Rightarrow^*_{rm} \alpha A w \Rightarrow_{rm} \alpha\beta w$

In this definition, $\Rightarrow_{rm}$ specifies a rightmost derivation step, and $\Rightarrow^*_{rm}$ specifies zero or more rightmost derivation steps. Although the definition of a handle is mathematically concise, it provides little help in finding the handle of a given right sentential form. In the following, we provide the definitions of several substrings of sentential forms that are related to handles. The purpose of these is to provide some intuition about handles.

> Definition: β is a **phrase** of the right sentential form γ if and only if $S \Rightarrow^* \gamma = \alpha_1 A \alpha_2 \Rightarrow + \alpha_1\beta\alpha_2$

In this definition, $\Rightarrow+$ means one or more derivation steps.

> Definition: β is a **simple phrase** of the right sentential form γ if and only if $S \Rightarrow^* \gamma = \alpha_1 A \alpha_2 \Rightarrow \alpha_1\beta\alpha_2$

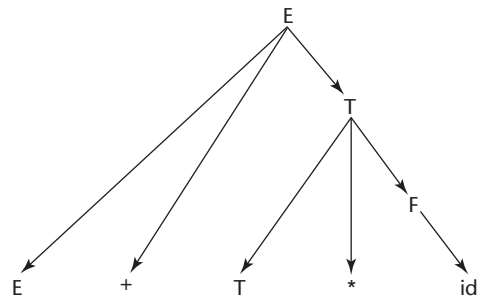
If these two definitions are compared carefully, it is clear that they differ only in the last derivation specification. The definition of phrase uses one or more steps, while the definition of simple phrase uses exactly one step.

The definitions of phrase and simple phrase may appear to have the same lack of practical value as that of a handle, but that is not true. Consider what a phrase is relative to a parse tree. It is the string of all of the leaves of the partial parse tree that is rooted at one particular internal node of the whole parse tree. A simple phrase is just a phrase that takes a single derivation step from its

root nonterminal node. In terms of a parse tree, a phrase can be derived from a single nonterminal in one or more tree levels, but a simple phrase can be derived in just a single tree level. Consider the parse tree shown in Figure 4.3.

Figure 4.3

A parse tree for E + T * id



The leaves of the parse tree in Figure 4.3 comprise the sentential form E + T * id. Because there are three internal nodes, there are three phrases. Each internal node is the root of a subtree, whose leaves are a phrase. The root node of the whole parse tree, E, generates all of the resulting sentential form, E + T * id, which is a phrase. The internal node, T, generates the leaves T * id, which is another phrase. Finally, the internal node, F, generates id, which is also a phrase. So, the phrases of the sentential form E + T * id are E + T * id, T * id, and id. Notice that phrases are not necessarily RHSs in the underlying grammar.

The simple phrases are a subset of the phrases. In the previous example, the only simple phrase is id. A simple phrase is always an RHS in the grammar.

The reason for discussing phrases and simple phrases is this: The handle of any rightmost sentential form is its leftmost simple phrase. So now we have a highly intuitive way to find the handle of any right sentential form, assuming we have the grammar and can draw a parse tree. This approach to finding handles is of course not practical for a parser. (If you already have a parse tree, why do you need a parser?) Its only purpose is to provide the reader with some intuitive feel for what a handle is, relative to a parse tree, which is easier than trying to think about handles in terms of sentential forms.

We can now consider bottom-up parsing in terms of parse trees, although the purpose of a parser is to produce a parse tree. Given the parse tree for an entire sentence, you easily can find the handle, which is the first thing to rewrite in the sentence to get the previous sentential form. Then the handle can be pruned from the parse tree and the process repeated. Continuing to the root of the parse tree, the entire rightmost derivation can be constructed.

4.5.2 Shift-Reduce Algorithms

Bottom-up parsers are often called **shift-reduce algorithms**, because shift and reduce are the two most common actions they specify. An integral part of every bottom-up parser is a stack. As with other parsers, the input to a

bottom-up parser is the stream of tokens of a program and the output is a sequence of grammar rules. The shift action moves the next input token onto the parser's stack. A reduce action replaces an RHS (the handle) on top of the parser's stack by its corresponding LHS. Every parser for a programming language is a **pushdown automaton** (**PDA**), because a PDA is a recognizer for a context-free language. You need not be intimate with PDAs to understand how a bottom-up parser works, although it helps. A PDA is a very simple mathematical machine that scans strings of symbols from left to right. A PDA is so named because it uses a pushdown stack as its memory. PDAs can be used as recognizers for context-free languages. Given a string of symbols over the alphabet of a context-free language, a PDA that is designed for the purpose can determine whether the string is or is not a sentence in the language. In the process, the PDA can produce the information needed to construct a parse tree for the sentence.

With a PDA, the input string is examined, one symbol at a time, left to right. The input is treated very much as if it were stored in another stack, because the PDA never sees more than the leftmost symbol of the input.

Note that a recursive-descent parser is also a PDA. In that case, the stack is that of the run-time system, which records subprogram calls (among other things), which correspond to the nonterminals of the grammar.

4.5.3 LR Parsers

Many different bottom-up parsing algorithms have been devised. Most of them are variations of a process called LR. LR parsers use a relatively small program and a parsing table that is built for a specific programming language. The original LR algorithm was designed by Donald Knuth (Knuth, 1965). This algorithm, which is sometimes called **canonical LR**, was not used in the years immediately following its publication because producing the required parsing table required large amounts of computer time and memory. Subsequently, several variations on the canonical LR table construction process were developed (DeRemer, 1971; DeRemer and Pennello, 1982). These are characterized by two properties: (1) They require far less computer resources to produce the required parsing table than the canonical LR algorithm, and (2) they work on smaller classes of grammars than the canonical LR algorithm.

There are three advantages to LR parsers:

1. They can be built for all programming languages.
2. They can detect syntax errors as soon as it is possible in a left-to-right scan.
3. The LR class of grammars is a proper superset of the class parsable by LL parsers (for example, many left recursive grammars are LR, but none are LL).

The only disadvantage of LR parsing is that it is difficult to produce by hand the parsing table for a given grammar for a complete programming language.

This is not a serious disadvantage, however, for there are several programs available that take a grammar as input and produce the parsing table, as discussed later in this section.

Prior to the appearance of the LR parsing algorithm, there were a number of parsing algorithms that found handles of right sentential forms by looking both to the left and to the right of the substring of the sentential form that was suspected of being the handle. Knuth's insight was that one could effectively look to the left of the suspected handle all the way to the bottom of the parse stack to determine whether it was the handle. But all of the information in the parse stack that was relevant to the parsing process could be represented by a single state, which could be stored on the top of the stack. In other words, Knuth discovered that regardless of the length of the input string, the length of the sentential form, or the depth of the parse stack, there were only a relatively small number of different situations, as far as the parsing process is concerned. Each situation could be represented by a state and stored in the parse stack, one state symbol for each grammar symbol on the stack. At the top of the stack would always be a state symbol, which represented the relevant information from the entire history of the parse, up to the current time. We will use subscripted uppercase S's to represent the parser states.

Figure 4.4 shows the structure of an LR parser. The contents of the parse stack for an LR parser have the following form:

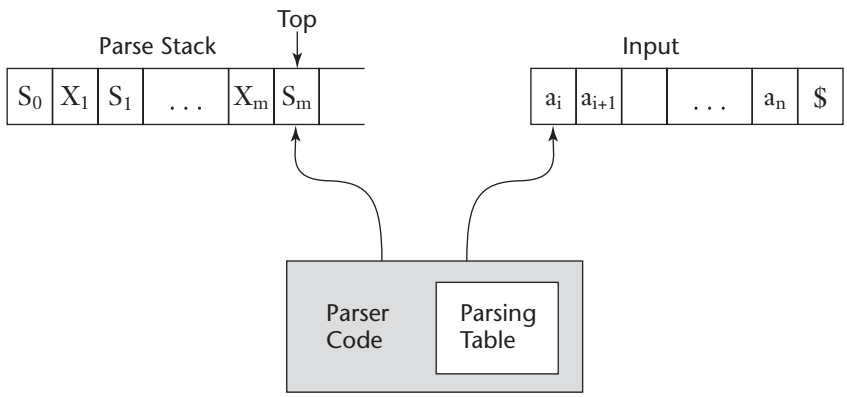
$$S_0X_1S_1X_2 \ \ldots \ X_mS_m \text{ (top)}$$

where the S's are state symbols and the X's are grammar symbols. An LR parser configuration is a pair of strings (stack, input), with the detailed form

$$(S_0X_1S_1X_2S_2 \ \ldots \ X_mS_m, a_ia_{i+1} \ \ldots \ a_n\$)$$

Figure 4.4

The structure of an LR parser



Notice that the input string has a dollar sign at its right end. This sign is put there during initialization of the parser. It is used for normal termination of the parser. Using this parser configuration, we can formally define the LR parser process, which is based on the parsing table.

An LR parsing table has two parts, named ACTION and GOTO. The ACTION part of the table specifies most of what the parser does. It has state symbols as its row labels and the terminal symbols of the grammar as its column labels. Given a current parser state, which is represented by the state symbol on top of the parse stack, and the next symbol (token) of input, the parse table specifies what the parser should do. The two primary parser actions are shift and reduce. Either the parser shifts the next input symbol onto the parse stack or it already has the handle on top of the stack, which it reduces to the LHS of the rule whose RHS is the same as the handle. Two other actions are possible: accept, which means the parser has successfully completed the parse of the input, and error, which means the parser has detected a syntax error.

The rows of the GOTO part of the LR parsing table have state symbols as labels. This part of the table has nonterminals as column labels. The values in the GOTO part of the table indicate which state symbol should be pushed onto the parse stack after a reduction has been completed, which means the handle has been removed from the parse stack and the new nonterminal has been pushed onto the parse stack. The specific symbol is found at the row whose label is the state symbol on top of the parse stack after the handle and its associated state symbols have been removed. The column of the GOTO table that is used is the one with the label that is the LHS of the rule used in the reduction.

Consider the traditional grammar for arithmetic expressions that follows:

1. $E \rightarrow E + T$
2. $E \rightarrow T$
3. $T \rightarrow T * F$
4. $T \rightarrow F$
5. $F \rightarrow (E)$
6. $F \rightarrow id$

The rules of this grammar are numbered to provide a simple way to reference them in a parsing table.

Figure 4.5 shows the LR parsing table for this grammar. Abbreviations are used for the actions: R for reduce and S for shift. R4 means reduce using rule 4; S6 means shift the next symbol of input onto the stack and push state S6 onto the stack. Empty positions in the ACTION table indicate syntax errors. In a complete parser, these could have calls to error-handling routines.

LR parsing tables can easily be constructed using a software tool, such as yacc (Johnson, 1975), which takes the grammar as input. Although LR parsing tables can be produced by hand, for a grammar of a real programming language, the task would be lengthy, tedious, and error prone. For real compilers, LR parsing tables are always generated with software tools.

The initial configuration of an LR parser is

$(S_0, a_1 \ \ldots \ a_n\$)$

Figure 4.5

The LR parsing table for an arithmetic expression grammar

	Action						Goto		
State	id	+	*	(	)	$	E	T	F
0	S5			S4			1	2	3
1		S6				accept			
2		R2	S7		R2	R2			
3		R4	R4		R4	R4			
4	S5			S4			8	2	3
5		R6	R6		R6	R6			
6	S5			S4				9	3
7	S5			S4					10
8		S6			S11				
9		R1	S7		R1	R1			
10		R3	R3		R3	R3			
11		R5	R5		R5	R5			

The parser actions are informally defined as follows:

1. The Shift process is simple: The next symbol of input is pushed onto the stack, along with the state symbol that is part of the Shift specification in the ACTION table.
2. For a Reduce action, the handle must be removed from the stack. Because for every grammar symbol on the stack there is a state symbol, the number of symbols removed from the stack is twice the number of symbols in the handle. After removing the handle and its associated state symbols, the LHS of the rule is pushed onto the stack. Finally, the GOTO table is used, with the row label being the symbol that was exposed when the handle and its state symbols were removed from the stack, and the column label being the nonterminal that is the LHS of the rule used in the reduction.
3. When the action is Accept, the parse is complete and no errors were found.
4. When the action is Error, the parser calls an error-handling routine.

Although there are many parsing algorithms based on the LR concept, they differ only in the construction of the parsing table. All LR parsers use this same parsing algorithm.

Perhaps the best way to become familiar with the LR parsing process is through an example. Initially, the parse stack has the single symbol 0, which

represents state 0 of the parser. The input contains the input string with an end marker, in this case a dollar sign, attached to its right end. At each step, the parser actions are dictated by the top (rightmost in Figure 4.4) symbol of the parse stack and the next (leftmost in Figure 4.4) token of input. The correct action is chosen from the corresponding cell of the ACTION part of the parse table. The GOTO part of the parse table is used after a reduction action. Recall that GOTO is used to determine which state symbol is placed on the parse stack after a reduction.

Following is a trace of a parse of the string id + id * id, using the LR parsing algorithm and the parsing table shown in Figure 4.5.

Stack	*Input*	*Action*
0	id + id * id $	Shift 5
0id5	+ id * id $	Reduce 6 (use GOTO[0, F])
0F3	+ id * id $	Reduce 4 (use GOTO[0, T])
0T2	+ id * id $	Reduce 2 (use GOTO[0, E])
0E1	+ id * id $	Shift 6
0E1+6	id * id $	Shift 5
0E1+6id5	* id $	Reduce 6 (use GOTO[6, F])
0E1+6F3	* id $	Reduce 4 (use GOTO[6, T])
0E1+6T9	* id $	Shift 7
0E1+6T9*7	id $	Shift 5
0E1+6T9*7id5	$	Reduce 6 (use GOTO[7, F])
0E1+6T9*7F10	$	Reduce 3 (use GOTO[6, T])
0E1+6T9	$	Reduce 1 (use GOTO[0, E])
0E1	$	Accept

The algorithms to generate LR parsing tables from given grammars, which are described in Aho et al. (2006), are not overly complex but are beyond the scope of a book on programming languages. As stated previously, there are a number of different software systems available to generate LR parsing tables.

SUMMARY

Syntax analysis is a common part of language implementation, regardless of the implementation approach used. Syntax analysis is normally based on a formal syntax description of the language being implemented. A context-free grammar, which is also called BNF, is the most common approach for describing syntax. The task of syntax analysis is usually divided into two parts: lexical analysis and syntax analysis. There are several reasons for separating lexical analysis—namely, simplicity, efficiency, and portability.

A lexical analyzer is a pattern matcher that isolates the small-scale parts of a program, which are called lexemes. Lexemes occur in categories, such as integer literals and names. These categories are called tokens. Each token is assigned a numeric code, which along with the lexeme is what the lexical analyzer produces. There are three distinct approaches to constructing a lexical analyzer: using a software tool to generate a table for a table-driven analyzer, building such a table by hand, and writing code to implement a state diagram description of the tokens of the language being implemented. The state diagram for tokens can be reasonably small if character classes are used for transitions, rather than having transitions for every possible character from every state node. Also, the state diagram can be simplified by using a table lookup to recognize reserved words.

Syntax analyzers have two goals: to detect syntax errors in a given program and to produce a parse tree, or possibly only the information required to build such a tree, for a given program. Syntax analyzers are either top-down, meaning they construct leftmost derivations and a parse tree in top-down order, or bottom-up, in which case they construct the reverse of a rightmost derivation and a parse tree in bottom-up order. Parsers that work for all unambiguous grammars have complexity $O(n^3)$. However, parsers used for implementing syntax analyzers for programming languages work on subclasses of unambiguous grammars and have complexity $O(n)$.

A recursive-descent parser is an LL parser that is implemented by writing code directly from the grammar of the source language. EBNF is ideal as the basis for recursive-descent parsers. A recursive-descent parser has a subprogram for each nonterminal in the grammar. The code for a given grammar rule is simple if the rule has a single RHS. The RHS is examined left to right. For each nonterminal, the code calls the associated subprogram for that nonterminal, which parses whatever the nonterminal generates. For each terminal, the code compares the terminal with the next token of input. If they match, the code simply calls the lexical analyzer to get the next token. If they do not, the subprogram reports a syntax error. If a rule has more than one RHS, the subprogram must first determine which RHS it should parse. It must be possible to make this determination on the basis of the next token of input.

Two distinct grammar characteristics prevent the construction of a recursive-descent parser based on the grammar. One of these is left recursion. The process of eliminating direct left recursion from a grammar is relatively simple. Although we do not cover it, an algorithm exists to remove both direct and indirect left recursion from a grammar. The other problem is detected with the pairwise disjointness test, which tests whether a parsing subprogram can determine which RHS is being parsed on the basis of the next token of input. Some grammars that fail the pairwise disjointness test often can be modified to pass it, using left factoring.

The parsing problem for bottom-up parsers is to find the substring of the current sentential form that must be reduced to its associated LHS to get the next (previous) sentential form in the rightmost derivation. This substring is called the handle of the sentential form. A parse tree can provide an intuitive

basis for recognizing a handle. A bottom-up parser is a shift-reduce algorithm, because in most cases it either shifts the next lexeme of input onto the parse stack or reduces the handle that is on top of the stack.

The LR family of shift-reduce parsers is the most commonly used bottom-up parsing approach for programming languages, because parsers in this family have several advantages over alternatives. An LR parser uses a parse stack, which contains grammar symbols and state symbols to maintain the state of the parser. The top symbol on the parse stack is always a state symbol that represents all of the information in the parse stack that is relevant to the parsing process. LR parsers use two parsing tables: ACTION and GOTO. The ACTION part specifies what the parser should do, given the state symbol on top of the parse stack and the next token of input. The GOTO table is used to determine which state symbol should be placed on the parse stack after a reduction has been done.

REVIEW QUESTIONS

1. What are three reasons why syntax analyzers are based on grammars?
2. Explain the three reasons why lexical analysis is separated from syntax analysis.
3. Define *lexeme* and *token*.
4. What are the primary tasks of a lexical analyzer?
5. Describe briefly the three approaches to building a lexical analyzer.
6. What is a state transition diagram?
7. Why are character classes used, rather than individual characters, for the letter and digit transitions of a state diagram for a lexical analyzer?
8. What are the two distinct goals of syntax analysis?
9. Describe the differences between top-down and bottom-up parsers.
10. Describe the parsing problem for a top-down parser.
11. Describe the parsing problem for a bottom-up parser.
12. Explain why compilers use parsing algorithms that work on only a subset of all grammars.
13. Why are named constants used, rather than numbers, for token codes?
14. Describe how a recursive-descent parsing subprogram is written for a rule with a single RHS.
15. Explain the two grammar characteristics that prohibit them from being used as the basis for a top-down parser.
16. What is the FIRST set for a given grammar and sentential form?
17. Describe the pairwise disjointness test.
18. What is left factoring?

19. What is a phrase of a sentential form?
20. What is a simple phrase of a sentential form?
21. What is the handle of a sentential form?
22. What is the mathematical machine on which both top-down and bottom-up parsers are based?
23. Describe three advantages of LR parsers.
24. What was Knuth's insight in developing the LR parsing technique?
25. Describe the purpose of the ACTION table of an LR parser.
26. Describe the purpose of the GOTO table of an LR parser.
27. Is left recursion a problem for LR parsers?

PROBLEM SET

1. Perform the pairwise disjointness test for the following grammar rules.
   a. $A \rightarrow aB \mid b \mid cBB$
   b. $B \rightarrow aB \mid bA \mid aBb$
   c. $C \rightarrow aaA \mid b \mid caB$
2. Perform the pairwise disjointness test for the following grammar rules.
   a. $S \rightarrow aSb \mid bAA$
   b. $A \rightarrow b\{aB\} \mid a$
   c. $B \rightarrow aB \mid a$
3. Show a trace of the recursive descent parser given in Section 4.4.1 for the string `a + b * c`.
4. Show a trace of the recursive descent parser given in Section 4.4.1 for the string `a * (b + c)`.
5. Given the following grammar and the right sentential form, draw a parse tree and show the phrases and simple phrases, as well as the handle.

   $S \rightarrow aAb \mid bBA \quad A \rightarrow ab \mid aAB \quad B \rightarrow aB \mid b$

   a. aaAbb
   b. bBab
   c. aaAbBb
6. Given the following grammar and the right sentential form, draw a parse tree and show the phrases and simple phrases, as well as the handle.

   $S \rightarrow AbB \mid bAc \quad A \rightarrow Ab \mid aBB \quad B \rightarrow Ac \mid cBb \mid c$

   a. aAcccbbc
   b. AbcaBccb
   c. baBcBbbc

7. Show a complete parse, including the parse stack contents, input string, and action for the string id * (id + id), using the grammar and parse table in Section 4.5.3.
8. Show a complete parse, including the parse stack contents, input string, and action for the string (id + id) * id, using the grammar and parse table in Section 4.5.3.
9. Write an EBNF rule that describes the **`while`** statement of Java or C++. Write the recursive-descent subprogram in Java or C++ for this rule.
10. Write an EBNF rule that describes the **`for`** statement of Java or C++. Write the recursive-descent subprogram in Java or C++ for this rule.
11. Get the algorithm to remove the indirect left recursion from a grammar from Aho et al. (2006). Use this algorithm to remove all left recursion from the following grammar:
    $S \rightarrow Aa \mid Bb$ $A \rightarrow Aa \mid Abc \mid c \mid Sb$ $B \rightarrow bb$

PROGRAMMING EXERCISES

1. Design a state diagram to recognize one form of the comments of the C-based programming languages, those that begin with `/*` and end with `*/`.
2. Design a state diagram to recognize the floating-point literals of your favorite programming language.
3. Write and test the code to implement the state diagram of Problem 1.
4. Write and test the code to implement the state diagram of Problem 2.
5. Modify the lexical analyzer given in Section 4.2 to recognize the following list of reserved words and return their respective token codes: **`for`** (`FOR_CODE`, `30`), **`if`** (`IF_CODE`, `31`), **`else`** (`ELSE_CODE`, `32`), **`while`** (`WHILE_CODE`, `33`), **`do`** (`DO_CODE`, `34`), **`int`** (`INT_CODE`, `35`), **`float`** (`FLOAT_CODE`, `36`), **`switch`** (`SWITCH_CODE`, `37`).
6. Convert the lexical analyzer (which is written in C) given in Section 4.2 to Java.
7. Convert the recursive descent parser routines for <expr>, <term>, and <factor> given in Section 4.4.1 to Java.
8. For those rules that pass the test in Problem 1, write a recursive-descent parsing subprogram that parses the language generated by the rules. Assume you have a lexical analyzer named `lex` and an error-handling subprogram named `error`, which is called whenever a syntax error is detected.
9. For those rules that pass the test in Problem 2, write a recursive-descent parsing subprogram that parses the language generated by the rules. Assume you have a lexical analyzer named `lex` and an error-handling subprogram named `error`, which is called whenever a syntax error is detected.
10. Implement and test the LR parsing algorithm given in Section 4.5.3.

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5

Names, Bindings, and Scopes

5.1 Introduction
5.2 Names
5.3 Variables
5.4 The Concept of Binding
5.5 Scope
5.6 Scope and Lifetime
5.7 Referencing Environments
5.8 Named Constants

This chapter introduces the fundamental semantic issues of variables. It begins by describing the nature of names and special words in programming languages. The attributes of variables, including type, address, and value, are then discussed, including the issue of aliases. The important concepts of binding and binding times are introduced next, including the different possible binding times for variable attributes and how they define four different categories of variables. Following that, two very different scoping rules for names, static and dynamic, are described, along with the concept of a referencing environment of a statement. Finally, named constants and variable initialization are discussed.

5.1 Introduction

Imperative programming languages are, to varying degrees, abstractions of the underlying von Neumann computer architecture. The architecture's two primary components are its memory, which stores both instructions and data, and its processor, which provides operations for modifying the contents of the memory. The abstractions in a language for the memory cells of the machine are variables. In some cases, the characteristics of the abstractions are very close to the characteristics of the cells; an example of this is an integer variable, which is usually represented directly in one or more bytes of memory. In other cases, the abstractions are far removed from the organization of the hardware memory, as with a three-dimensional array, which requires a software mapping function to support the abstraction.

A variable can be characterized by a collection of properties, or attributes, the most important of which is type, a fundamental concept in programming languages. Designing the data types of a language requires that a variety of issues be considered. (Data types are discussed in Chapter 6.) Among the most important of these issues are the scope and lifetime of variables.

Functional programming languages allow expressions to be named. These named expressions appear like assignments to variable names in imperative languages, but are fundamentally different in that they cannot be changed. So, they are like the named constants of the imperative languages. Pure functional languages do not have variables that are like those of the imperative languages. However, many functional languages do include such variables.

In the remainder of this book, families of languages will often be referred to as if they were single languages. For example, Fortran will mean all of the versions of Fortran. This is also the case for Ada. Likewise, a reference to C will mean the original version of C, as well as C89 and C99. When a specific version of a language is named, it is because it is different from the other family members within the topic being discussed. If we add a plus sign (+) to the name of a version of a language, we mean all versions of the language beginning with the one named. For example, Fortran 95+ means all versions of Fortran beginning with Fortran 95. The phrase **C-based languages** will be used to refer to C, Objective-C, C++, Java, and C#.[1]

1. We were tempted to include the scripting languages JavaScript and PHP as C-based languages, but decided they were just a bit too different from their ancestors.

5.2 Names

Before beginning our discussion of variables, the design of one of the fundamental attributes of variables, names, must be covered. Names are also associated with subprograms, formal parameters, and other program constructs. The term *identifier* is often used interchangeably with *name*.

5.2.1 Design Issues

The following are the primary design issues for names:

- Are names case sensitive?
- Are the special words of the language reserved words or keywords?

These issues are discussed in the following two subsections, which also include examples of several design choices.

5.2.2 Name Forms

A **name** is a string of characters used to identify some entity in a program.

Fortran 95+ allows up to 31 characters in its names. C99 has no length limitation on its internal names, but only the first 63 are significant. External names in C99 (those defined outside functions, which must be handled by the linker) are restricted to 31 characters. Names in Java, C#, and Ada have no length limit, and all characters in them are significant. C++ does not specify a length limit on names, although implementors sometimes do.

history note

The earliest programming languages used single-character names. This notation was natural because early programming was primarily mathematical, and mathematicians have long used single-character names for unknowns in their formal notations.

Fortran I broke with the tradition of the single-character name, allowing up to six characters in its names.

Names in most programming languages have the same form: a letter followed by a string consisting of letters, digits, and underscore characters (_). Although the use of underscore characters to form names was widely used in the 1970s and 1980s, that practice is now far less popular. In the C-based languages, it has to a large extent been replaced by the so-called *camel notation*, in which all of the words of a multiple-word name except the first are capitalized, as in `myStack`.[2] Note that the use of underscores and mixed case in names is a programming style issue, not a language design issue.

All variable names in PHP must begin with a dollar sign. In Perl, the special character at the beginning of a variable's name, `$`, `@`, or `%`, specifies its type (although in a different sense than in other languages). In Ruby, special characters at the beginning of a variable's name, `@` or `@@`, indicate that the variable is an instance or a class variable, respectively.

2. It is called "camel" because words written in it often have embedded uppercase letters, which look like a camel's humps.

In many languages, notably the C-based languages, uppercase and lowercase letters in names are distinct; that is, names in these languages are **case sensitive**. For example, the following three names are distinct in C++: `rose`, `ROSE`, and `Rose`. To some people, this is a serious detriment to readability, because names that look very similar in fact denote different entities. In that sense, case sensitivity violates the design principle that language constructs that look similar should have similar meanings. But in languages whose variable names are case-sensitive, although `Rose` and `rose` look similar, there is no connection between them.

Obviously, not everyone agrees that case sensitivity is bad for names. In C, the problems of case sensitivity are avoided by the convention that variable names do not include uppercase letters. In Java and C#, however, the problem cannot be escaped because many of the predefined names include both uppercase and lowercase letters. For example, the Java method for converting a string to an integer value is `parseInt`, and spellings such as `ParseInt` and `parseint` are not recognized. This is a problem of writability rather than readability, because the need to remember specific case usage makes it more difficult to write correct programs. It is a kind of intolerance on the part of the language designer, which is enforced by the compiler.

5.2.3 Special Words

Special words in programming languages are used to make programs more readable by naming actions to be performed. They also are used to separate the syntactic parts of statements and programs. In most languages, special words are classified as reserved words, which means they cannot be redefined by programmers, but in some they are only keywords, which means they can be redefined.

A **keyword** is a word of a programming language that is special only in certain contexts. Fortran is the only remaining widely used language whose special words are keywords. In Fortran, the word `Integer`, when found at the beginning of a statement and followed by a name, is considered a keyword that indicates the statement is a declarative statement. However, if the word `Integer` is followed by the assignment operator, it is considered a variable name. These two uses are illustrated in the following:

```
Integer Apple
Integer = 4
```

Fortran compilers and people reading Fortran programs must distinguish between names and special words by context.

A **reserved word** is a special word of a programming language that cannot be used as a name. As a language design choice, reserved words are better than keywords because the ability to redefine keywords can be confusing. For example, in Fortran, one could have the following statements:

```
Integer Real
Real Integer
```

These statements declare the program variable `Real` to be of `Integer` type and the variable `Integer` to be of `Real` type.[3] In addition to the strange appearance of these declaration statements, the appearance of `Real` and `Integer` as variable names elsewhere in the program could be misleading to program readers.

There is one potential problem with reserved words: If the language includes a large number of reserved words, the user may have difficulty making up names that are not reserved. The best example of this is COBOL, which has 300 reserved words. Unfortunately, some of the most commonly chosen names by programmers are in the list of reserved words—for example, `LENGTH`, `BOTTOM`, `DESTINATION`, and `COUNT`.

In program code examples in this book, reserved words are presented in boldface.

In most languages, names that are defined in other program units, such as Java packages and C and C++ libraries, can be made visible to a program. These names are predefined, but visible only if explicitly imported. Once imported, they cannot be redefined.

5.3 Variables

A program variable is an abstraction of a computer memory cell or collection of cells. Programmers often think of variable names as names for memory locations, but there is much more to a variable than just a name.

The move from machine languages to assembly languages was largely one of replacing absolute numeric memory addresses for data with names, making programs far more readable and therefore easier to write and maintain. That step also provided an escape from the problem of manual absolute addressing, because the translator that converted the names to actual addresses also chose those addresses.

A variable can be characterized as a sextuple of attributes: (name, address, value, type, lifetime, and scope). Although this may seem too complicated for such an apparently simple concept, it provides the clearest way to explain the various aspects of variables.

Our discussion of variable attributes will lead to examinations of the important related concepts of aliases, binding, binding times, declarations, scoping rules, and referencing environments.

The name, address, type, and value attributes of variables are discussed in the following subsections. The lifetime and scope attributes are discussed in Sections 5.4.3 and 5.5, respectively.

3. Of course, any professional programmer who would write such code should not expect job security.

5.3.1 Name

Variable names are the most common names in programs. They were discussed at length in Section 5.2 in the general context of entity names in programs. Most variables have names. The ones that do not are discussed in Section 5.4.3.3.

5.3.2 Address

The **address** of a variable is the machine memory address with which it is associated. This association is not as simple as it may at first appear. In many languages, it is possible for the same variable to be associated with different addresses at different times in the program. For example, if a subprogram has a local variable that is allocated from the run-time stack when the subprogram is called, different calls may result in that variable having different addresses. These are in a sense different instantiations of the same variable.

The process of associating variables with addresses is further discussed in Section 5.4.3. An implementation model for subprograms and their activations is discussed in Chapter 10.

The address of a variable is sometimes called its **l-value**, because the address is what is required when the name of a variable appears in the left side of an assignment.

It is possible to have multiple variables that have the same address. When more than one variable name can be used to access the same memory location, the variables are called **aliases**. Aliasing is a hindrance to readability because it allows a variable to have its value changed by an assignment to a different variable. For example, if variables named `total` and `sum` are aliases, any change to the value of `total` also changes the value of `sum` and vice versa. A reader of the program must always remember that `total` and `sum` are different names for the same memory cell. Because there can be any number of aliases in a program, this may be very difficult in practice. Aliasing also makes program verification more difficult.

Aliases can be created in programs in several different ways. One common way in C and C++ is with their union types. Unions are discussed at length in Chapter 6.

Two pointer variables are aliases when they point to the same memory location. The same is true for reference variables. This kind of aliasing is simply a side effect of the nature of pointers and references. When a C++ pointer is set to point at a named variable, the pointer, when dereferenced, and the variable's name are aliases.

Aliasing can be created in many languages through subprogram parameters. These kinds of aliases are discussed in Chapter 9.

The time when a variable becomes associated with an address is very important to an understanding of programming languages. This subject is discussed in Section 5.4.3.

5.3.3 Type

The **type** of a variable determines the range of values the variable can store and the set of operations that are defined for values of the type. For example, the `int` type in Java specifies a value range of −2147483648 to 2147483647 and arithmetic operations for addition, subtraction, multiplication, division, and modulus.

5.3.4 Value

The **value** of a variable is the contents of the memory cell or cells associated with the variable. It is convenient to think of computer memory in terms of *abstract* cells, rather than physical cells. The physical cells, or individually addressable units, of most contemporary computer memories are byte-size, with a byte usually being eight bits in length. This size is too small for most program variables. An abstract memory cell has the size required by the variable with which it is associated. For example, although floating-point values may occupy four physical bytes in a particular implementation of a particular language, a floating-point value is thought of as occupying a single abstract memory cell. The value of each simple nonstructured type is considered to occupy a single abstract cell. Henceforth, the term *memory cell* means abstract memory cell.

A variable's value is sometimes called its **r-value** because it is what is required when the name of the variable appears in the right side of an assignment statement. To access the *r*-value, the *l*-value must be determined first. Such determinations are not always simple. For example, scoping rules can greatly complicate matters, as is discussed in Section 5.5.

5.4 The Concept of Binding

A **binding** is an association between an attribute and an entity, such as between a variable and its type or value, or between an operation and a symbol. The time at which a binding takes place is called **binding time**. Binding and binding times are prominent concepts in the semantics of programming languages. Bindings can take place at language design time, language implementation time, compile time, load time, link time, or run time. For example, the asterisk symbol (`*`) is usually bound to the multiplication operation at language design time. A data type, such as `int` in C, is bound to a range of possible values at language implementation time. At compile time, a variable in a Java program is bound to a particular data type. A variable may be bound to a storage cell when the program is loaded into memory. That same binding does not happen until run time in some cases, as with variables declared in Java methods. A call to a library subprogram is bound to the subprogram code at link time.

Consider the following Java assignment statement:

```
count = count + 5;
```

Some of the bindings and their binding times for the parts of this assignment statement are as follows:

- The type of `count` is bound at compile time.
- The set of possible values of `count` is bound at compiler design time.
- The meaning of the operator symbol + is bound at compile time, when the types of its operands have been determined.
- The internal representation of the literal `5` is bound at compiler design time.
- The value of `count` is bound at execution time with this statement.

A complete understanding of the binding times for the attributes of program entities is a prerequisite for understanding the semantics of a programming language. For example, to understand what a subprogram does, one must understand how the actual parameters in a call are bound to the formal parameters in its definition. To determine the current value of a variable, it may be necessary to know when the variable was bound to storage and with which statement or statements.

5.4.1 Binding of Attributes to Variables

A binding is **static** if it first occurs before run time begins and remains unchanged throughout program execution. If the binding first occurs during run time or can change in the course of program execution, it is called **dynamic**. The physical binding of a variable to a storage cell in a virtual memory environment is complex, because the page or segment of the address space in which the cell resides may be moved in and out of memory many times during program execution. In a sense, such variables are bound and unbound repeatedly. These bindings, however, are maintained by computer hardware, and the changes are invisible to the program and the user. Because they are not important to the discussion, we are not concerned with these hardware bindings. The essential point is to distinguish between static and dynamic bindings.

5.4.2 Type Bindings

Before a variable can be referenced in a program, it must be bound to a data type. The two important aspects of this binding are how the type is specified and when the binding takes place. Types can be specified statically through some form of explicit or implicit declaration.

5.4.2.1 Static Type Binding

An **explicit declaration** is a statement in a program that lists variable names and specifies that they are a particular type. An **implicit declaration** is a means of associating variables with types through default conventions, rather than declaration statements. In this case, the first appearance of a variable name in a program constitutes its implicit declaration. Both explicit and implicit declarations create static bindings to types.

Most widely used programming languages that use static type binding exclusively and were designed since the mid-1960s require explicit declarations of all variables (Perl, JavaScript, Ruby, and ML are some exceptions).

Implicit variable type binding is done by the language processor, either a compiler or an interpreter. There are several different bases for implicit variable type bindings. The simplest of these is naming conventions. In this case, the compiler or interpreter binds a variable to a type based on the syntactic form of the variable's name. For example, in Fortran, an identifier that appears in a program that is not explicitly declared is implicitly declared according to the following convention: If the identifier begins with one of the letters `I`, `J`, `K`, `L`, `M`, or `N`, or their lowercase versions, it is implicitly declared to be `Integer` type; otherwise, it is implicitly declared to be `Real` type.

Although they are a minor convenience to programmers, implicit declarations can be detrimental to reliability because they prevent the compilation process from detecting some typographical and programmer errors. In Fortran, variables that are accidentally left undeclared by the programmer are given default types and possibly unexpected attributes, which could cause subtle errors that are difficult to diagnose. Many Fortran programmers now include the declaration `Implicit none` in their programs. This declaration instructs the compiler to not implicitly declare any variables, thereby avoiding the potential problems of accidentally undeclared variables.

Some of the problems with implicit declarations can be avoided by requiring names for specific types to begin with particular special characters. For example, in Perl any name that begins with `$` is a scalar, which can store either a string or a numeric value. If a name begins with `@`, it is an array; if it begins with a `%`, it is a hash structure.[4] This creates different namespaces for different type variables. In this scenario, the names `@apple` and `%apple` are unrelated, because each is from a different namespace. Furthermore, a program reader always knows the type of a variable when reading its name. Note that this design is different from Fortran, because Fortran has both implicit and explicit declarations, so the type of a variable cannot necessarily be determined from the spelling of its name.

Another kind of implicit type declarations uses context. This is sometimes called **type inference**. In the simpler case, the context is the type of the value assigned to the variable in a declaration statement. For example, in C# a **`var`**

4. Both arrays and hashes are considered types—both can store any scalar in their elements.

declaration of a variable must include an initial value, whose type is made the type of the variable. Consider the following declarations:

```
var sum = 0;
var total = 0.0;
var name = "Fred";
```

The types of `sum`, `total`, and `name` are **`int`**, **`float`**, and **`string`**, respectively. Keep in mind that these are statically typed variables—their types are fixed for the lifetime of the unit in which they are declared.

Visual BASIC 9.0+, Go, and the functional languages ML, Haskell, OCaml, and F# also use type inferencing. In these functional languages, the context of the appearance of a name is the basis for determining its type. This kind of type inferencing is discussed in detail in Chapter 15.

5.4.2.2 Dynamic Type Binding

With dynamic type binding, the type of a variable is not specified by a declaration statement, nor can it be determined by the spelling of its name. Instead, the variable is bound to a type when it is assigned a value in an assignment statement. When the assignment statement is executed, the variable being assigned is bound to the type of the value of the expression on the right side of the assignment. Such an assignment may also bind the variable to an address and a memory cell, because different type values may require different amounts of storage. Any variable can be assigned any type value. Furthermore, a variable's type can change any number of times during program execution. It is important to realize that the type of a variable whose type is dynamically bound may be temporary.

When the type of a variable is statically bound, the name of the variable can be thought of being bound to a type, in the sense that the type and name of a variable are simultaneously bound. However, when a variable's type is dynamically bound, its name can be thought of as being only temporarily bound to a type. In reality, the names of variables are never bound to types. Names can be bound to variables and variables can be bound to types.

Languages in which types are dynamically bound are dramatically different from those in which types are statically bound. The primary advantage of dynamic binding of variables to types is that it provides more programming flexibility. For example, a program to process numeric data in a language that uses dynamic type binding can be written as a generic program, meaning that it is capable of dealing with data of any numeric type. Whatever type data is input will be acceptable, because the variables in which the data are to be stored can be bound to the correct type when the data is assigned to the variables after input. By contrast, because of static binding of types, one cannot write a C program to process data without knowing the type of that data.

Before the mid-1990s, the most commonly used programming languages used static type binding, the primary exceptions being some functional

languages such as LISP. However, since then there has been a significant shift to languages that use dynamic type binding. In Python, Ruby, JavaScript, and PHP, type binding is dynamic. For example, a JavaScript script may contain the following statement:

```
list = [10.2, 3.5];
```

Regardless of the previous type of the variable named `list`, this assignment causes it to become the name of a single-dimensioned array of length 2. If the statement

```
list = 47;
```

followed the previous example assignment, `list` would become the name of a scalar variable.

The option of dynamic type binding was introduced in C# 2010. A variable can be declared to use dynamic type binding by including the **`dynamic`** reserved word in its declaration, as in the following example:

```
dynamic any;
```

This is similar, although also different from declaring `any` to have type **`object`**. It is similar in that `any` can be assigned a value of any type, just as if it were declared **`object`**. It is different in that it is not useful for several different situations of interoperation; for example, with dynamically typed languages such as IronPython and IronRuby (.NET versions of Python and Ruby, respectively). However, it is useful when data of unknown type come into a program from an external source. Class members, properties, method parameters, method return values, and local variables can all be declared **`dynamic`**.

In pure object-oriented languages—for example, Ruby—all variables are references and do not have types; all data are objects and any variable can reference any object. Variables in such languages are, in a sense, all the same type—they are references. However, unlike the references in Java, which are restricted to referencing one specific type of value, variables in Ruby can reference any object.

There are two disadvantages to dynamic type binding. First, it causes programs to be less reliable, because the error-detection capability of the compiler is diminished relative to a compiler for a language with static type bindings. Dynamic type binding allows any variable to be assigned a value of any type. Incorrect types of right sides of assignments are not detected as errors; rather, the type of the left side is simply changed to the incorrect type. For example, suppose that in a particular JavaScript program, `i` and `x` are currently the names of scalar numeric variables and `y` is currently the name of an array. Furthermore, suppose that the program needs the assignment statement

```
i = x;
```

but because of a keying error, it has the assignment statement

```
i = y;
```

In JavaScript (or any other language that uses dynamic type binding), no error is detected in this statement by the interpreter—the type of the variable named `i` is simply changed to an array. But later uses of `i` will expect it to be a scalar, and correct results will be impossible. In a language with static type binding, such as Java, the compiler would detect the error in the assignment `i = y`, and the program would not get to execution.

Note that this disadvantage is also present to some extent in some languages that use static type binding, such as Fortran, C, and C++, which in many cases automatically convert the type of the RHS of an assignment to the type of the LHS.

Perhaps the greatest disadvantage of dynamic type binding is cost. The cost of implementing dynamic attribute binding is considerable, particularly in execution time. Type checking must be done at run time. Furthermore, every variable must have a run-time descriptor associated with it to maintain the current type. The storage used for the value of a variable must be of varying size, because different type values require different amounts of storage.

Finally, languages that have dynamic type binding for variables are usually implemented using pure interpreters rather than compilers. Computers do not have instructions whose operand types are not known at compile time. Therefore, a compiler cannot build machine instructions for the expression A + B if the types of A and B are not known at compile time. Pure interpretation typically takes at least 10 times as long as it does to execute equivalent machine code. Of course, if a language is implemented with a pure interpreter, the time to do dynamic type binding is hidden by the overall time of interpretation, so it seems less costly in that environment. On the other hand, languages with static type bindings are seldom implemented by pure interpretation, because programs in these languages can be easily translated to very efficient machine code versions.

5.4.3 Storage Bindings and Lifetime

The fundamental character of an imperative programming language is in large part determined by the design of the storage bindings for its variables. It is therefore important to have a clear understanding of these bindings.

The memory cell to which a variable is bound somehow must be taken from a pool of available memory. This process is called **allocation**. **Deallocation** is the process of placing a memory cell that has been unbound from a variable back into the pool of available memory.

The **lifetime** of a variable is the time during which the variable is bound to a specific memory location. So, the lifetime of a variable begins when it is bound to a specific cell and ends when it is unbound from that cell. To investigate storage bindings of variables, it is convenient to separate scalar

(unstructured) variables into four categories, according to their lifetimes. These categories are named static, stack-dynamic, explicit heap-dynamic, and implicit heap-dynamic. In the following sections, we discuss the definitions of these four categories, along with their purposes, advantages, and disadvantages.

5.4.3.1 Static Variables

Static variables are those that are bound to memory cells before program execution begins and remain bound to those same memory cells until program execution terminates. Statically bound variables have several valuable applications in programming. Globally accessible variables are often used throughout the execution of a program, thus making it necessary to have them bound to the same storage during that execution. Sometimes it is convenient to have subprograms that are history sensitive. Such a subprogram must have local static variables.

One advantage of static variables is efficiency. All addressing of static variables can be direct;[5] other kinds of variables often require indirect addressing, which is slower. Also, no run-time overhead is incurred for allocation and deallocation of static variables, although this time is often negligible.

One disadvantage of static binding to storage is reduced flexibility; in particular, a language that has only static variables cannot support recursive subprograms. Another disadvantage is that storage cannot be shared among variables. For example, suppose a program has two subprograms, both of which require large arrays. Furthermore, suppose that the two subprograms are never active at the same time. If the arrays are static, they cannot share the same storage for their arrays.

C and C++ allow programmers to include the **`static`** specifier on a variable definition in a function, making the variables it defines static. Note that when the **`static`** modifier appears in the declaration of a variable in a class definition in C++, Java, and C#, it also implies that the variable is a class variable, rather than an instance variable. Class variables are created statically some time before the class is first instantiated.

5.4.3.2 Stack-Dynamic Variables

Stack-dynamic variables are those whose storage bindings are created when their declaration statements are elaborated, but whose types are statically bound. **Elaboration** of such a declaration refers to the storage allocation and binding process indicated by the declaration, which takes place when execution reaches the code to which the declaration is attached. Therefore, elaboration occurs during run time. For example, the variable declarations that appear at the beginning of a Java method are elaborated when the method is called and the variables defined by those declarations are deallocated when the method completes its execution.

---

5. In some implementations, static variables are addressed through a base register, making accesses to them as costly as for stack-allocated variables.

As their name indicates, stack-dynamic variables are allocated from the run-time stack.

Some languages—for example, C++ and Java—allow variable declarations to occur anywhere a statement can appear. In some implementations of these languages, all of the stack-dynamic variables declared in a function or method (not including those declared in nested blocks) may be bound to storage at the beginning of execution of the function or method, even though the declarations of some of these variables do not appear at the beginning. In such cases, the variable becomes visible at the declaration, but the storage binding (and initialization, if it is specified in the declaration) occurs when the function or method begins execution. The fact that storage binding of a variable takes place before it becomes visible does not affect the semantics of the language.

The advantages of stack-dynamic variables are as follows: To be useful, at least in most cases, recursive subprograms require some form of dynamic local storage so that each active copy of the recursive subprogram has its own version of the local variables. These needs are conveniently met by stack-dynamic variables. Even in the absence of recursion, having stack-dynamic local storage for subprograms is not without merit, because all subprograms share the same memory space for their locals.

The disadvantages, relative to static variables, of stack-dynamic variables are the run-time overhead of allocation and deallocation, possibly slower accesses because indirect addressing is required, and the fact that subprograms cannot be history sensitive. The time required to allocate and deallocate stack-dynamic variables is not significant, because all of the stack-dynamic variables that are declared at the beginning of a subprogram are allocated and deallocated together, rather than by separate operations.

Fortran 95+ allows implementors to use stack-dynamic variables for locals, but includes the following statement:

`Save` list

This declaration allows the programmer to specify that some or all of the variables (those in the list) in the subprogram in which `Save` is placed will be static.

In Java, C++, and C#, variables defined in methods are by default stack dynamic. In Ada, all non-heap variables defined in subprograms are stack dynamic.

All attributes other than storage are statically bound to stack-dynamic scalar variables. That is not the case for some structured types, as is discussed in Chapter 6. Implementation of allocation/deallocation processes for stack-dynamic variables is discussed in Chapter 10.

5.4.3.3 Explicit Heap-Dynamic Variables

Explicit heap-dynamic variables are nameless (abstract) memory cells that are allocated and deallocated by explicit run-time instructions written by the programmer. These variables, which are allocated from and deallocated to the heap, can only be referenced through pointer or reference variables. The heap is a collection of storage cells whose organization is highly disorganized because of the

unpredictability of its use. The pointer or reference variable that is used to access an explicit heap-dynamic variable is created as any other scalar variable. An explicit heap-dynamic variable is created by either an operator (for example, in C++) or a call to a system subprogram provided for that purpose (for example, in C).

In C++, the allocation operator, named **new**, uses a type name as its operand. When executed, an explicit heap-dynamic variable of the operand type is created and its address is returned. Because an explicit heap-dynamic variable is bound to a type at compile time, that binding is static. However, such variables are bound to storage at the time they are created, which is during run time.

In addition to a subprogram or operator for creating explicit heap-dynamic variables, some languages include a subprogram or operator for explicitly destroying them.

As an example of explicit heap-dynamic variables, consider the following C++ code segment:

```
int *intnode;       // Create a pointer
intnode = new int;  // Create the heap-dynamic variable
...
delete intnode;     // Deallocate the heap-dynamic variable
                              // to which intnode points
```

In this example, an explicit heap-dynamic variable of **int** type is created by the **new** operator. This variable can then be referenced through the pointer, `intnode`. Later, the variable is deallocated by the **delete** operator. C++ requires the explicit deallocation operator **delete**, because it does not use implicit storage reclamation, such as garbage collection.

In Java, all data except the primitive scalars are objects. Java objects are explicitly heap dynamic and are accessed through reference variables. Java has no way of explicitly destroying a heap-dynamic variable; rather, implicit garbage collection is used. Garbage collection is discussed in Chapter 6.

C# has both explicit heap-dynamic and stack-dynamic objects, all of which are implicitly deallocated. In addition, C# supports C++-style pointers. Such pointers are used to reference heap, stack, and even static variables and objects. These pointers have the same dangers as those of C++, and the objects they reference on the heap are not implicitly deallocated. Pointers are included in C# to allow C# components to interoperate with C and C++ components. To discourage their use, and also to make clear to any program reader that the code uses pointers, the header of any method that defines a pointer must include the reserved word **unsafe**.

Explicit heap-dynamic variables are often used to construct dynamic structures, such as linked lists and trees, that need to grow and/or shrink during execution. Such structures can be built conveniently using pointers or references and explicit heap-dynamic variables.

The disadvantages of explicit heap-dynamic variables are the difficulty of using pointer and reference variables correctly, the cost of references to the

variables, and the complexity of the required storage management implementation. This is essentially the problem of heap management, which is costly and complicated. Implementation methods for explicit heap-dynamic variables are discussed at length in Chapter 6.

5.4.3.4 Implicit Heap-Dynamic Variables

Implicit heap-dynamic variables are bound to heap storage only when they are assigned values. In fact, all their attributes are bound every time they are assigned. For example, consider the following JavaScript assignment statement:

```
highs = [74, 84, 86, 90, 71];
```

Regardless of whether the variable named `highs` was previously used in the program or what it was used for, it is now an array of five numeric values.

The advantage of such variables is that they have the highest degree of flexibility, allowing highly generic code to be written. One disadvantage of implicit heap-dynamic variables is the run-time overhead of maintaining all the dynamic attributes, which could include array subscript types and ranges, among others. Another disadvantage is the loss of some error detection by the compiler, as discussed in Section 5.4.2.2. Examples of implicit heap-dynamic variables in JavaScript appear in Section 5.4.2.2.

5.5 Scope

One of the important factors in understanding variables is scope. The **scope** of a variable is the range of statements in which the variable is visible. A variable is **visible** in a statement if it can be referenced in that statement.

The scope rules of a language determine how a particular occurrence of a name is associated with a variable, or in the case of a functional language, how a name is associated with an expression. In particular, scope rules determine how references to variables declared outside the currently executing subprogram or block are associated with their declarations and thus their attributes (blocks are discussed in Section 5.5.2). A clear understanding of these rules for a language is therefore essential to the ability to write or read programs in that language.

A variable is **local** in a program unit or block if it is declared there. The nonlocal variables of a program unit or block are those that are visible within the program unit or block but are not declared there. Global variables are a special category of nonlocal variables. They are discussed in Section 5.5.4.

Scoping issues of classes, packages, and namespaces are discussed in Chapter 11.

5.5.1 Static Scope

ALGOL 60 introduced the method of binding names to nonlocal variables called **static scoping**,[6] which has been copied by many subsequent imperative languages and many nonimperative languages as well. Static scoping is so named because the scope of a variable can be statically determined—that is, prior to execution. This permits a human program reader (and a compiler) to determine the type of every variable in the program simply by examining its source code.

There are two categories of static-scoped languages: those in which subprograms can be nested, which creates nested static scopes, and those in which subprograms cannot be nested. In the latter category, static scopes are also created by subprograms but nested scopes are created only by nested class definitions and blocks.

Ada, JavaScript, Common LISP, Scheme, Fortran 2003+, F#, and Python allow nested subprograms, but the C-based languages do not.

Our discussion of static scoping in this section focuses on those languages that allow nested subprograms. Initially, we assume that *all* scopes are associated with program units and that all referenced nonlocal variables are declared in other program units.[7] In this chapter, it is assumed that scoping is the only method of accessing nonlocal variables in the languages under discussion. This is not true for all languages. It is not even true for all languages that use static scoping, but the assumption simplifies the discussion here.

When the reader of a program finds a reference to a variable, the attributes of the variable can be determined by finding the statement in which it is declared (either explicitly or implicitly). In static-scoped languages with nested subprograms, this process can be thought of in the following way. Suppose a reference is made to a variable `x` in subprogram `sub1`. The correct declaration is found by first searching the declarations of subprogram `sub1`. If no declaration is found for the variable there, the search continues in the declarations of the subprogram that declared subprogram `sub1`, which is called its **static parent**. If a declaration of `x` is not found there, the search continues to the next-larger enclosing unit (the unit that declared `sub1`'s parent), and so forth, until a declaration for `x` is found or the largest unit's declarations have been searched without success. In that case, an undeclared variable error is reported. The static parent of subprogram `sub1`, and its static parent, and so forth up to and including the largest enclosing subprogram, are called the **static ancestors** of `sub1`. Actual implementation techniques for static scoping, which are discussed in Chapter 10, are usually much more efficient than the process just described.

---

6. Static scoping is sometimes called *lexical scoping*.

7. Nonlocal variables not defined in other program units are discussed in Section 5.5.4.

Consider the following JavaScript function, `big`, in which the two functions `sub1` and `sub2` are nested:

```
function big() {
  function sub1() {
    var x = 7;
    sub2();
  }
  function sub2() {
    var y = x;
  }
  var x = 3;
  sub1();
}
```

Under static scoping, the reference to the variable `x` in `sub2` is to the `x` declared in the procedure `big`. This is true because the search for `x` begins in the procedure in which the reference occurs, `sub2`, but no declaration for `x` is found there. The search continues in the static parent of `sub2`, `big`, where the declaration of `x` is found. The `x` declared in `sub1` is ignored, because it is not in the static ancestry of `sub2`.

In some languages that use static scoping, regardless of whether nested subprograms are allowed, some variable declarations can be hidden from some other code segments. For example, consider again the JavaScript function `big`. The variable `x` is declared in both `big` and in `sub1`, which is nested inside `big`. Within `sub1`, every simple reference to `x` is to the local `x`. Therefore, the outer `x` is hidden from `sub1`.

In Ada, hidden variables from ancestor scopes can be accessed with selective references, which include the ancestor scope's name. For example, if our previous example function `big` were written in Ada, the `x` declared in `big` could be accessed in `sub1` by the reference `big.x`.

5.5.2 Blocks

Many languages allow new static scopes to be defined in the midst of executable code. This powerful concept, introduced in ALGOL 60, allows a section of code to have its own local variables whose scope is minimized. Such variables are typically stack dynamic, so their storage is allocated when the section is entered and deallocated when the section is exited. Such a section of code is called a **block**. Blocks provide the origin of the phrase **block-structured language**.

The C-based languages allow any compound statement (a statement sequence surrounded by matched braces) to have declarations and thereby define a new scope. Such compound statements are called blocks. For example, if `list` were an integer array, one could write

```
if (list[i] < list[j]) {
  int temp;
  temp = list[i];
  list[i] = list[j];
  list[j] = temp;
}
```

The scopes created by blocks, which could be nested in larger blocks, are treated exactly like those created by subprograms. References to variables in a block that are not declared there are connected to declarations by searching enclosing scopes (blocks or subprograms) in order of increasing size.

Consider the following skeletal C function:

```
void sub() {
  int count;
  ...
  while (...) {
    int count;
    count++;
    ...
  }
  ...
}
```

The reference to `count` in the **`while`** loop is to that loop's local `count`. In this case, the `count` of `sub` is hidden from the code inside the **`while`** loop. In general, a declaration for a variable effectively hides any declaration of a variable with the same name in a larger enclosing scope.[8] Note that this code is legal in C and C++ but illegal in Java and C#. The designers of Java and C# believed that the reuse of names in nested blocks was too error prone to be allowed.

Although JavaScript uses static scoping for its nested functions, nonfunction blocks cannot be defined in the language.

Most functional programming languages include a construct that is related to the blocks of the imperative languages, usually named *let*. These constructs have two parts, the first of which is to bind names to values, usually specified as expressions. The second part is an expression that uses the names defined in the first part. Programs in functional languages are comprised of expressions, rather than statements. Therefore, the final part of a `let` construct is an expression,

8. As discussed in Section 5.5.4, in C++, such hidden global variables can be accessed in the inner scope using the scope operator (`::`).

rather than a statement. In Scheme, a `let` construct is a call to the function `LET` with the following form:

```
(LET (
   (name1 expression1)
   . . .
   (namen expressionn))
   expression
)
```

The semantics of the call to `LET` is as follows: The first *n* expressions are evaluated and the values are assigned to the associated names. Then, the final expression is evaluated and the return value of `LET` is that value. This differs from a block in an imperative language in that the names are of values; they are not variables in the imperative sense. Once set, they cannot be changed. However, they are like local variables in a block in an imperative language in that their scope is local to the call to `LET`. Consider the following call to `LET`:

```
(LET (
  (top (+ a b))
  (bottom (- c d)))
  (/ top bottom)
)
```

This call computes and returns the value of the expression `(a + b) / (c - d)`.

In ML, the form of a **`let`** construct is as follows:

```
let
 val name1 = expression1
 ...
 val namen = expressionn
in
 expression
end;
```

Each **`val`** statement binds a name to an expression. As with Scheme, the names in the first part are like the named constants of imperative languages; once set, they cannot be changed.[9] Consider the following **`let`** construct:

```
let
  val top = a + b
  val bottom = c - d
in
  top / bottom
end;
```

9. In Chapter 15, we will see that they can be reset, but that the process actually creates a new name.

The general form of a **let** construct in F# is as follows:

let left_side = expression

The left_side of **let** can be a name or a tuple pattern (a sequence of names separated by commas).

The scope of a name defined with **let** inside a function definition is from the end of the defining expression to the end of the function. The scope of **let** can be limited by indenting the following code, which creates a new local scope. Although any indentation will work, the convention is that the indentation is four spaces. Consider the following code:

```
let n1 =
    let n2 = 7
    let n3 = n2 + 3
    n3;;
let n4 = n3 + n1;;
```

The scope of `n1` extends over all of the code. However, the scope of `n2` and `n3` ends when the indentation ends. So, the use of `n3` in the last **let** causes an error. The last line of the **let** `n1` scope is the value bound to `n1`; it could be any expression.

Chapter 15, includes more details of the let constructs in Scheme, ML, Haskell, and F#.

5.5.3 Declaration Order

In C89, as well as in some other languages, all data declarations in a function except those in nested blocks must appear at the beginning of the function. However, some languages—for example, C99, C++, Java, JavaScript, and C#—allow variable declarations to appear anywhere a statement can appear in a program unit. Declarations may create scopes that are not associated with compound statements or subprograms. For example, in C99, C++, and Java, the scope of all local variables is from their declarations to the ends of the blocks in which those declarations appear. However, in C#, the scope of any variable declared in a block is the whole block, regardless of the position of the declaration in the block, as long as it is not in a nested block. The same is true for methods. Note that C# still requires that all variables be declared before they are used. Therefore, although the scope of a variable extends from the declaration to the top of the block or subprogram in which that declaration appears, the variable still cannot be used above its declaration.

In JavaScript, local variables can be declared anywhere in a function, but the scope of such a variable is always the entire function. If used before its declaration in the function, such a variable has the value `undefined`.

The **for** statements of C++, Java, and C# allow variable definitions in their initialization expressions. In early versions of C++, the scope of such a variable was from its definition to the end of the smallest enclosing block. In the standard version, however, the scope is restricted to the **for** construct, as is the case with Java and C#. Consider the following skeletal method:

```
void fun() {
    . . .
    for (int count = 0; count < 10; count++){
        . . .
    }
    . . .
}
```

In later versions of C++, as well as in Java and C#, the scope of `count` is from the **for** statement to the end of its body.

5.5.4 Global Scope

Some languages, including C, C++, PHP, JavaScript, and Python, allow a program structure that is a sequence of function definitions, in which variable definitions can appear outside the functions. Definitions outside functions in a file create global variables, which potentially can be visible to those functions.

C and C++ have both declarations and definitions of global data. Declarations specify types and other attributes but do not cause allocation of storage. Definitions specify attributes *and* cause storage allocation. For a specific global name, a C program can have any number of compatible declarations, but only a single definition.

A declaration of a variable outside function definitions specifies that the variable is defined in a different file. A global variable in C is implicitly visible in all subsequent functions in the file, except those that include a declaration of a local variable with the same name. A global variable that is defined after a function can be made visible in the function by declaring it to be external, as in the following:

```
extern int sum;
```

In C99, definitions of global variables usually have initial values. Declarations of global variables never have initial values. If the declaration is outside function definitions, it need not include the **extern** qualifier.

This idea of declarations and definitions carries over to the functions of C and C++, where prototypes declare names and interfaces of functions but do not provide their code. Function definitions, on the other hand, are complete.

In C++, a global variable that is hidden by a local with the same name can be accessed using the scope operator (`::`). For example, if `x` is a global that is hidden in a function by a local named `x`, the global could be referenced as `::x`.

PHP statements can be interspersed with function definitions. Variables in PHP are implicitly declared when they appear in statements. Any variable that is implicitly declared outside any function is a global variable; variables implicitly declared in functions are local variables. The scope of global variables extends from their declarations to the end of the program but skips over any subsequent function definitions. So, global variables are not implicitly visible in any function. Global variables can be made visible in functions in their scope in two ways: (1) If the function includes a local variable with the same name as a global, that global can be accessed through the `$GLOBALS` array, using the name of the global as a string literal subscript, and (2) if there is no local variable in the function with the same name as the global, the global can be made visible by including it in a `global` declaration statement. Consider the following example:

```
$day = "Monday";
$month = "January";

function calendar() {
  $day = "Tuesday";
  global $month;
  print "local day is $day <br />";
  $gday = $GLOBALS['day'];
  print "global day is $gday <br \>";
  print "global month is $month <br />";
}

calendar();
```

Interpretation of this code produces the following:

```
local day is Tuesday
global day is Monday
global month is January
```

The global variables of JavaScript are very similar to those of PHP, except that there is no way to access a global variable in a function that has declared a local variable with the same name.

The visibility rules for global variables in Python are unusual. Variables are not normally declared, as in PHP. They are implicitly declared when they appear as the targets of assignment statements. A global variable can be referenced in a function, but a global variable can be assigned in a function only

if it has been declared to be global in the function. Consider the following examples:

```
day = "Monday"

def tester():
  print "The global day is:", day

tester()
```

The output of this script, because globals can be referenced directly in functions, is as follows:

```
The global day is: Monday
```

The following script attempts to assign a new value to the global `day`:

```
day = "Monday"

def tester():
  print "The global day is:", day
  day = "Tuesday"
  print "The new value of day is:", day

tester()
```

This script creates an `UnboundLocalError` error message, because the assignment to `day` in the second line of the body of the function makes `day` a local variable, which makes the reference to `day` in the first line of the body of the function an illegal forward reference to the local.

The assignment to `day` can be to the global variable if `day` is declared to be global at the beginning of the function. This prevents the assignment to `day` from creating a local variable. This is shown in the following script:

```
day = "Monday"

def tester():
  global day
  print "The global day is:", day
  day = "Tuesday"
  print "The new value of day is:", day

tester()
```

The output of this script is as follows:

```
The global day is: Monday
The new value of day is: Tuesday
```

Functions can be nested in Python. Variables defined in nesting functions are accessible in a nested function through static scoping, but such variables must be declared **`nonlocal`** in the nested function.[10] An example skeletal program in Section 5.7 illustrates accesses to nonlocal variables.

All names defined outside function definitions in F# are globals. Their scope extends from their definitions to the end of the file.

Declaration order and global variables are also issues in the class and member declarations in object-oriented languages. These are discussed in Chapter 12.

5.5.5 Evaluation of Static Scoping

Static scoping provides a method of nonlocal access that works well in many situations. However, it is not without its problems. First, in most cases it allows more access to both variables and subprograms than is necessary. It is simply too crude a tool for concisely specifying such restrictions. Second, and perhaps more important, is a problem related to program evolution. Software is highly dynamic—programs that are used regularly continually change. These changes often result in restructuring, thereby destroying the initial structure that restricted variable and subprogram access. To avoid the complexity of maintaining these access restrictions, developers often discard structure when it gets in the way. Thus, getting around the restrictions of static scoping can lead to program designs that bear little resemblance to the original, even in areas of the program in which changes have not been made. Designers are encouraged to use far more globals than are necessary. All subprograms can end up being nested at the same level, in the main program, using globals instead of deeper levels of nesting.[11] Moreover, the final design may be awkward and contrived, and it may not reflect the underlying conceptual design. These and other defects of static scoping are discussed in detail in Clarke, Wileden, and Wolf (1980). An alternative to the use of static scoping to control access to variables and subprograms is an encapsulation construct, which is included in many newer languages. Encapsulation constructs are discussed in Chapter 11.

5.5.6 Dynamic Scope

The scope of variables in APL, SNOBOL4, and the early versions of LISP is dynamic. Perl and Common LISP also allow variables to be declared to have dynamic scope, although the default scoping mechanism in these languages is static. **Dynamic scoping** is based on the calling sequence of subprograms, not on their spatial relationship to each other. Thus, the scope can be determined only at run time.

---

10. The **`nonlocal`** reserved word was introduced in Python 3.

11. Sounds like the structure of a C program, doesn't it?

Consider again the function `big` from Section 5.5.1, which is reproduced here, minus the function calls:

```
function big() {
  function sub1() {
    var x = 7;
  }
  function sub2() {
    var y = x;
    var z = 3;
  }
  var x = 3;
}
```

Assume that dynamic-scoping rules apply to nonlocal references. The meaning of the identifier `x` referenced in `sub2` is dynamic—it cannot be determined at compile time. It may reference the variable from either declaration of `x`, depending on the calling sequence.

One way the correct meaning of `x` can be determined during execution is to begin the search with the local declarations. This is also the way the process begins with static scoping, but that is where the similarity between the two techniques ends. When the search of local declarations fails, the declarations of the dynamic parent, or calling function, are searched. If a declaration for `x` is not found there, the search continues in that function's dynamic parent, and so forth, until a declaration for `x` is found. If none is found in any dynamic ancestor, it is a run-time error.

Consider the two different call sequences for `sub2` in the earlier example. First, `big` calls `sub1`, which calls `sub2`. In this case, the search proceeds from the local procedure, `sub2`, to its caller, `sub1`, where a declaration for `x` is found. So, the reference to `x` in `sub2` in this case is to the `x` declared in `sub1`. Next, `sub2` is called directly from `big`. In this case, the dynamic parent of `sub2` is `big`, and the reference is to the `x` declared in `big`.

Note that if static scoping were used, in either calling sequence discussed, the reference to `x` in `sub2` would be to `big`'s `x`.

Perl's dynamic scoping is unusual—in fact, it is not exactly like that discussed in this section, although the semantics are often that of traditional dynamic scoping (see Programming Exercise 1).

5.5.7 Evaluation of Dynamic Scoping

The effect of dynamic scoping on programming is profound. When dynamic scoping is used, the correct attributes of nonlocal variables visible to a program statement cannot be determined statically. Furthermore, a reference to the name of such a variable is not always to the same variable. A statement in a subprogram that contains a reference to a nonlocal variable can refer to different nonlocal variables during different executions of the subprogam. Several kinds of programming problems follow directly from dynamic scoping.

First, during the time span beginning when a subprogram begins its execution and ending when that execution ends, the local variables of the subprogram are all visible to any other executing subprogram, regardless of its textual proximity or how execution got to the currently executing subprogram. There is no way to protect local variables from this accessibility. Subprograms are *always* executed in the environment of all previously called subprograms that have not yet completed their executions. As a result, dynamic scoping results in less reliable programs than static scoping.

A second problem with dynamic scoping is the inability to type check references to nonlocals statically. This problem results from the inability to statically find the declaration for a variable referenced as a nonlocal.

Dynamic scoping also makes programs much more difficult to read, because the calling sequence of subprograms must be known to determine the meaning of references to nonlocal variables. This task can be virtually impossible for a human reader.

Finally, accesses to nonlocal variables in dynamic-scoped languages take far longer than accesses to nonlocals when static scoping is used. The reason for this is explained in Chapter 10.

On the other hand, dynamic scoping is not without merit. In many cases, the parameters passed from one subprogram to another are variables that are defined in the caller. None of these needs to be passed in a dynamically scoped language, because they are implicitly visible in the called subprogram.

It is not difficult to understand why dynamic scoping is not as widely used as static scoping. Programs in static-scoped languages are easier to read, are more reliable, and execute faster than equivalent programs in dynamic-scoped languages. It was precisely for these reasons that dynamic scoping was replaced by static scoping in most current dialects of LISP. Implementation methods for both static and dynamic scoping are discussed in Chapter 10.

5.6 Scope and Lifetime

Sometimes the scope and lifetime of a variable appear to be related. For example, consider a variable that is declared in a Java method that contains no method calls. The scope of such a variable is from its declaration to the end of the method. The lifetime of that variable is the period of time beginning when the method is entered and ending when execution of the method terminates. Although the scope and lifetime of the variable are clearly not the same, because static scope is a textual, or spatial, concept whereas lifetime is a temporal concept, they at least appear to be related in this case.

This apparent relationship between scope and lifetime does not hold in other situations. In C and C++, for example, a variable that is declared in a function using the specifier **`static`** is statically bound to the scope of that function and is also statically bound to storage. So, its scope is static and local to the function, but its lifetime extends over the entire execution of the program of which it is a part.

Scope and lifetime are also unrelated when subprogram calls are involved. Consider the following C++ functions:

```
void printheader() {
  ...
 }  /* end of printheader */
void compute() {
  int sum;
  ...
  printheader();
 }  /* end of compute */
```

The scope of the variable `sum` is completely contained within the `compute` function. It does not extend to the body of the function `printheader`, although `printheader` executes in the midst of the execution of `compute`. However, the lifetime of `sum` extends over the time during which `printheader` executes. Whatever storage location `sum` is bound to before the call to `printheader`, that binding will continue during and after the execution of `printheader`.

5.7 Referencing Environments

The **referencing environment** of a statement is the collection of all variables that are visible in the statement. The referencing environment of a statement in a static-scoped language is the variables declared in its local scope plus the collection of all variables of its ancestor scopes that are visible. In such a language, the referencing environment of a statement is needed while that statement is being compiled, so code and data structures can be created to allow references to variables from other scopes during run time. Techniques for implementing references to nonlocal variables in both static- and dynamic-scoped languages are discussed in Chapter 10.

In Python, scopes can be created by function definitions. The referencing environment of a statement includes the local variables, plus all of the variables declared in the functions in which the statement is nested (excluding variables in nonlocal scopes that are hidden by declarations in nearer functions). Each function definition creates a new scope and thus a new environment. Consider the following Python skeletal program:

```
g = 3;  # A global

def sub1():
    a = 5;  # Creates a local
    b = 7;  # Creates another local
    ...  <------------------------------------ 1
  def sub2():
    global g;  # Global g is now assignable here
```

```
c = 9;  # Creates a new local
...  <------------------------------------- 2
def sub3():
  nonlocal c:  # Makes nonlocal c visible here
  g = 11;  # Creates a new local
  ...   <------------------------------- 3
```

The referencing environments of the indicated program points are as follows:

Point	*Referencing Environment*
1	local `a` and `b` (of `sub1`), global `g` for reference, but not for assignment
2	local `c` (of `sub2`), global `g` for both reference and for assignment
3	nonlocal `c` (of `sub2`), local `g` (of `sub3`)

Now consider the variable declarations of this skeletal program. First, note that, although the scope of `sub1` is at a higher level (it is less deeply nested) than `sub3`, the scope of `sub1` is not a static ancestor of `sub3`, so `sub3` does not have access to the variables declared in `sub1`. There is a good reason for this. The variables declared in `sub1` are stack dynamic, so they are not bound to storage if `sub1` is not in execution. Because `sub3` can be in execution when `sub1` is not, it cannot be allowed to access variables in `sub1`, which would not necessarily be bound to storage during the execution of `sub3`.

A subprogram is **active** if its execution has begun but has not yet terminated. The referencing environment of a statement in a dynamically scoped language is the locally declared variables, plus the variables of all other subprograms that are currently active. Once again, some variables in active subprograms can be hidden from the referencing environment. Recent subprogram activations can have declarations for variables that hide variables with the same names in previous subprogram activations.

Consider the following example program. Assume that the only function calls are the following: `main` calls `sub2`, which calls `sub1`.

```
void sub1() {
  int a, b;
  ...  <------------------ 1
}  /* end of sub1 */
void sub2() {
  int b, c;
  ....  <------------------ 2
  sub1();
}  /* end of sub2 */
void main() {
```

```
  int c, d;
  ...  <------------------ 3
  sub2();
}  /* end of main */
```

The referencing environments of the indicated program points are as follows:

Point	*Referencing Environment*
1	`a` and `b` of `sub1`, `c` of `sub2`, `d` of `main`, (`c` of `main` and `b` of `sub2` are hidden)
2	`b` and `c` of `sub2`, `d` of `main`, (`c` of `main` is hidden)
3	`c` and `d` of `main`

5.8 Named Constants

A **named constant** is a variable that is bound to a value only once. Named constants are useful as aids to readability and program reliability. Readability can be improved, for example, by using the name `pi` instead of the constant `3.14159265`.

Another important use of named constants is to parameterize a program. For example, consider a program that processes a fixed number of data values, say 100. Such a program usually uses the constant `100` in a number of locations for declaring array subscript ranges and for `loop` control limits. Consider the following skeletal Java program segment:

```
void example() {
  int[] intList = new int[100];
  String[] strList = new String[100];
  ...
  for (index = 0; index < 100; index++) {
    ...
  }
  ...
  for (index = 0; index < 100; index++) {
    ...
  }
  ...
  average = sum / 100;
  ...
}
```

When this program must be modified to deal with a different number of data values, all occurrences of `100` must be found and changed. On a large

program, this can be tedious and error prone. An easier and more reliable method is to use a named constant as a program parameter, as follows:

```
void example() {
  final int len = 100;
  int[] intList = new int[len];
  String[] strList = new String[len];
  ...
  for (index = 0; index < len; index++) {
    ...
  }
  ...
  for (index = 0; index < len; index++) {
    ...
  }
  ...
  average = sum / len;
  ...
}
```

Now, when the length must be changed, only one line must be changed (the variable `len`), regardless of the number of times it is used in the program. This is another example of the benefits of abstraction. The name `len` is an abstraction for the number of elements in some arrays and the number of iterations in some loops. This illustrates how named constants can aid modifiability.

Ada and C++ allow dynamic binding of values to named constants. This allows expressions containing variables to be assigned to constants in the declarations. For example, the C++ statement

```
const int result = 2 * width + 1;
```

declares `result` to be an integer type named constant whose value is set to the value of the expression `2 * width + 1`, where the value of the variable `width` must be visible when `result` is allocated and bound to its value.

Java also allows dynamic binding of values to named constants. In Java, named constants are defined with the **`final`** reserved word (as in the earlier example). The initial value can be given in the declaration statement or in a subsequent assignment statement. The assigned value can be specified with any expression.

C# has two kinds of named constants: those defined with **`const`** and those defined with **`readonly`**. The **`const`** named constants, which are implicitly **`static`**, are statically bound to values; that is, they are bound to values at compile time, which means those values can be specified only with literals or other **`const`** members. The **`readonly`** named constants, which are dynamically bound to values, can be assigned in the declaration or with a static

constructor.[12] So, if a program needs a constant-valued object whose value is the same on every use of the program, a **`const`** constant is used. However, if a program needs a constant-valued object whose value is determined only when the object is created and can be different for different executions of the program, then a **`readonly`** constant is used.

Ada allows named constants of enumeration and structured types, which are discussed in Chapter 6.

The discussion of binding values to named constants naturally leads to the topic of initialization, because binding a value to a named constant is the same process, except it is permanent.

In many instances, it is convenient for variables to have values before the code of the program or subprogram in which they are declared begins executing. The binding of a variable to a value at the time it is bound to storage is called **initialization**. If the variable is statically bound to storage, binding and initialization occur before run time. In these cases, the initial value must be specified as a literal or an expression whose only nonliteral operands are named constants that have already been defined. If the storage binding is dynamic, initialization is also dynamic and the initial values can be any expression.

In most languages, initialization is specified on the declaration that creates the variable. For example, in C++, we could have

```
int sum = 0;
int* ptrSum = &sum;
char name[] = "George Washington Carver";
```

SUMMARY

Case sensitivity and the relationship of names to special words, which are either reserved words or keywords, are the design issues for names.

Variables can be characterized by the sextuple of attributes: name, address, value, type, lifetime, and scope.

Aliases are two or more variables bound to the same storage address. They are regarded as detrimental to reliability but are difficult to eliminate entirely from a language.

Binding is the association of attributes with program entities. Knowledge of the binding times of attributes to entities is essential to understanding the semantics of programming languages. Binding can be static or dynamic. Declarations, either explicit or implicit, provide a means of specifying the static binding of variables to types. In general, dynamic binding allows greater flexibility but at the expense of readability, efficiency, and reliability.

12. Static constructors in C# run at some indeterminate time before the class is instantiated.

Scalar variables can be separated into four categories by considering their lifetimes: static, stack dynamic, explicit heap dynamic, and implicit heap dynamic.

Static scoping is a central feature of ALGOL 60 and some of its descendants. It provides a simple, reliable, and efficient method of allowing visibility of nonlocal variables in subprograms. Dynamic scoping provides more flexibility than static scoping but, again, at the expense of readability, reliability, and efficiency.

Most functional languages allow the user to create local scopes with `let` constructs, which limit the scope of their defined names.

The referencing environment of a statement is the collection of all of the variables that are visible to that statement.

Named constants are simply variables that are bound to values only once.

REVIEW QUESTIONS

1. What are the design issues for names?
2. What is the potential danger of case-sensitive names?
3. In what way are reserved words better than keywords?
4. What is an alias?
5. Which category of C++ reference variables is always aliases?
6. What is the *l*-value of a variable? What is the *r*-value?
7. Define *binding* and *binding time*.
8. After language design and implementation [what are the four times bindings can take place in a program?]
9. Define *static binding* and *dynamic binding*.
10. What are the advantages and disadvantages of implicit declarations?
11. What are the advantages and disadvantages of dynamic type binding?
12. Define *static*, *stack-dynamic*, *explicit heap-dynamic*, and *implicit heap-dynamic variables*. What are their advantages and disadvantages?
13. Define *lifetime*, *scope*, *static scope*, and *dynamic scope*.
14. How is a reference to a nonlocal variable in a static-scoped program connected to its definition?
15. What is the general problem with static scoping?
16. What is the referencing environment of a statement?
17. What is a static ancestor of a subprogram? What is a dynamic ancestor of a subprogram?
18. What is a block?
19. What is the purpose of the `let` constructs in functional languages?
20. What is the difference between the names defined in an ML `let` construct from the variables declared in a C block?

21. Describe the encapsulation of an F# `let` inside a function and outside all functions.
22. What are the advantages and disadvantages of dynamic scoping?
23. What are the advantages of named constants?

PROBLEM SET

1. Which of the following identifier forms is most readable? Support your decision.

   `SumOfSales`

   `sum_of_sales`

   `SUMOFSALES`
2. Some programming languages are typeless. What are the obvious advantages and disadvantages of having no types in a language?
3. Write a simple assignment statement with one arithmetic operator in some language you know. For each component of the statement, list the various bindings that are required to determine the semantics when the statement is executed. For each binding, indicate the binding time used for the language.
4. Dynamic type binding is closely related to implicit heap-dynamic variables. Explain this relationship.
5. Describe a situation when a history-sensitive variable in a subprogram is useful.
6. Consider the following JavaScript skeletal program:

```
// The main program
var x;
function sub1() {
  var x;
  function sub2() {
    ...
  }
}
function sub3() {
  ...
}
```

   Assume that the execution of this program is in the following unit order:

   `main` calls `sub1`

   `sub1` calls `sub2`

   `sub2` calls `sub3`

a. Assuming static scoping, in the following, which declaration of x is the correct one for a reference to x?

   i. sub1
   ii. sub2
   iii. sub3

b. Repeat part a, but assume dynamic scoping.

7. Assume the following JavaScript program was interpreted using static-scoping rules. What value of x is displayed in function sub1? Under dynamic-scoping rules, what value of x is displayed in function sub1?

```
var x;
function sub1() {
  document.write("x = " + x + "<br />");
}
function sub2() {
  var x;
  x = 10;
  sub1();
}
x = 5;
sub2();
```

8. Consider the following JavaScript program:

```
var x, y, z;
function sub1() {
  var a, y, z;
  function sub2() {
    var a, b, z;
    ...
  }
  ...
}
function sub3() {
  var a, x, w;
  ...
}
```

List all the variables, along with the program units where they are declared, that are visible in the bodies of `sub1`, `sub2`, and `sub3`, assuming static scoping is used.

9. Consider the following Python program:

```
x = 1;
y = 3;
z = 5;
def sub1():
  a = 7;
  y = 9;
  z = 11;
  ...
def sub2():
  global x;
  a = 13;
  x = 15;
  w = 17;
  ...
  def sub3():
    nonlocal a;
    a = 19;
    b = 21;
    z = 23;
    ...
...
```

List all the variables, along with the program units where they are declared, that are visible in the bodies of `sub1`, `sub2`, and `sub3`, assuming static scoping is used.

10. Consider the following C program:

```
void fun(void) {
  int a, b, c; /* definition 1 */
  ...
  while (...) {
    int b, c, d; /*definition 2 */
    ... <------------- 1
    while (...) {
```

```
      int c, d, e; /* definition 3 */
      ... <------------------ 2
      }
     ... <------------------- 3
     }
   ... <--------------------- 4
}
```

For each of the four marked points in this function, list each visible variable, along with the number of the definition statement that defines it.

11. Consider the following skeletal C program:

```
void fun1(void);  /* prototype */
void fun2(void);  /* prototype */
void fun3(void);  /* prototype */
void main() {
  int a, b, c;
  ...
 }
void fun1(void) {
  int b, c, d;
  ...
 }
void fun2(void) {
  int c, d, e;
  ...
 }
void fun3(void) {
  int d, e, f;
  ...
 }
```

Given the following calling sequences and assuming that dynamic scoping is used, what variables are visible during execution of the last function called? Include with each visible variable the name of the function in which it was defined.

a. `main` calls `fun1`; `fun1` calls `fun2`; `fun2` calls `fun3`.

b. `main` calls `fun1`; `fun1` calls `fun3`.

c. `main` calls `fun2`; `fun2` calls `fun3`; `fun3` calls `fun1`.

d. `main` calls `fun3`; `fun3` calls `fun1`.

e. `main` calls `fun1`; `fun1` calls `fun3`; `fun3` calls `fun2`.

f. `main` calls `fun3`; `fun3` calls `fun2`; `fun2` calls `fun1`.

12. Consider the following program, written in JavaScript-like syntax:

```
// main program
var x, y, z;

function sub1() {
  var a, y, z;
  ...
}
function sub2() {
  var a, b, z;
  ...
}
function sub3() {
  var a, x, w;
  ...
}
```

Given the following calling sequences and assuming that dynamic scoping is used, what variables are visible during execution of the last subprogram activated? Include with each visible variable the name of the unit where it is declared.

a. `main` calls `sub1`; `sub1` calls `sub2`; `sub2` calls `sub3`.

b. `main` calls `sub1`; `sub1` calls `sub3`.

c. `main` calls `sub2`; `sub2` calls `sub3`; `sub3` calls `sub1`.

d. `main` calls `sub3`; `sub3` calls `sub1`.

e. `main` calls `sub1`; `sub1` calls `sub3`; `sub3` calls `sub2`.

f. `main` calls `sub3`; `sub3` calls `sub2`; `sub2` calls `sub1`.

PROGRAMMING EXERCISES

1. Perl allows both static and a kind of dynamic scoping. Write a Perl program that uses both and clearly shows the difference in effect of the two. Explain clearly the difference between the dynamic scoping described in this chapter and that implemented in Perl.

2. Write a Common LISP program that clearly shows the difference between static and dynamic scoping.
3. Write a JavaScript script that has subprograms nested three deep and in which each nested subprogram references variables defined in all of its enclosing subprograms.
4. Repeat Programming Exercise 3 with Python.
5. Write a C function that includes the following sequence of statements:

```
x = 21;
int x;
x = 42;
```

   Run the program and explain the results. Rewrite the same code in C++ and Java and compare the results.
6. Write test programs in C++, Java, and C# to determine the scope of a variable declared in a **`for`** statement. Specifically, the code must determine whether such a variable is visible after the body of the **`for`** statement.
7. Write three functions in C or C++: one that declares a large array statically, one that declares the same large array on the stack, and one that creates the same large array from the heap. Call each of the subprograms a large number of times (at least 100,000) and output the time required by each. Explain the results.